STRUCTURAL SYSTEMS 1 *ARC 2713*

STEEL BUILDING: FLOOR & LATERAL

Lecture 10

1. STEEL corrugated deck and wide-flange beams & girders
2. STEEL FLOOR parameters & configuration options
3. STEEL FLOOR framing example
4. STEEL FLOOR SIZING for Deck, Beam, and Girder
5. STEEL FLOOR Template Examples

6. STEEL LATERAL MECHANISMS Horizontal Diaphragms & Concentric Braced Frames

STEEL MATERIAL measures of strength

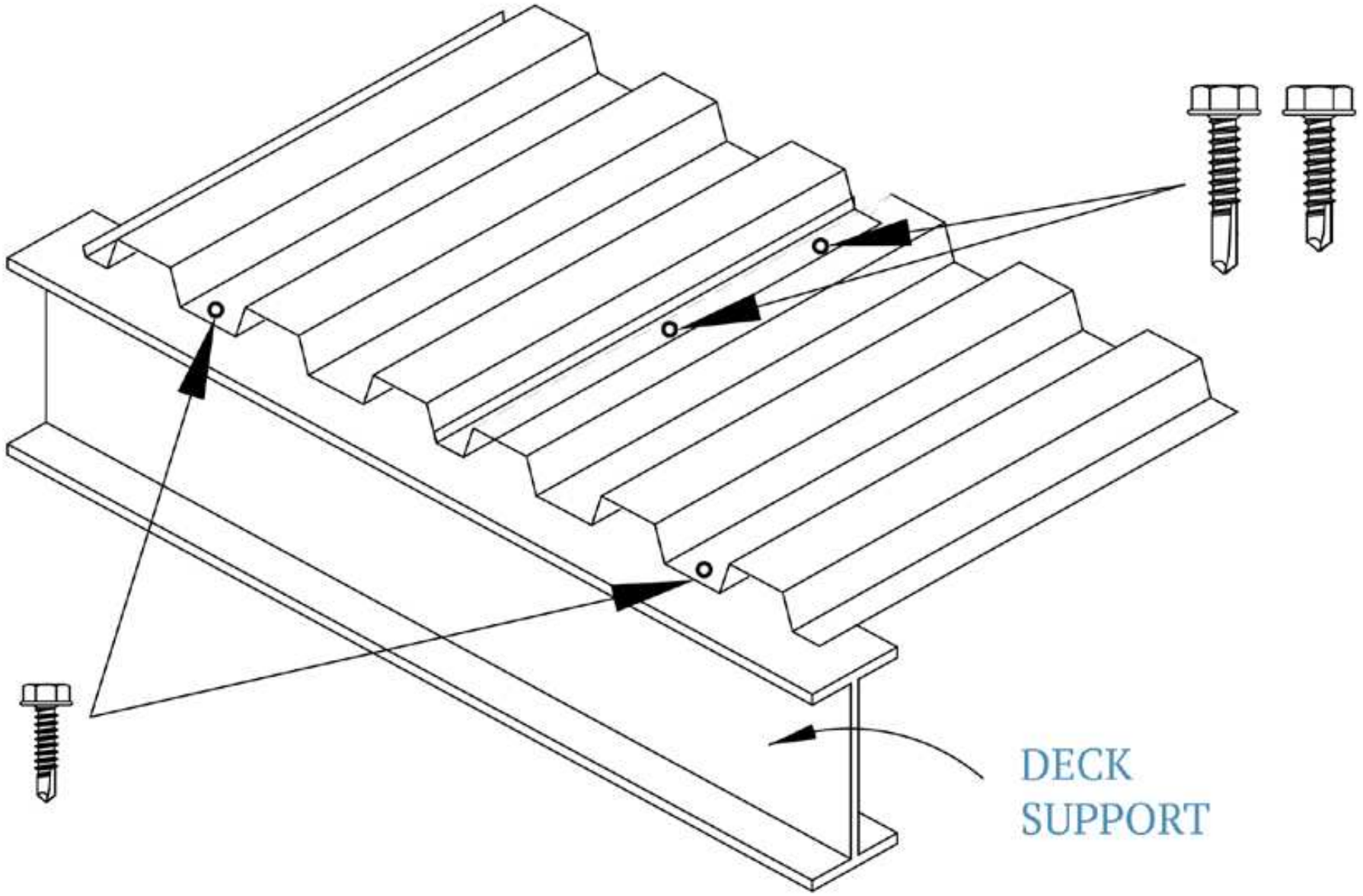
…how is steel optimized as structural shapes or elements?



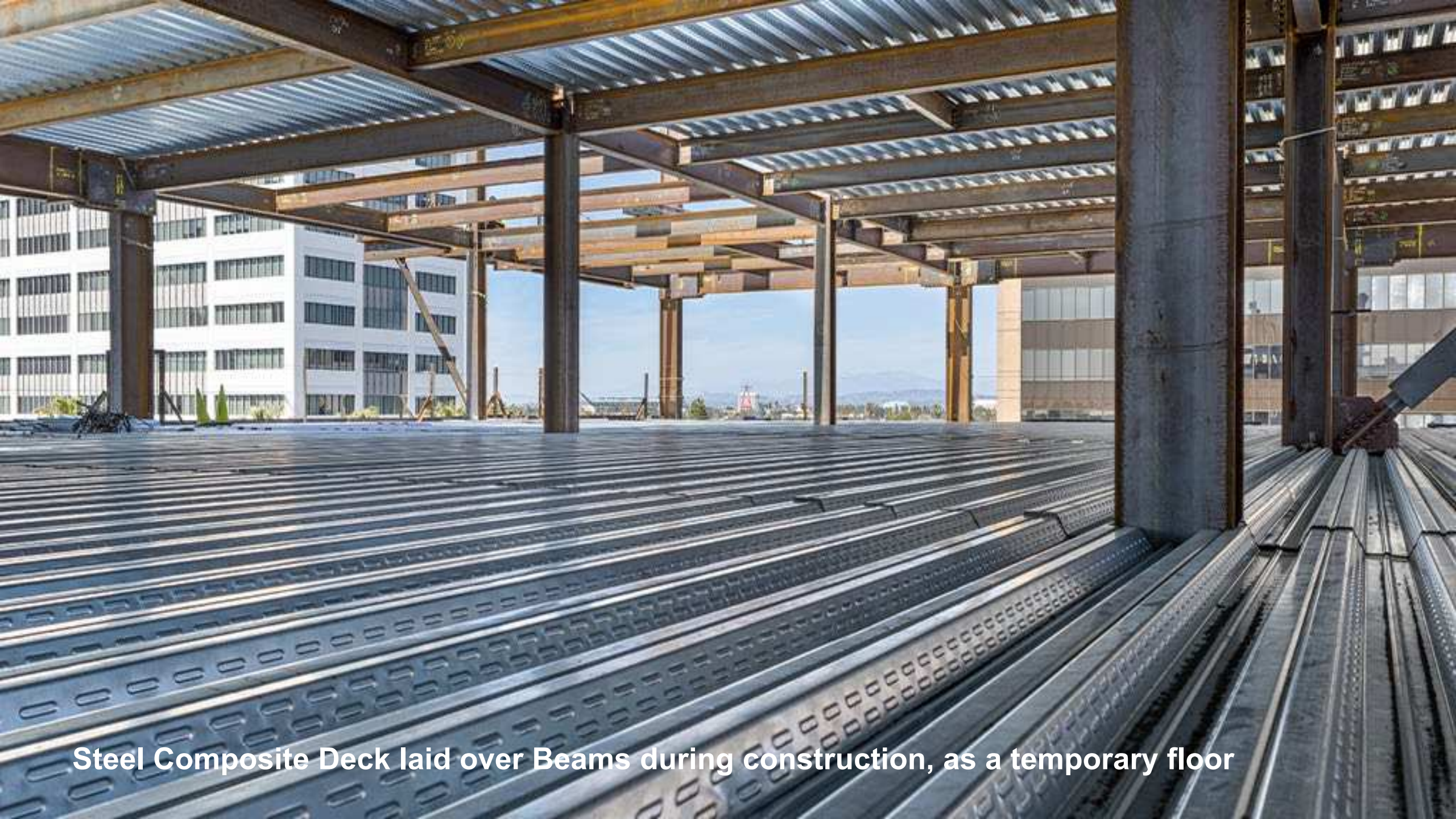


Structural Steel Corrugated Deck (B-Type 1.5” corrugated most common)

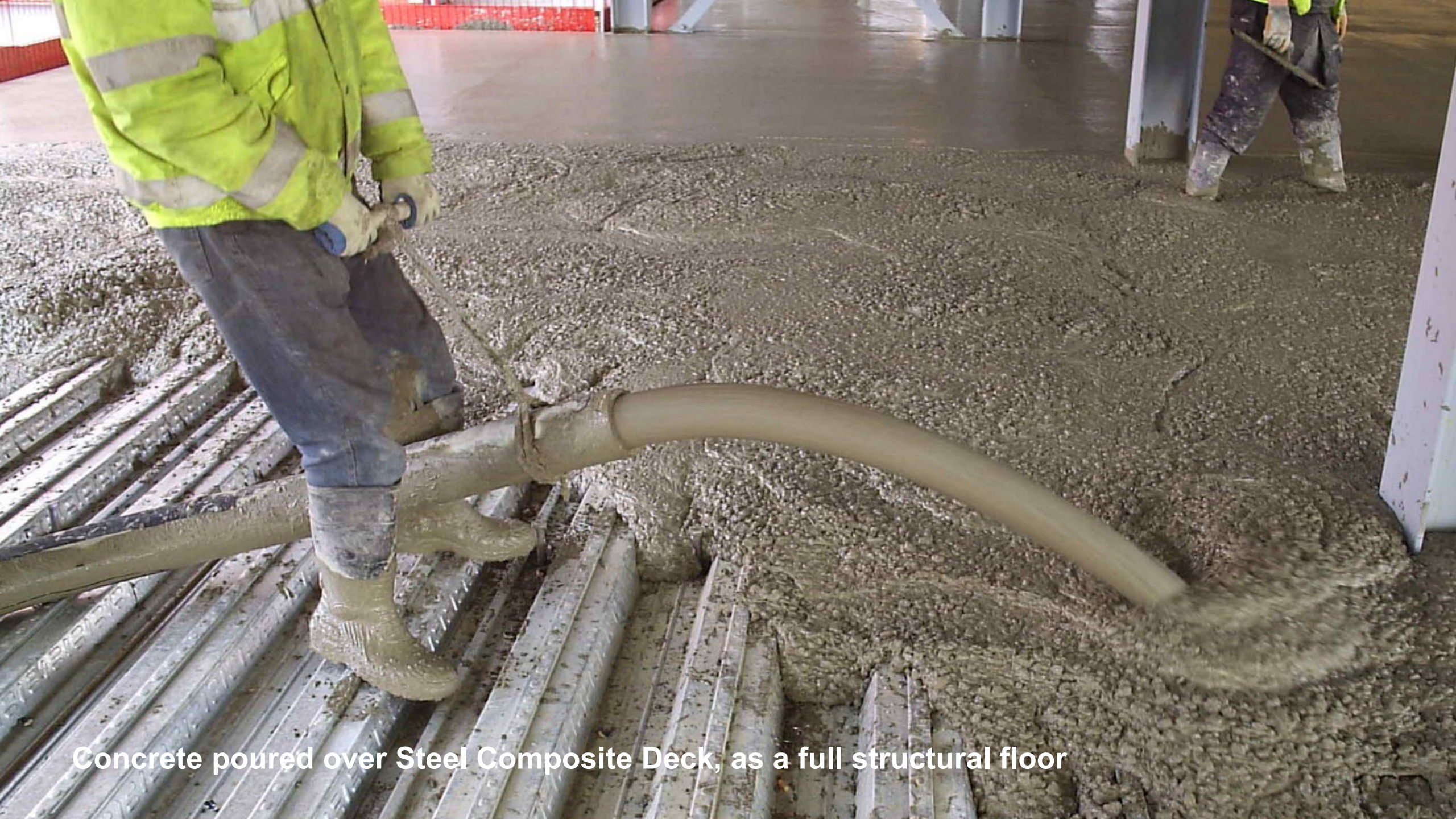
Deck anchored with self-drilling screws



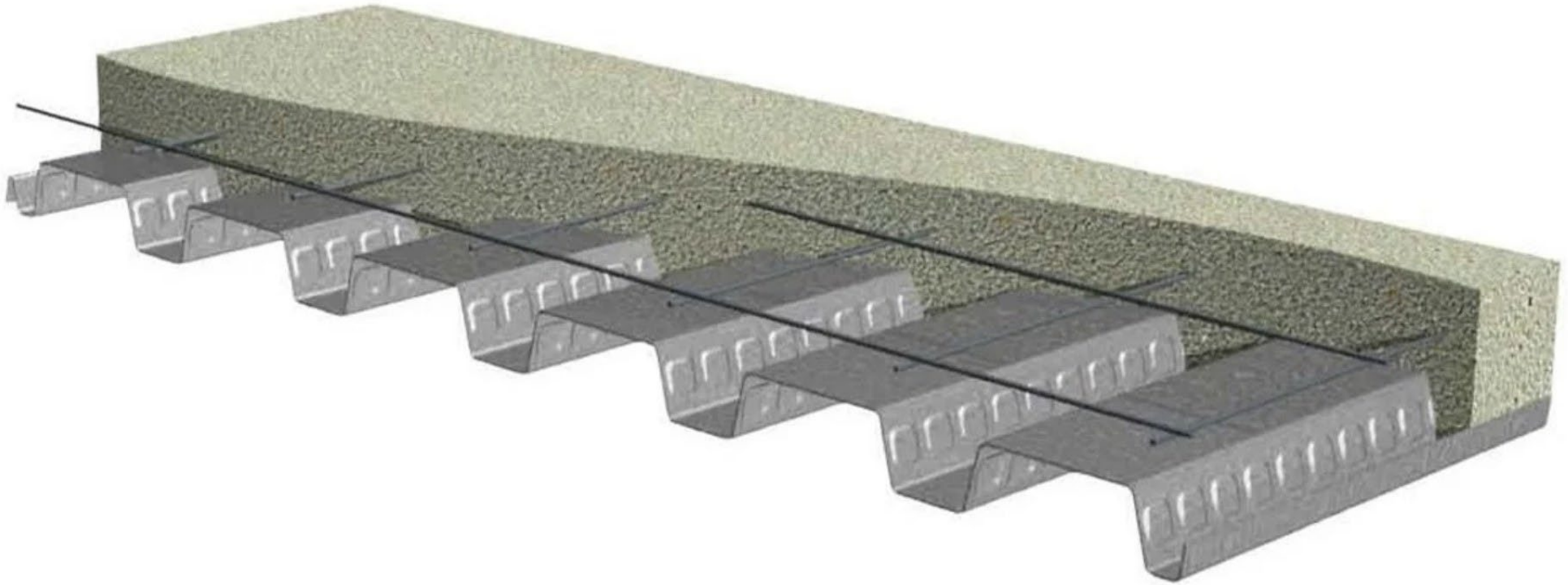
Steel Deck spans between Beams, always perpendicular except at edge conditions



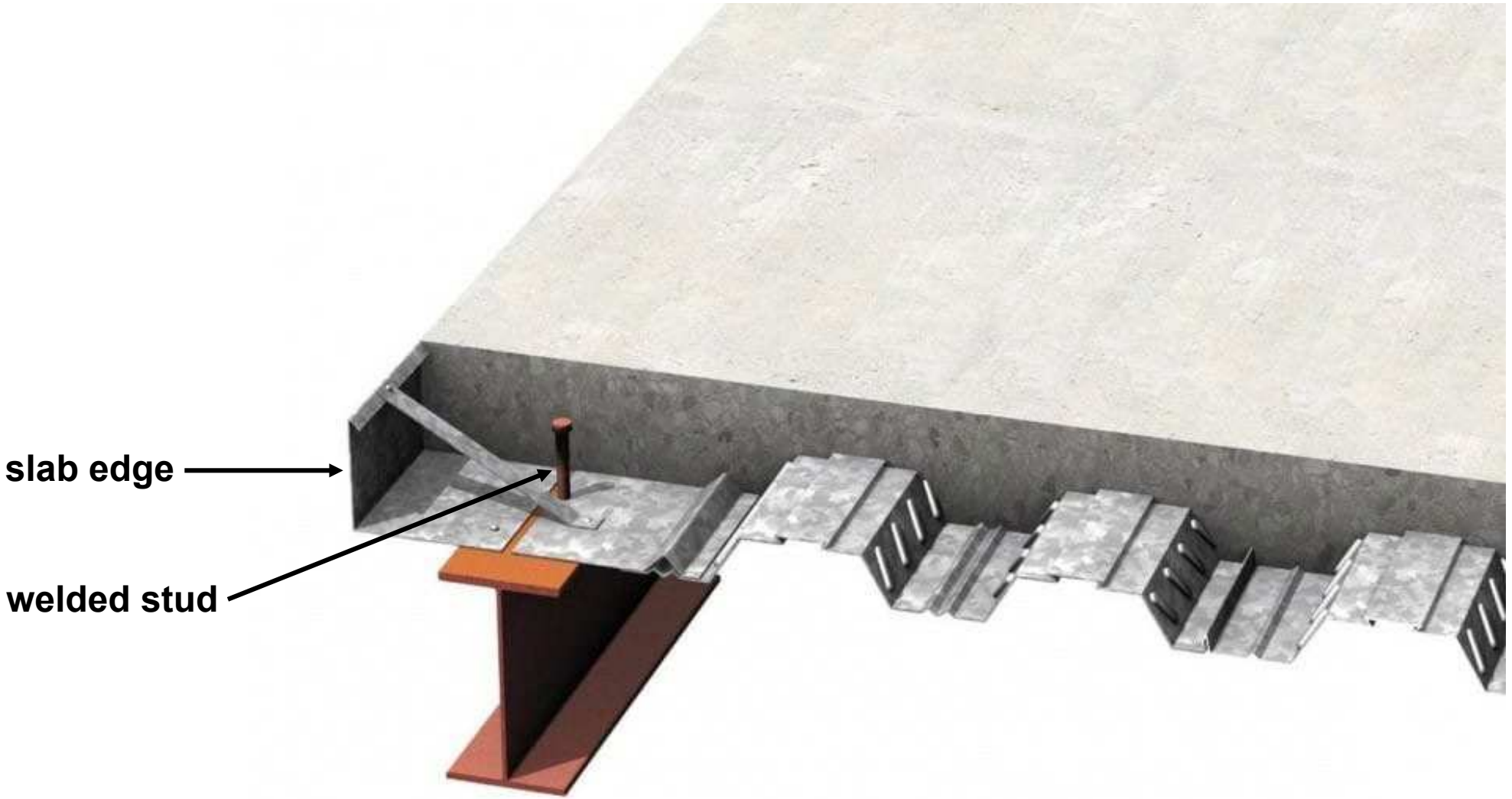
Steel Composite Deck laid over Beams during construction, as a temporary floor



Concrete poured over Steel Composite Deck, as a full structural floor

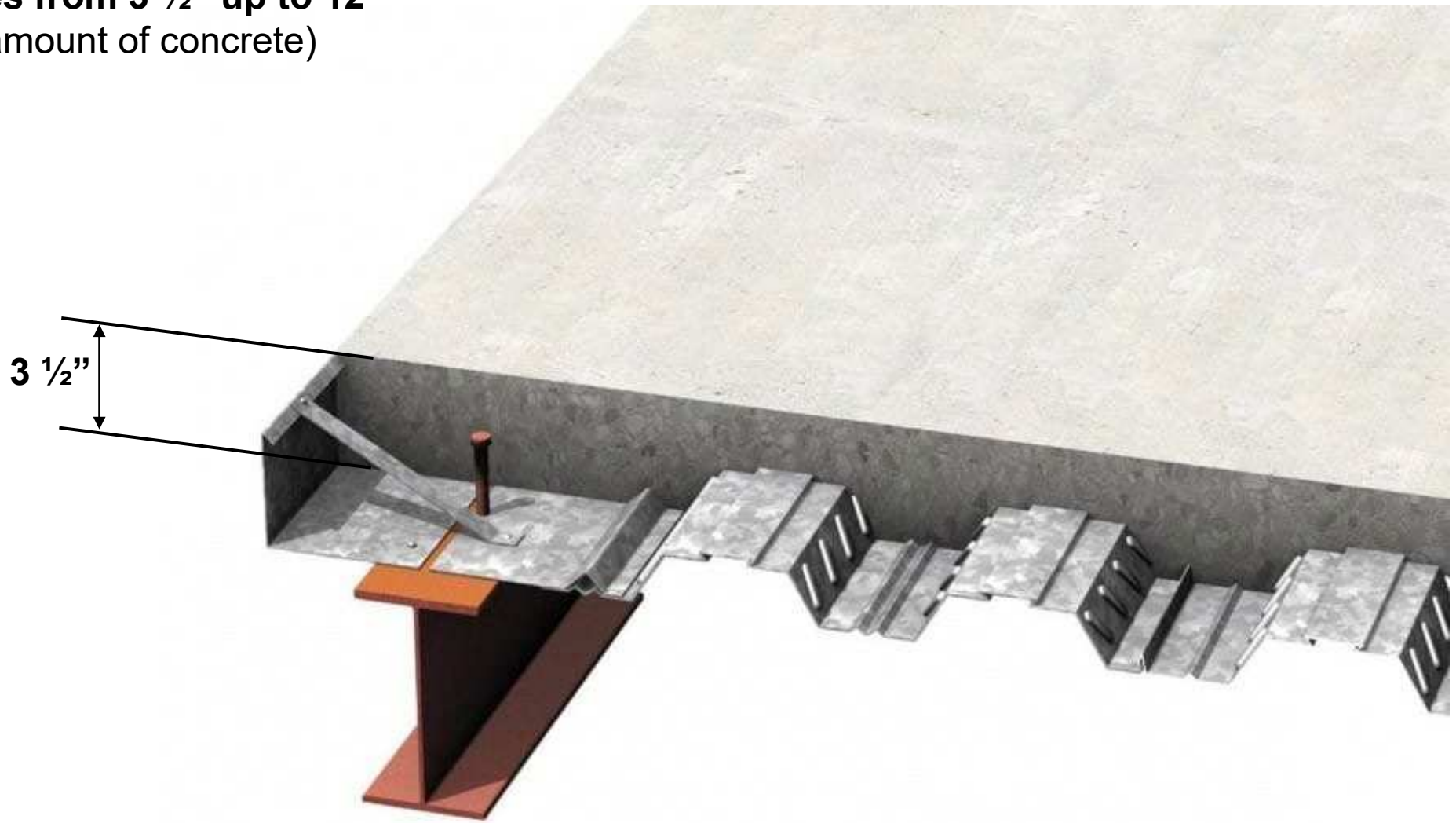


Composite Steel Deck has “keys” to form a composite structural bond with concrete



Composite Steel Deck at end-condition, with steel studs on Beam, to bond with concrete

Slab thickness varies from 3 ½” up to 12”
(3 ½” uses the least amount of concrete)



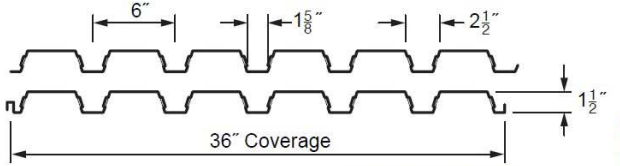
Composite Steel Deck at end-condition, with steel studs on Beam, to bond with concrete

1.5CD, 1.5CDI Composite Deck

NORMAL-WEIGHT CONCRETE

PROPERTIES

Gage	Thickness (in)	Cover (in)	Weight (psf)
22	0.0295	36	1.63
20	0.0358		1.98
18	0.0474		2.62
16	0.0598		3.30



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REFER TO ESR-3818

ASD SECTION PROPERTIES

DESIGN STRENGTHS (No Concrete Fill)

Gage	F_y (ksi)	A_s (in²/ft)	I_D 1 Span (in⁴/ft)	I_D 2+ Span (in⁴/ft)	I_p (in⁴/ft)	I_n (in⁴/ft)	S_p (in³/ft)	S_n (in³/ft)	Mn,p/Ω (in-lb/ft)	Mn,n/Ω (in-lb/ft)	Vn/Ω (lb/ft)	*Rbe/Ω (lb/ft)	*Rbi/Ω (lb/ft)
22	40	0.480	0.163	0.175	0.157	0.175	0.176	0.183	4215	4391	2009	654	1181
20	40	0.582	0.205	0.213	0.201	0.213	0.224	0.235	5375	5624	2428	932	1696
18	40	0.770	0.280	0.281	0.279	0.281	0.304	0.315	7291	7553	3192	1557	2858
16	40	0.971	0.355	0.355	0.355	0.355	0.390	0.395	9332	9454	3996	2381	4396

CONSTRUCTION CLEAR SPANS

Total Slab Depth (in)	Gage	Concrete Weight (psf)	Maximum Construction Clear Span (ft - in)				Concrete Volume (ft^3/ft^2)
			Single	Double	Triple	Cantilever	
3 1/2	22	31	5 - 9	6 - 9	6 - 10	2 - 0	0.213
	20		6 - 9	8 - 0	8 - 1	2 - 5	
	18		8 - 4	9 - 8	9 - 11	3 - 1	
	16		9 - 8	10 - 9	11 - 1	3 - 8	

SLAB DESIGN - ALLOWABLE SUPERIMPOSED UNIFORM LOADS AND MINIMUM SDI SLAB REINFORCEMENT

Normal-Weight Concrete (γc = 145 pcf, f'c = 3000 psi, n = 9)

Total Slab Depth (in)	Gage	Allowable Superimposed Uniform Load (psf) Clear Span (ft - in)									Recommended Welded Wire Fabric	Wire Area (in^2/ft)
		6 - 0	6 - 6	7 - 0	7 - 6	8 - 0	8 - 6	9 - 0	10 - 0	11 - 0		
3 1/2	22	294	246	207	176	151	130	113	85	65	6x6 - W1.4xW1.4	0.028
	20	354	297	251	215	185	160	139	106	82		
	18	400	386	328	281	243	212	185	139	105		
	16	400	400	400	349	301	251	212	154	116		
4	22	370	310	262	223	191	165	143	109	83	6x6 - W1.4xW1.4	0.028
	20	400	375	318	272	234	203	177	136	105		
	18	400	400	400	357	309	269	236	183	145		
	16	400	400	400	400	385	336	295	229	172		
4 1/2	22	400	377	319	272	234	202	175	133	103	6x6 - W1.4xW1.4	0.028
	20	400	400	387	331	286	248	216	167	130		
	18	400	400	400	400	378	330	289	225	178		
	16	400	400	400	400	400	400	363	285	228		
5	22	400	400	378	322	277	240	208	159	123	6x6 - W2.0xW2.0	0.040
	20	400	400	400	393	339	295	258	199	156		
	18	400	400	400	400	400	392	344	269	213		
	16	400	400	400	400	400	400	400	341	273		
5 1/2	22	400	400	400	374	322	279	242	186	144	6x6 - W2.0xW2.0	0.040
	20	400	400	400	400	394	343	300	232	182		
	18	400	400	400	400	400	400	400	314	249		
	16	400	400	400	400	400	400	400	398	319		
6	22	400	400	400	400	367	318	277	212	165	4x4 - W1.4xW1.4	0.042
	20	400	400	400	400	400	392	342	265	208		
	18	400	400	400	400	400	400	400	360	286		
	16	400	400	400	400	400	400	400	400	366		

Notes:

7. Welded wire fabric shown in the table above is the SDI minimum required for temperature and shrinkage (0.00075 x concrete area above top of deck).
8. Allowable Superimposed Uniform Loads shown in table above are based on the following criteria:
   - Unfactored service level loads, determined using SDI design method per ANSI/SDI C-2017 Standard for Composite Steel Floor Deck-Slabs.
   - Single span conditions without negative bending reinforcing over supports.
   - The presence of shear studs have not been considered, design strength may be increased if shear studs are used.
   - Slab deflection is limited to minimum of Clear Span/360 or 1" under service level superimposed loading.

SPAN & LOAD TABLE: Composite Steel Deck

Span & Load Table

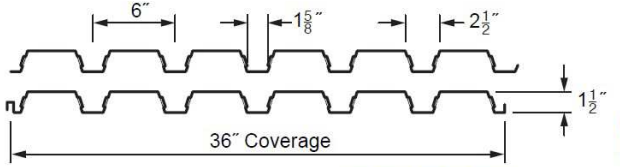
3 ½” Slab Depth (total depth)
20-gauge steel Composite Deck
with normal-weight concrete
(Composite steel deck self-weight: 2 psf)
(Concrete weight: 31 psf)

1.5CD, 1.5CDI Composite Deck

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PROPERTIES

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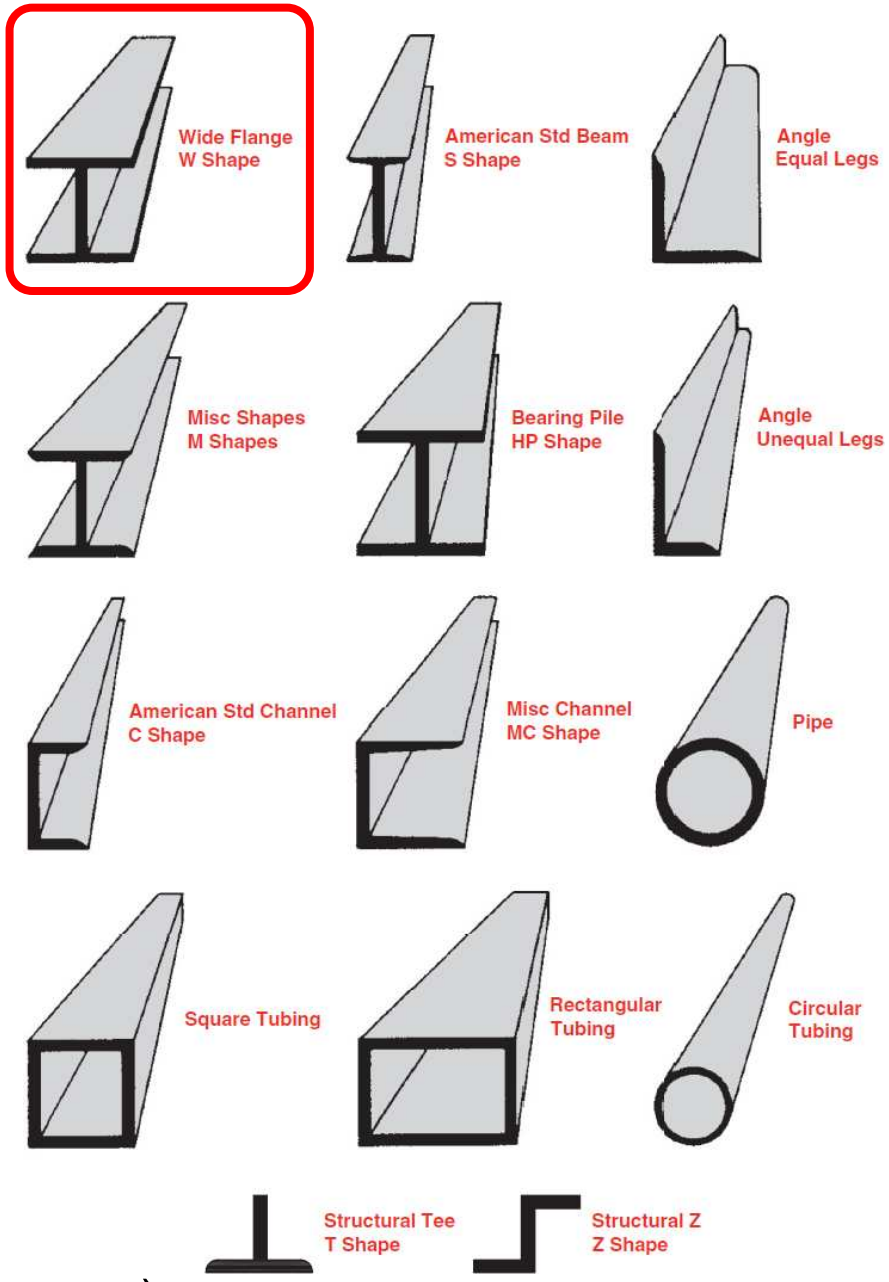
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3 1/2	22	294	246	207	176	151	130	113	85	65	6x6 - W1.4xW1.4	0.028
	20	354	297	251	215	185	160	139	106	82		
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	18	400	400	400	400	400	400	400	360	286		
	16	400	400	400	400	400	400	400	400	366		

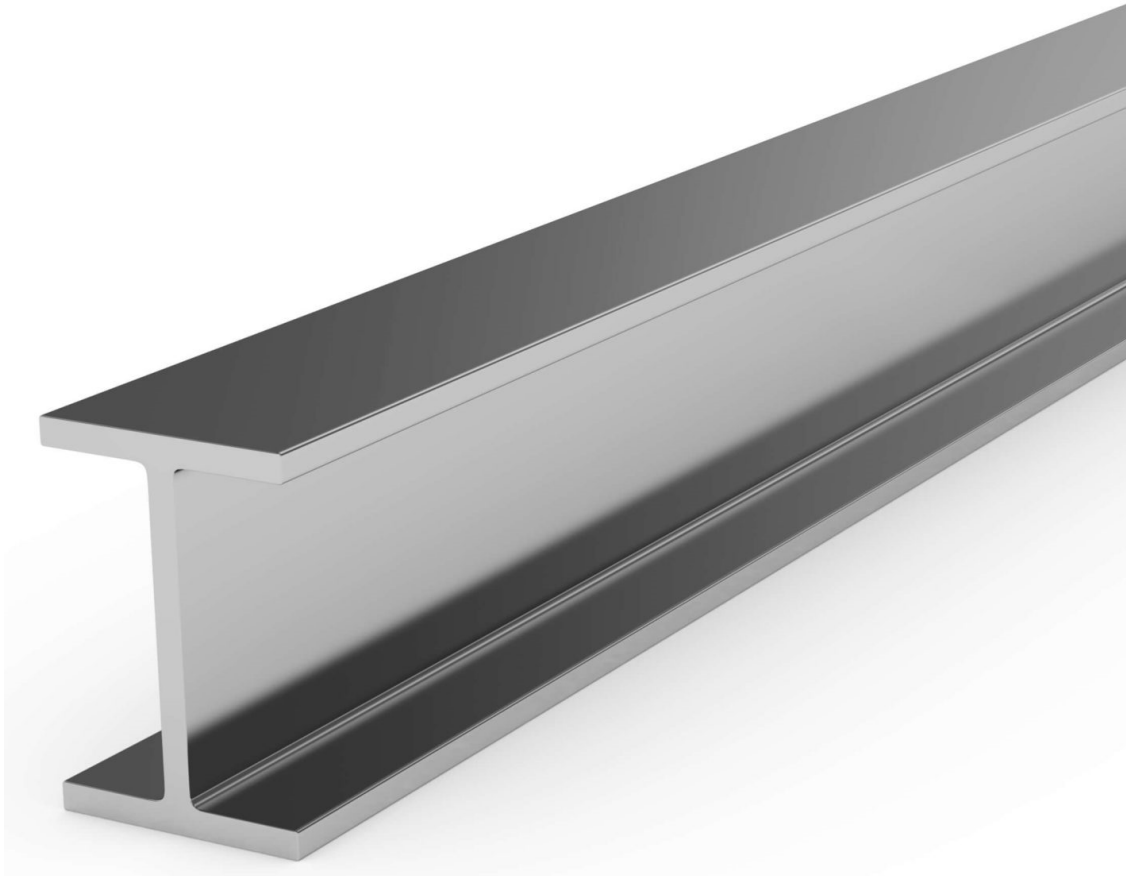
Notes:
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Composite Steel Deck for Steel Building Project





Structural Steel sections (wide-flange and angle most common)



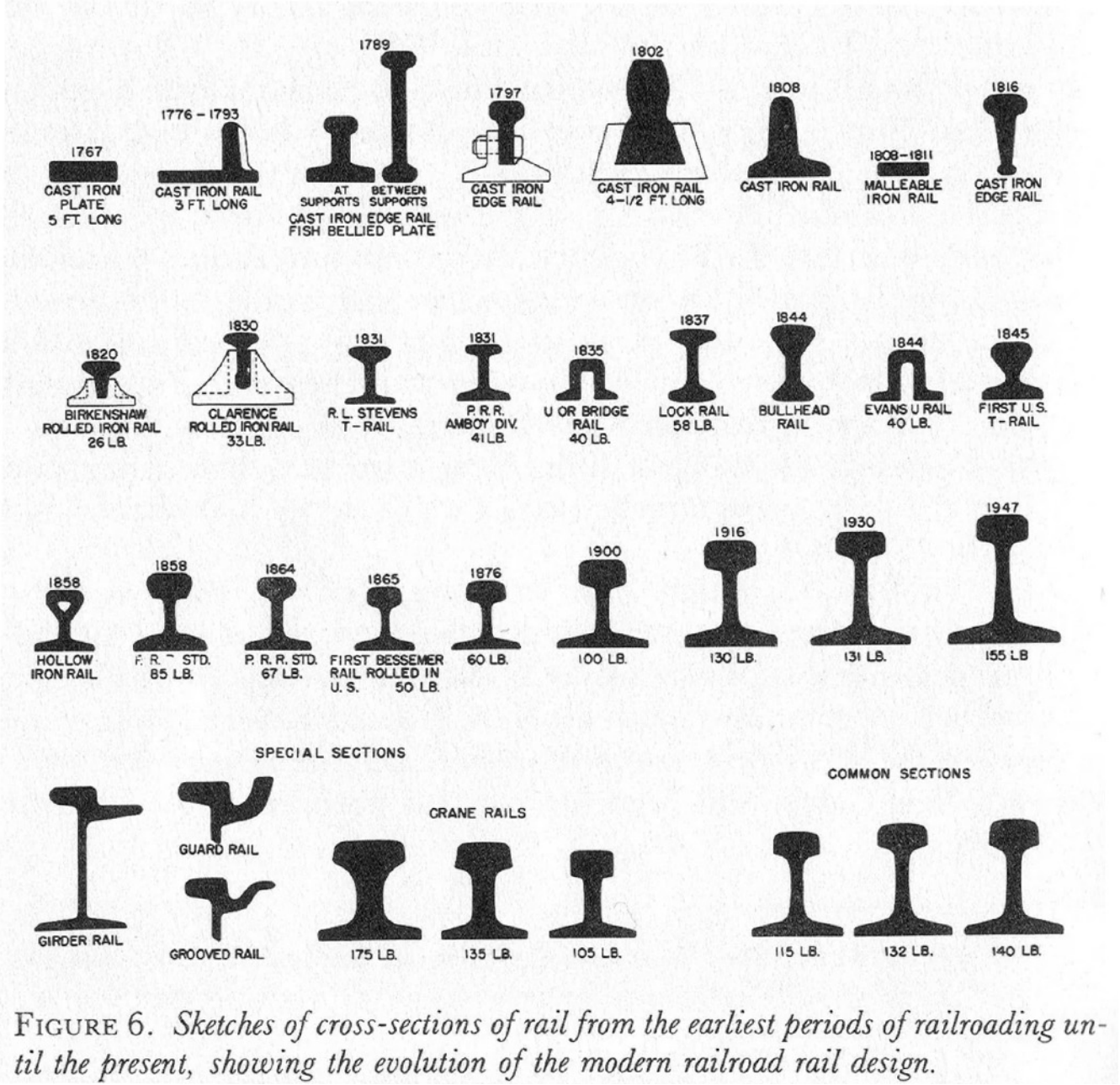
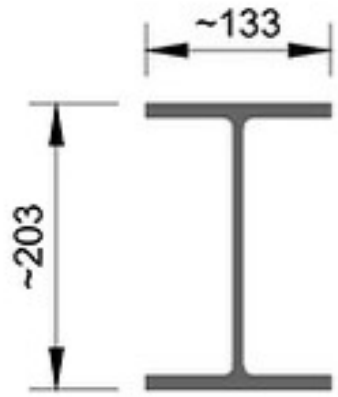


FIGURE 6. *Sketches of cross-sections of rail from the earliest periods of railroading until the present, showing the evolution of the modern railroad rail design.*

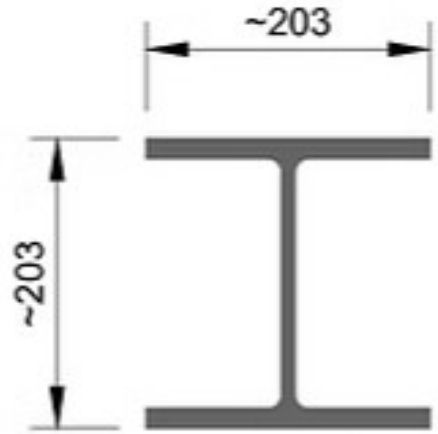
Wide Flange Steel Section origins in railroad track

Universal Beam [UB]



eg. 203 x 133 x 30 UB
UB's are always taller than they are wide

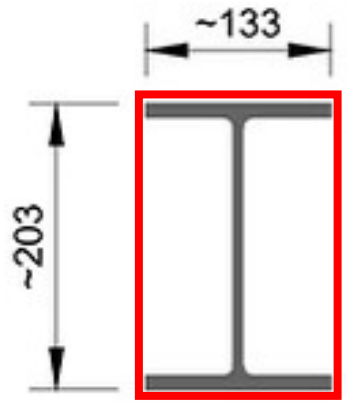
Universal Column [UC]



eg. 203 x 203 x 46 UC
UC's are approximately equal in height and width

Wide Flange as Beam/Girder, and as Column

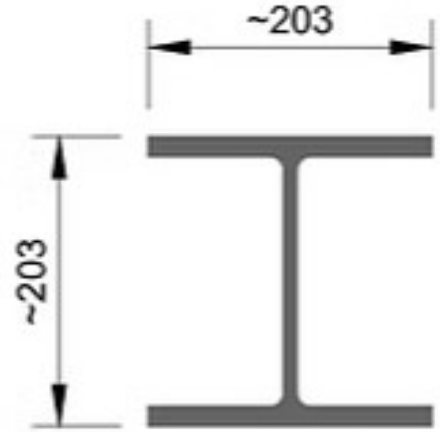
Universal Beam [UB]



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Universal Column [UC]

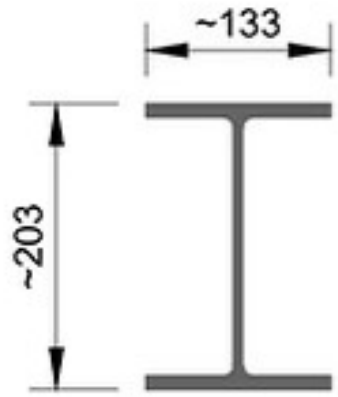


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Wide Flange as Beam/Girder, and as Column

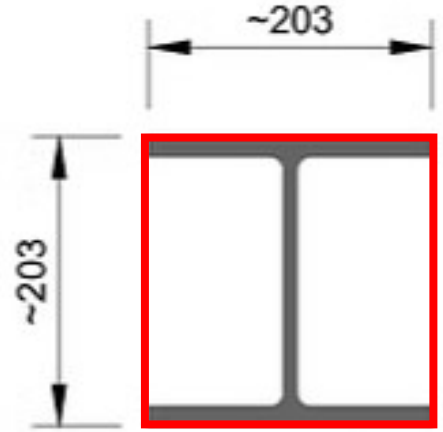
Universal Beam [UB]



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Universal Column [UC]

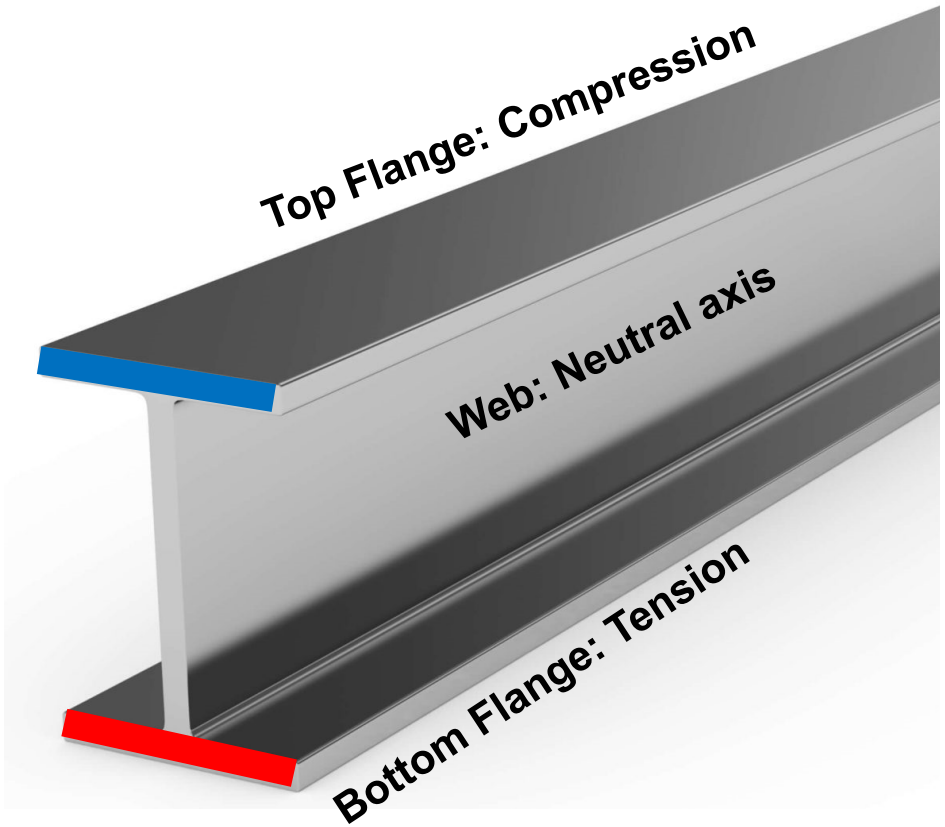


eg. 203 x 203 x 46 UC

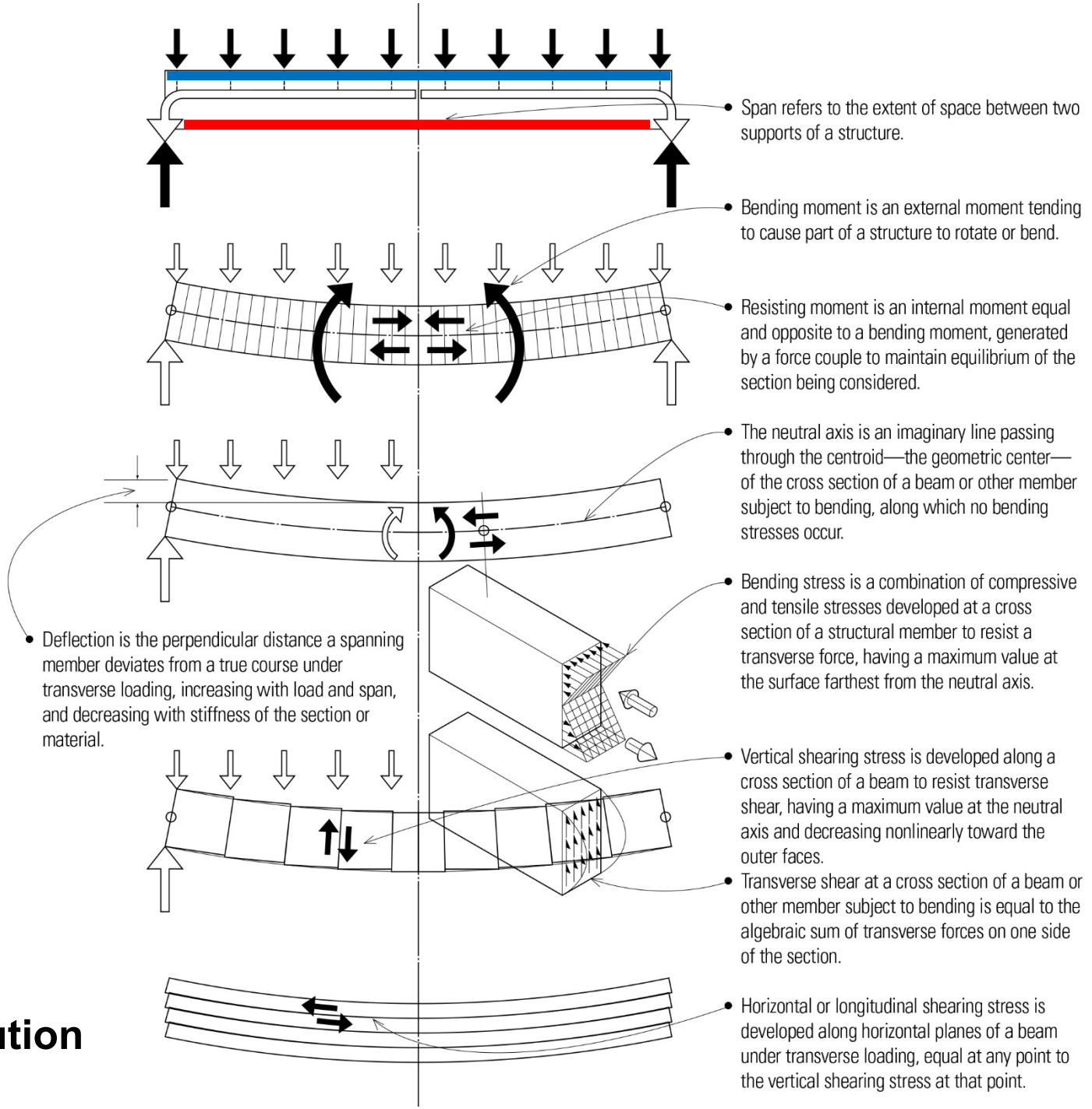
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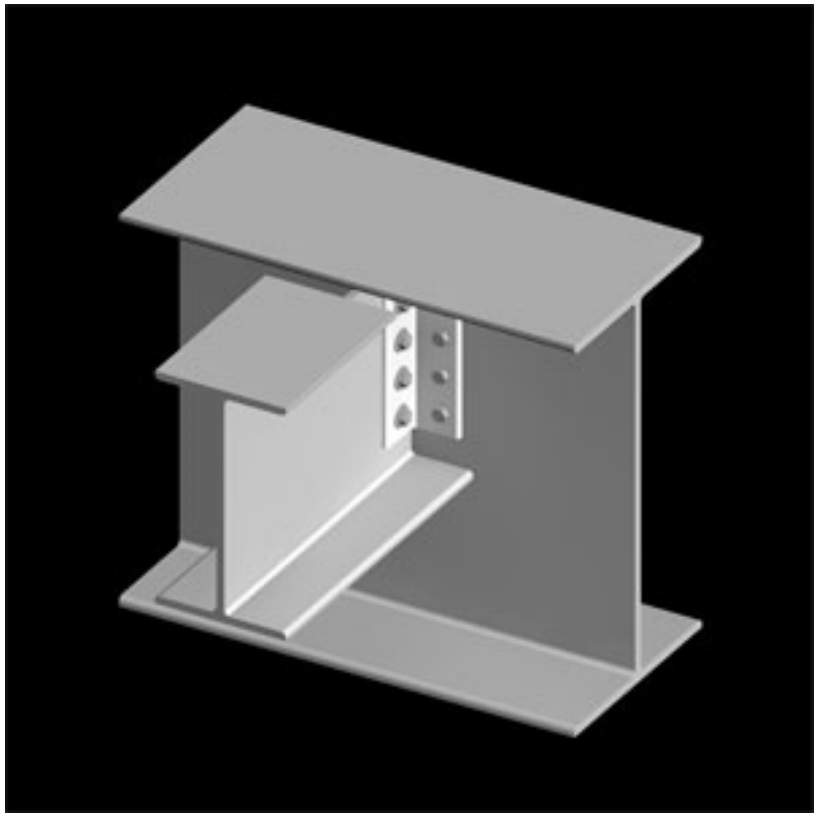
Wide Flange as Beam/Girder, and as Column

Wide Flange as Beam or Girder

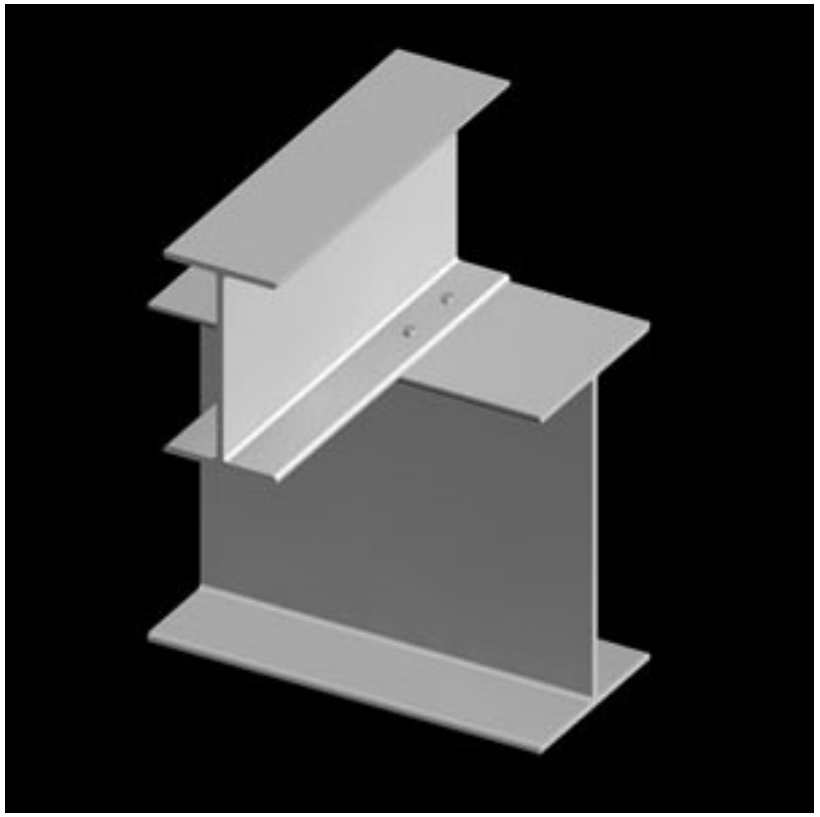


Wide Flange optimizes force distribution





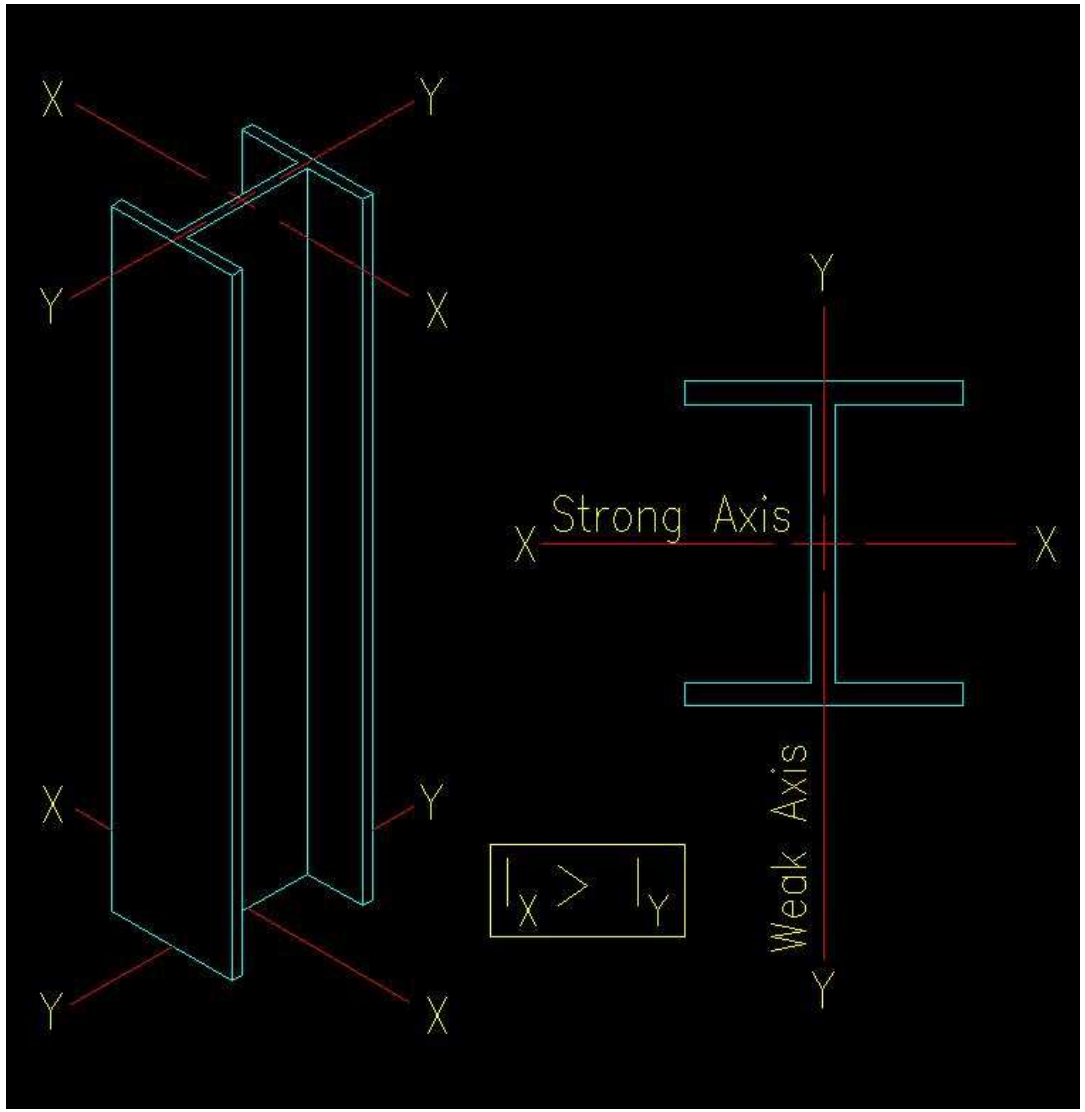
Top Flange framing

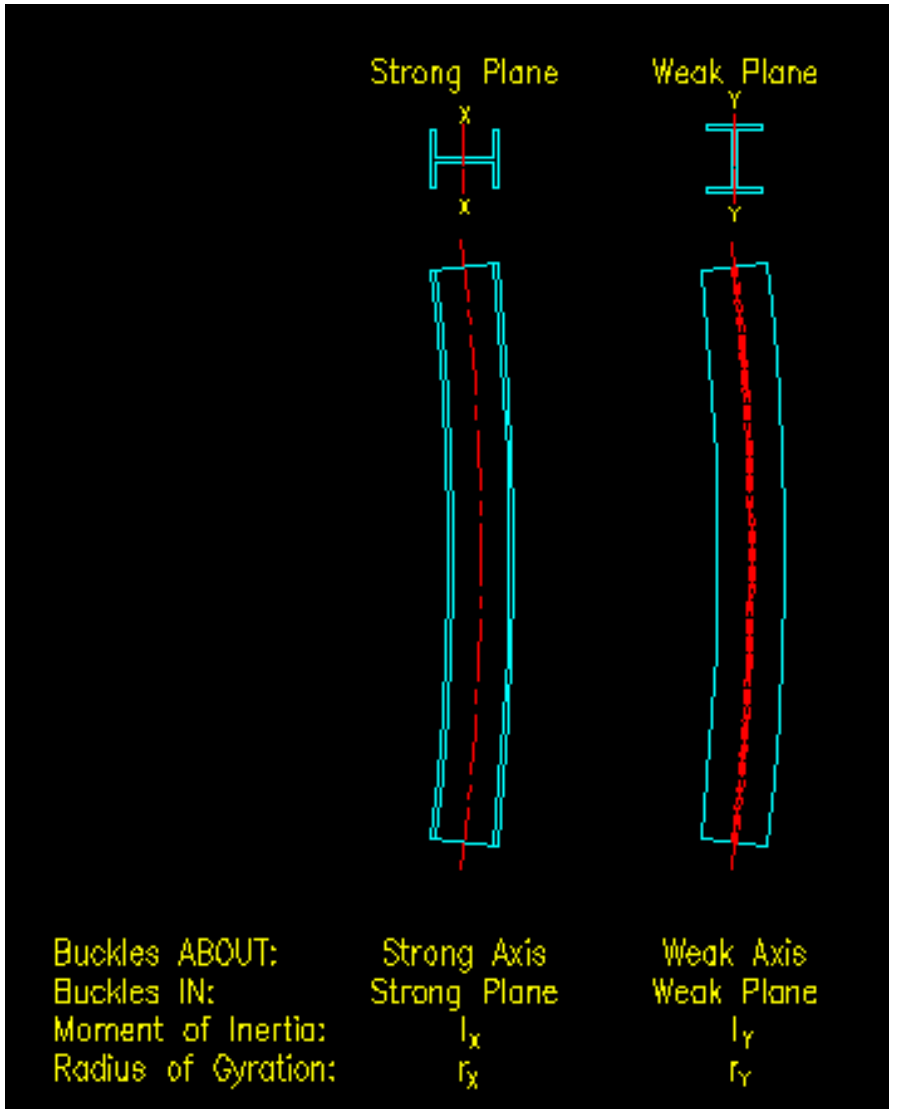


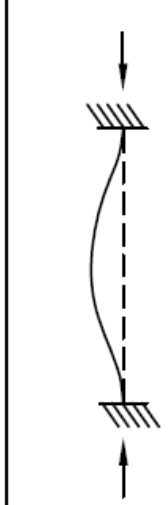
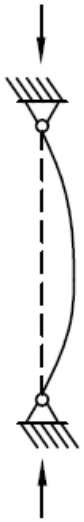
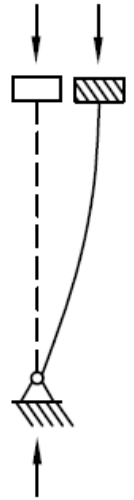
Beam on Girder

Beam to Girder connections*…choose one type for your floor structure*

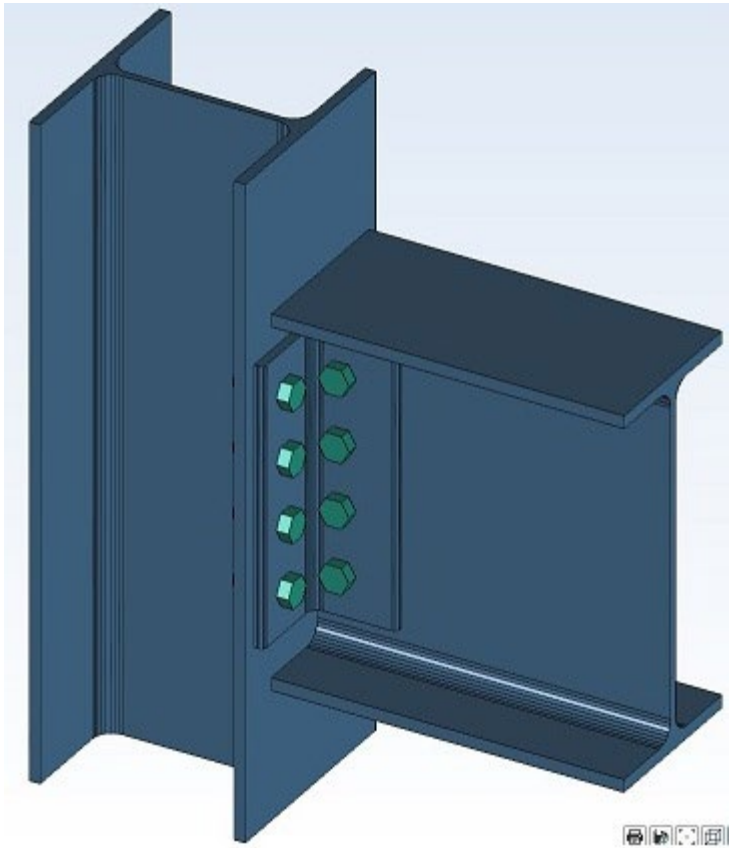
Wide Flange as Column



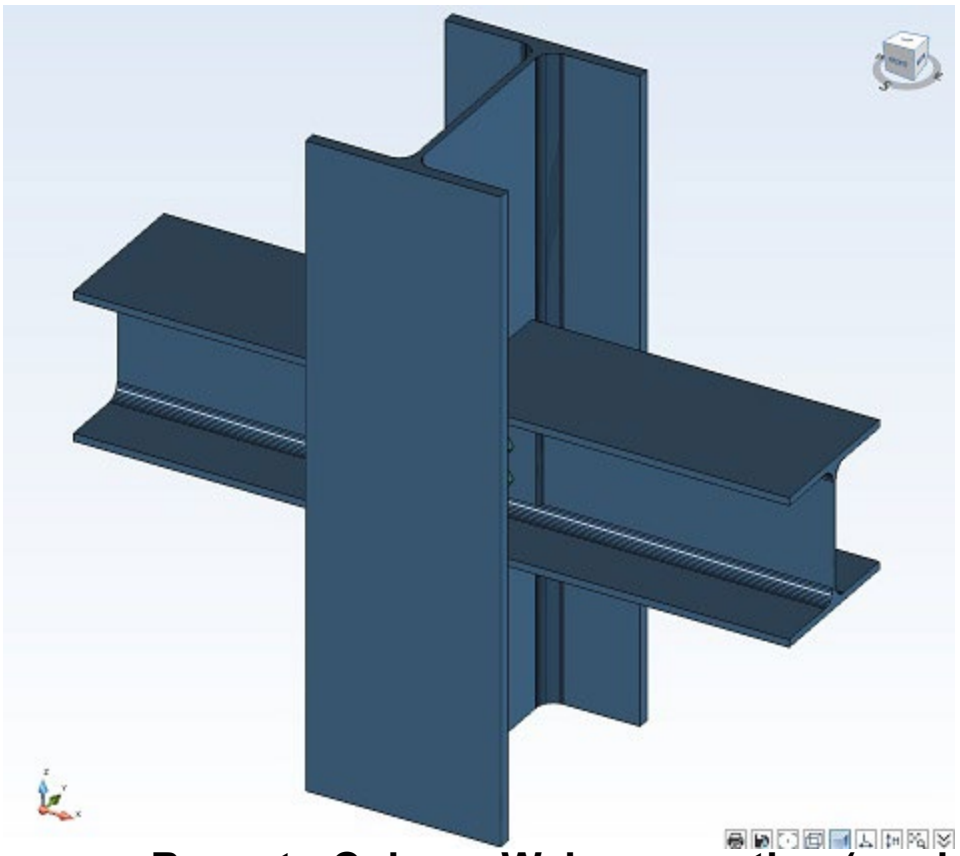


	Braced column			Unbraced column		
Buckled shape						
Effective length factor (k)	0.70	0.85	1.00	1.20	2.20	2.20
Symbols for end restraint conditions	 = Rotation fixed, translation fixed			 = Rotation fixed, translation free		
	 = Rotation free, translation fixed			= Rotation free, translation free		

Types of Column buckling



Girder to Column Flange connection (strong axis)

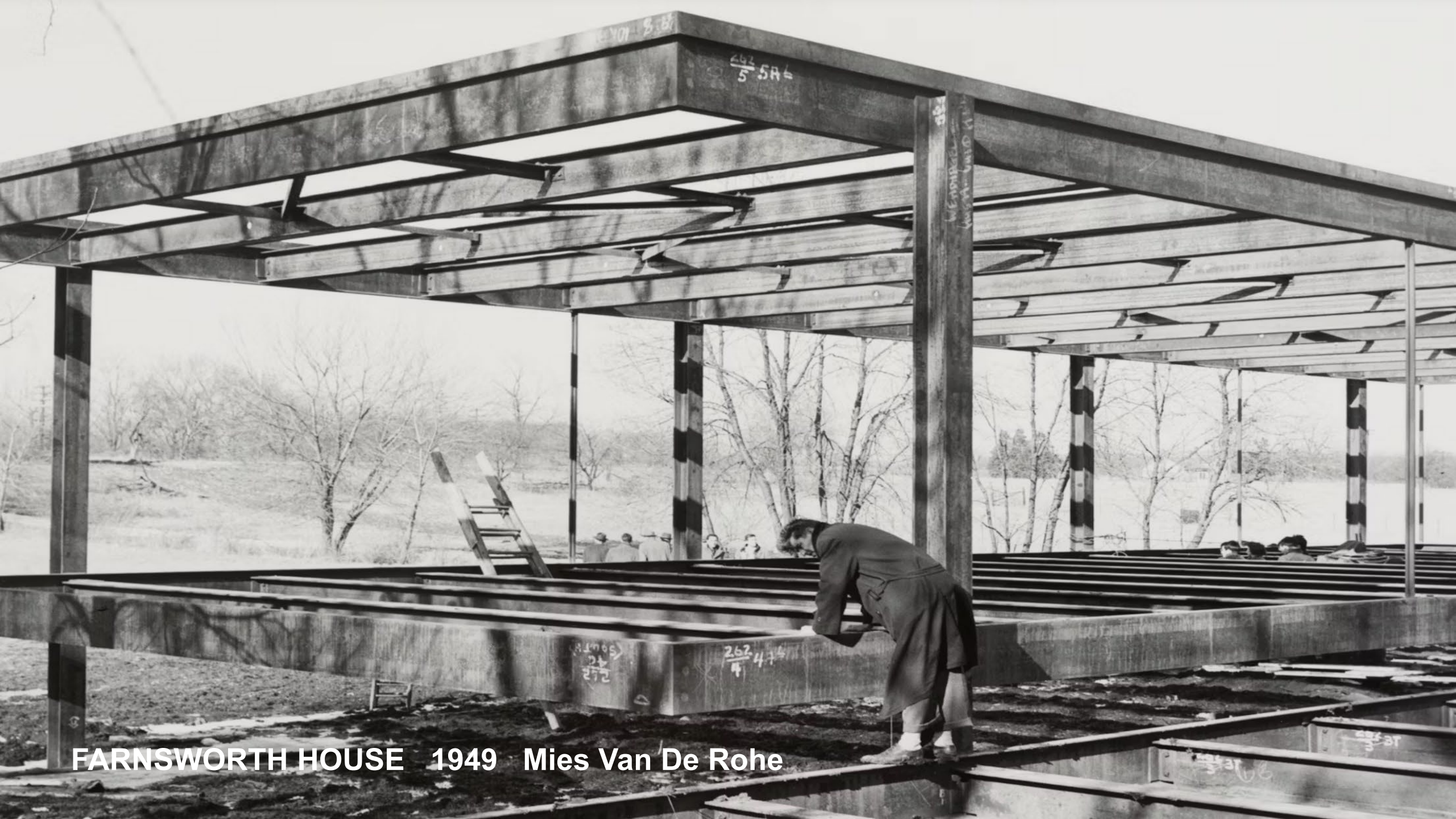


Beam to Column Web connection (weak axis)

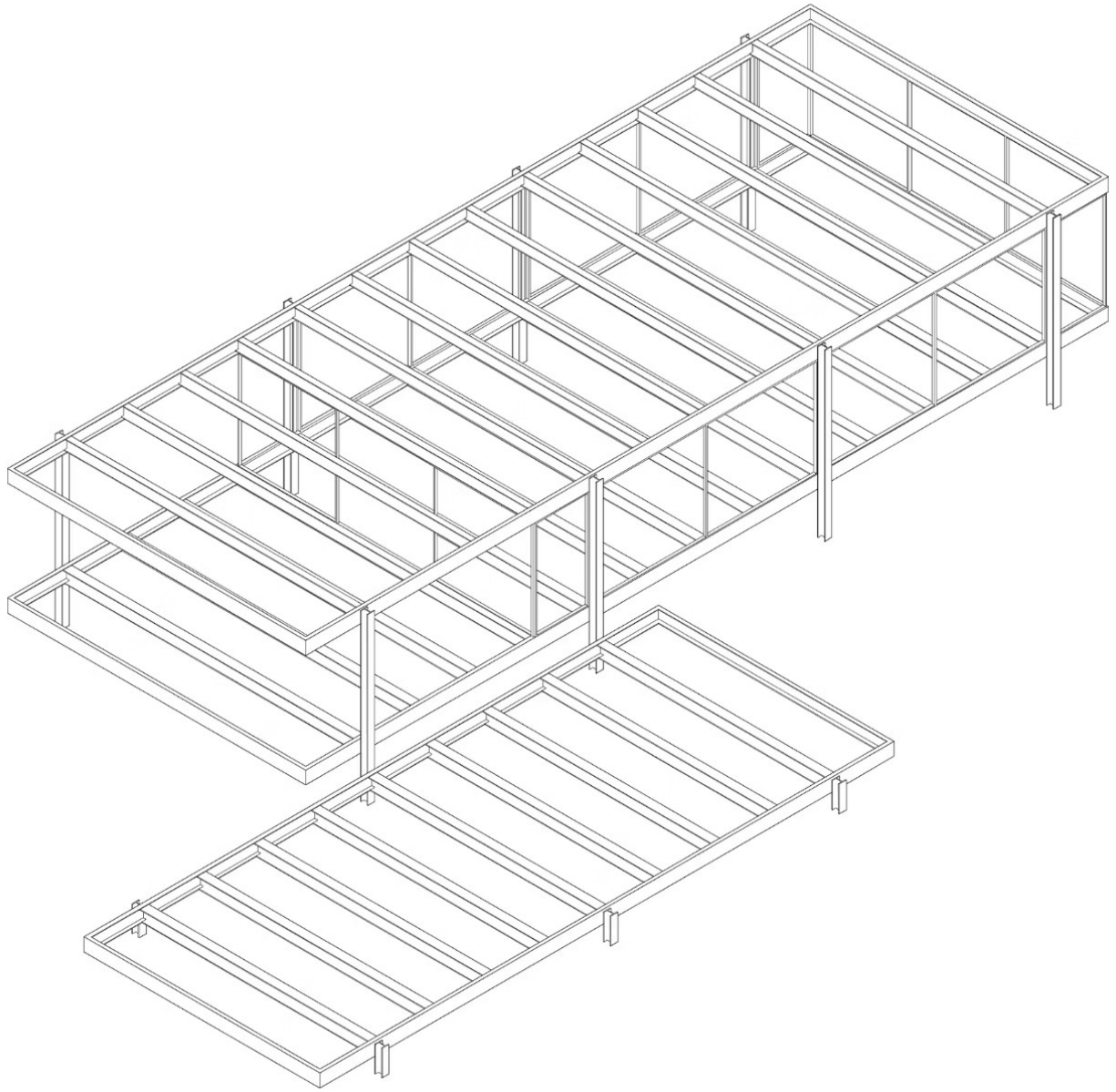
Beam or Girder to Column connections



FARNSWORTH HOUSE 1949 Mies Van De Rohe



FARNSWORTH HOUSE 1949 Mies Van De Rohe



FARNSWORTH HOUSE 1949 Mies Van De Rohe

STEEL BUILDING project

PROJECT PARAMETERS: Floor

ENGINEERING OFFICE: Ypsilanti, Michigan
Floor LL: 65 psf Floor DL: 15 psf
Second-Floor: 4,200 sf to 5,400 sf (see layout options)
Composite Deck with concrete: 33 psf (31+2)

GROCERY STORE: Pinconning, Michigan
Floor LL: 75 psf Floor DL: 15 psf
Second-Floor: 4,200 sf to 5,400 sf (see layout options)
Composite Deck with concrete: 33 psf (31+2)

PUBLIC LIBRARY: Houghton, Michigan
Floor LL: 150 psf Floor DL: 15 psf
Second-Floor: 4,200 sf to 5,400 sf (see layout options)
Composite Deck with concrete: 33 psf (31+2)

Note: Do students want to work in small groups on floor structure?

STEEL BUILDING project

PROJECT PARAMETERS: Floor

Tertiary Elements: (span determined by floor load)

Type-B Composite Deck: 3.5” total depth/ slab thickness, 36” width (see span & load table)

Secondary Elements: (spacing determined by composite floor deck span)

Wide-Flange Steel Beams (size using Tributary Load and OCHSHORN Steel Beam Calculator)
Diagonal Bracing: HSS Square Tube, Angle, or Tension Rods

Primary Elements: (span and load determined by secondary elements)

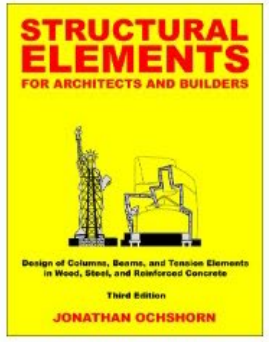
Wide-Flange Steel Girders (size using Tributary Load and OCHSHORN Steel Beam Calculator)
Wide-Flange Steel Columns (size with OCHSHORN Steel Column Calculator after Beam & Girder tributary load determined)

STEEL BUILDING project

PROJECT CALCULATORS: Steel Beams and Columns

OCHSHORN Steel Beam Calculator: https://jonochshorn.com/scholarship/calculators-st/example8.6/index.html

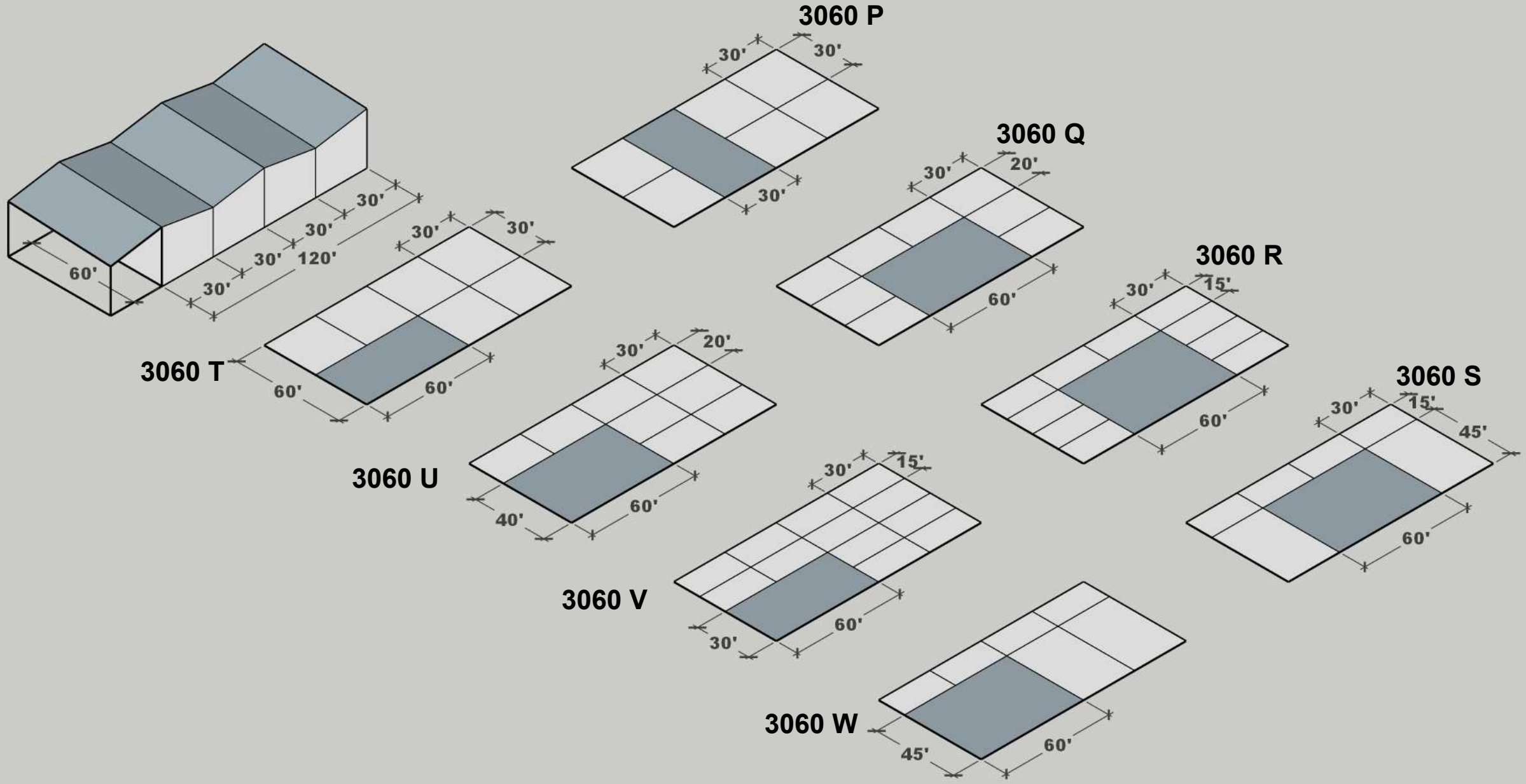
OCHSHORN Steel Column Calculator: https://jonochshorn.com/scholarship/calculators-st/steel-column-design/index.html



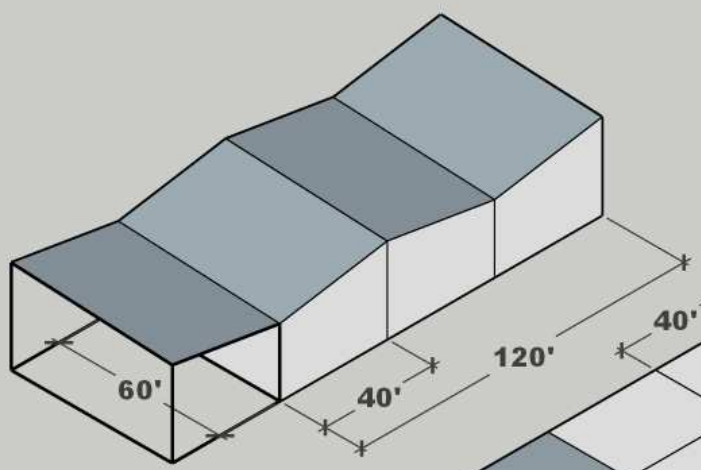
Laterally-Supported Steel Floor Beam Calculator

Jonathan Ochshorn

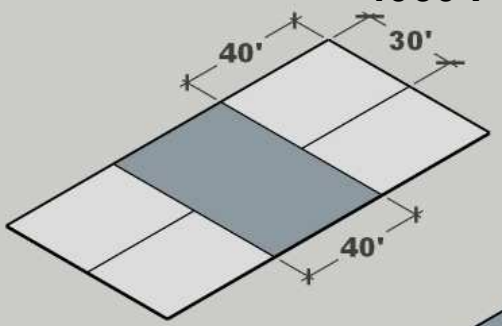
© 2009 Jonathan Ochshorn.



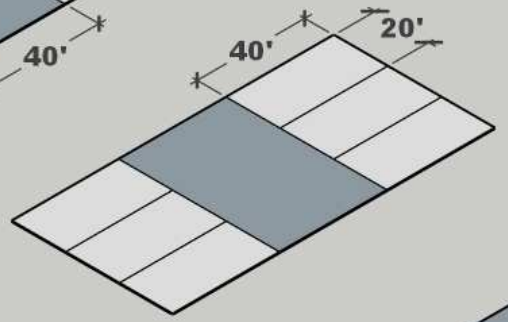
3060 Second-Floor Plan options (blue grey area is open to roof in plan)



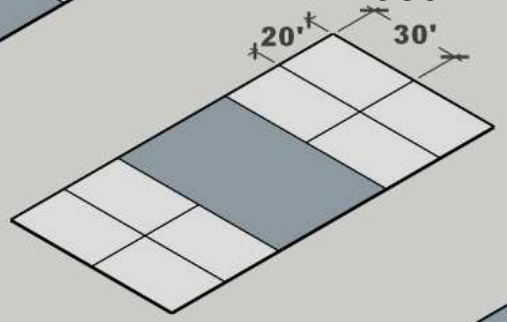
4060 P



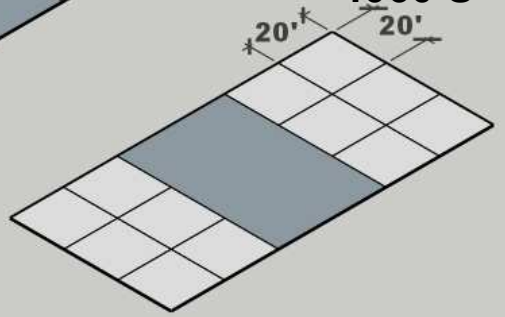
4060 Q



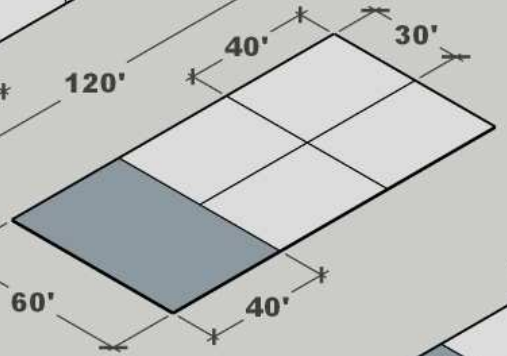
4060 R



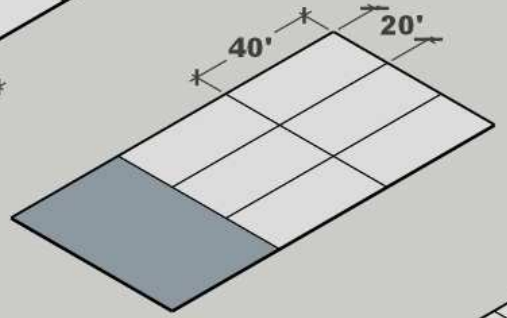
4060 S



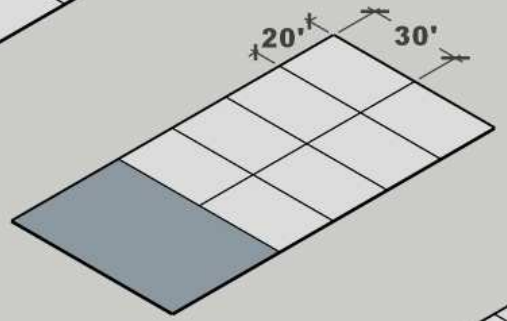
4060 T



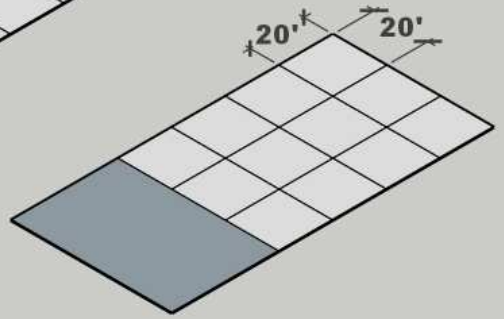
4060 U



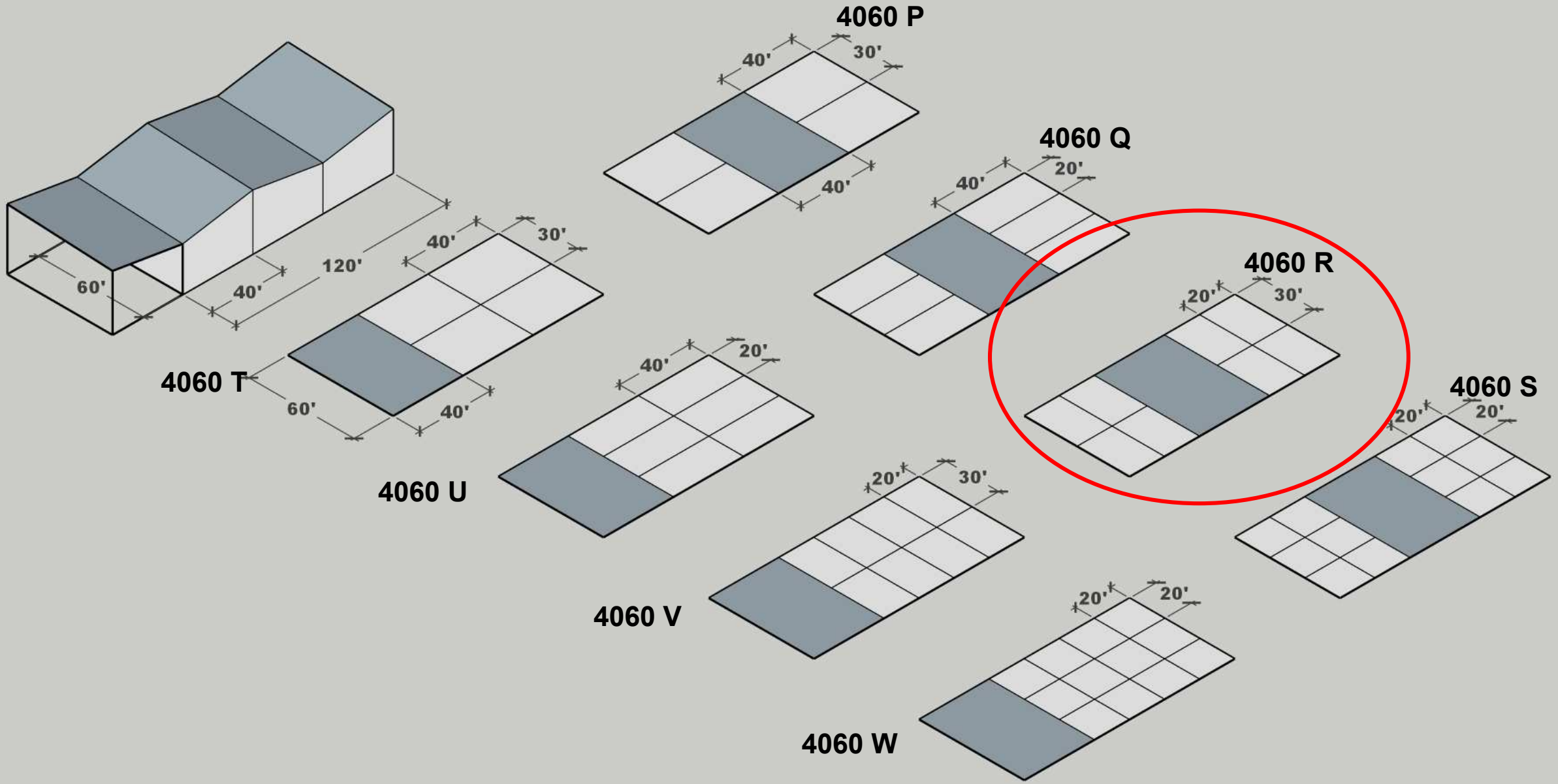
4060 V



4060 W



4060 Second-Floor Plan options (blue grey area is open to roof in plan)

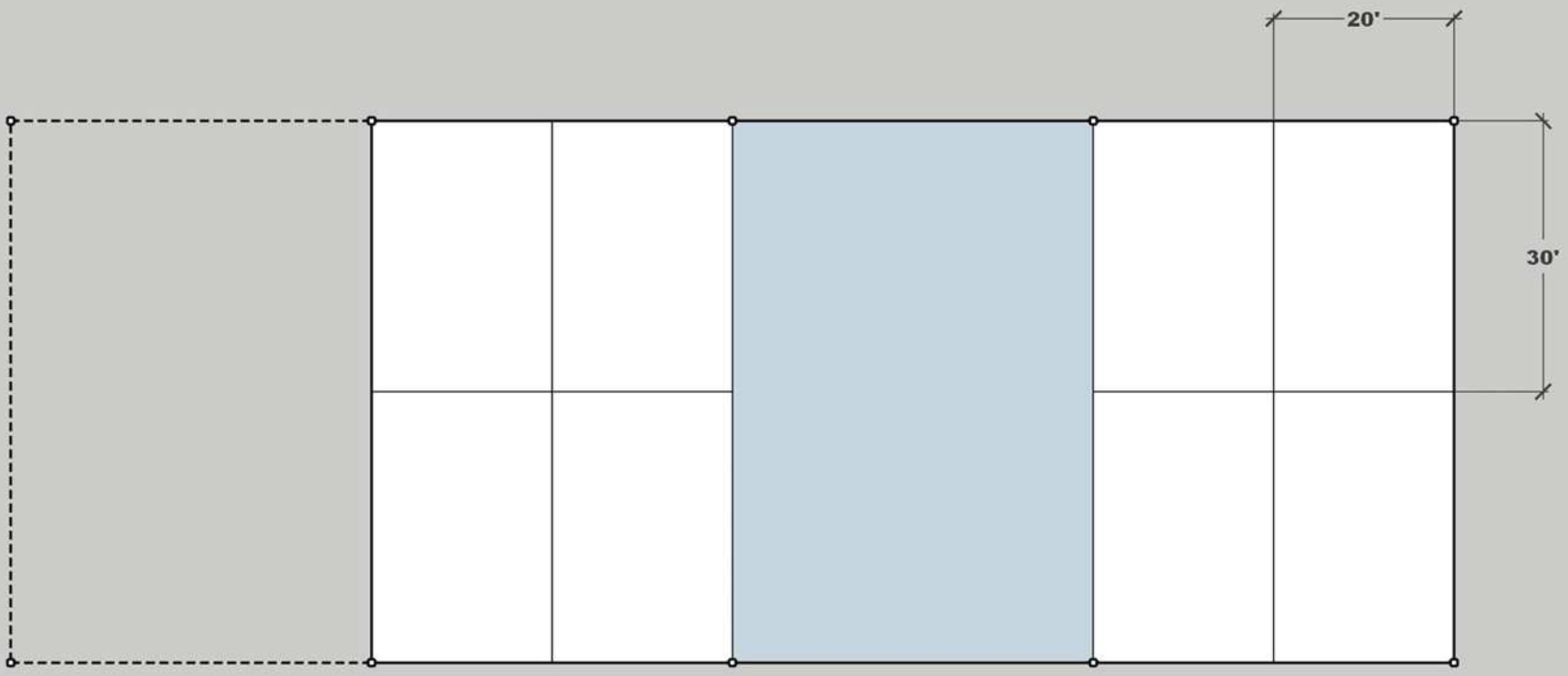


4060 Second-Floor Plan options (blue grey area is open to roof in plan)

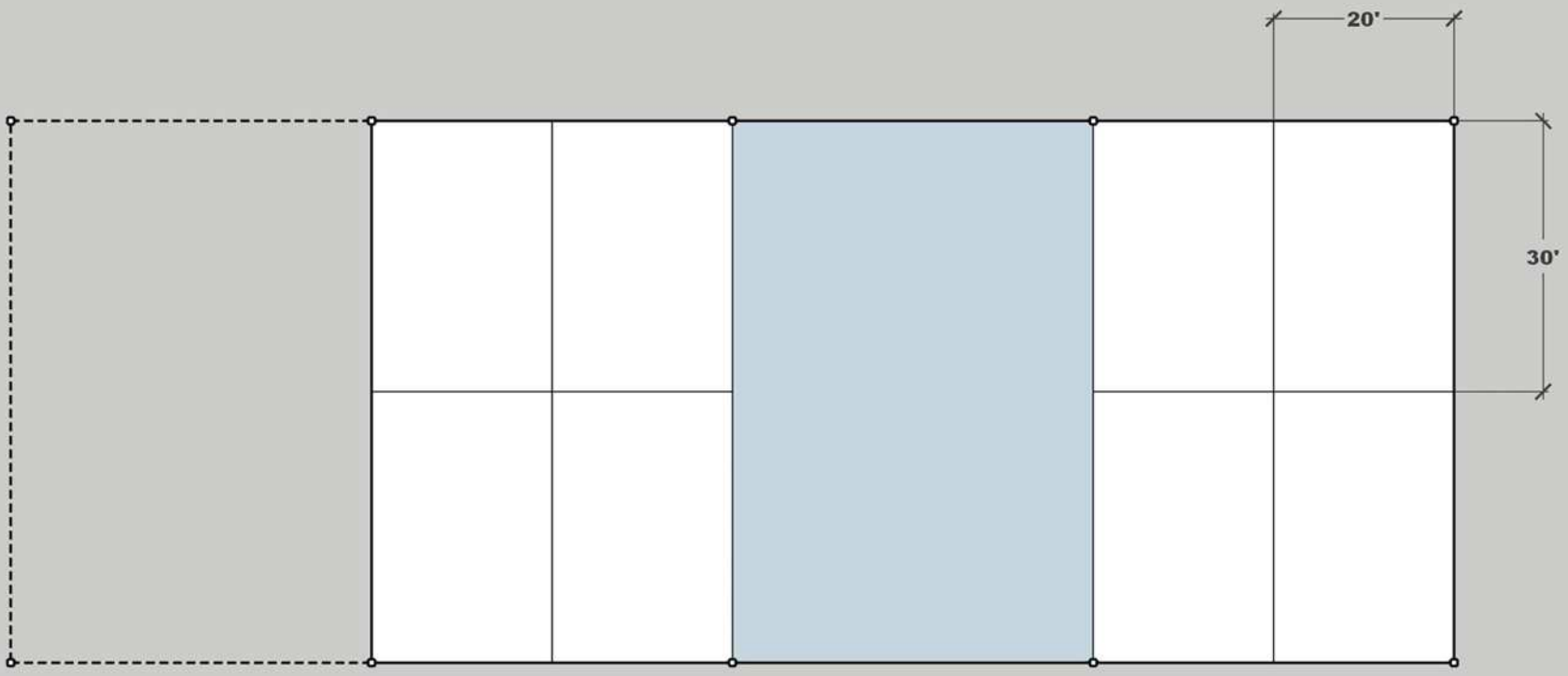
STEEL BUILDING as LIBRARY

PROJECT PARAMETER SELECTIONS:

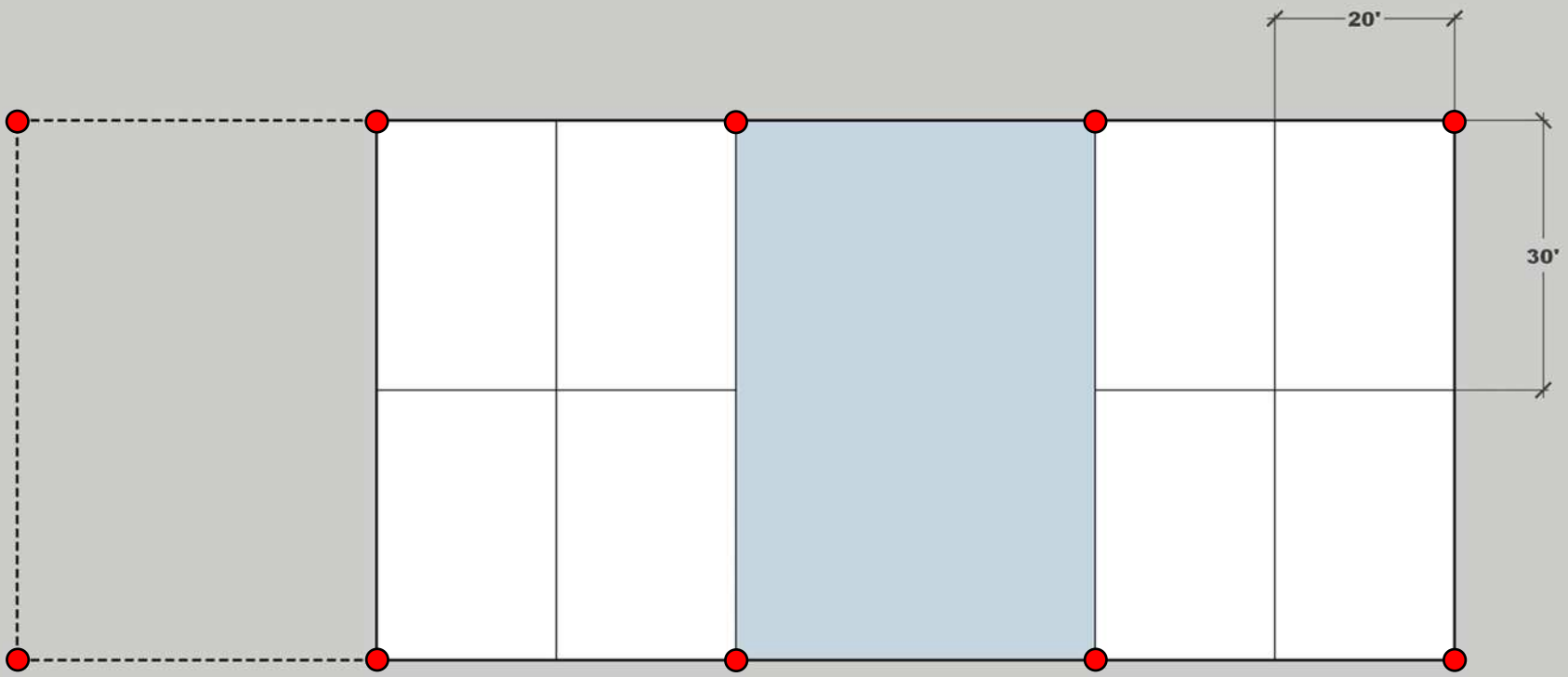
ROOF: MODEL 4060 C
FLOOR: MODEL 4060 R



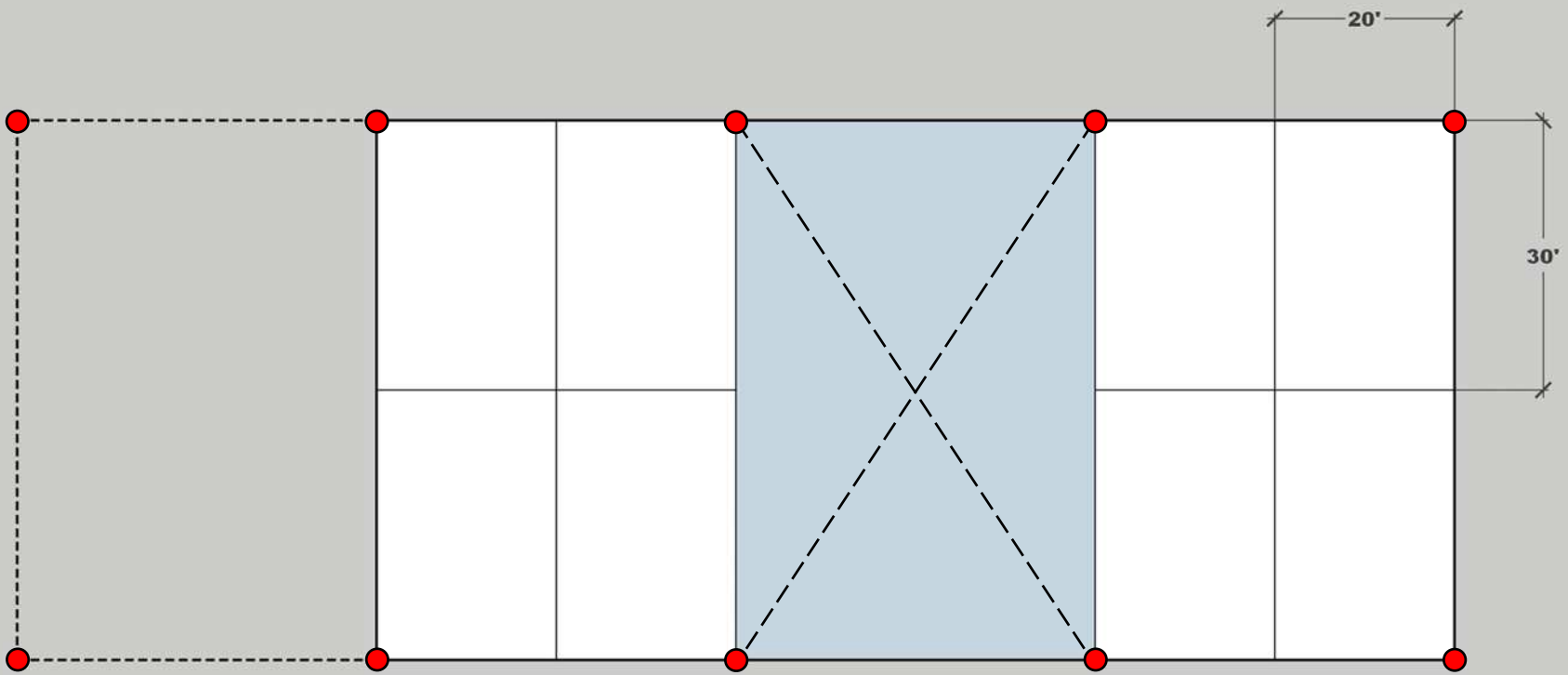
4060 R 2nd Floor Plan option selected



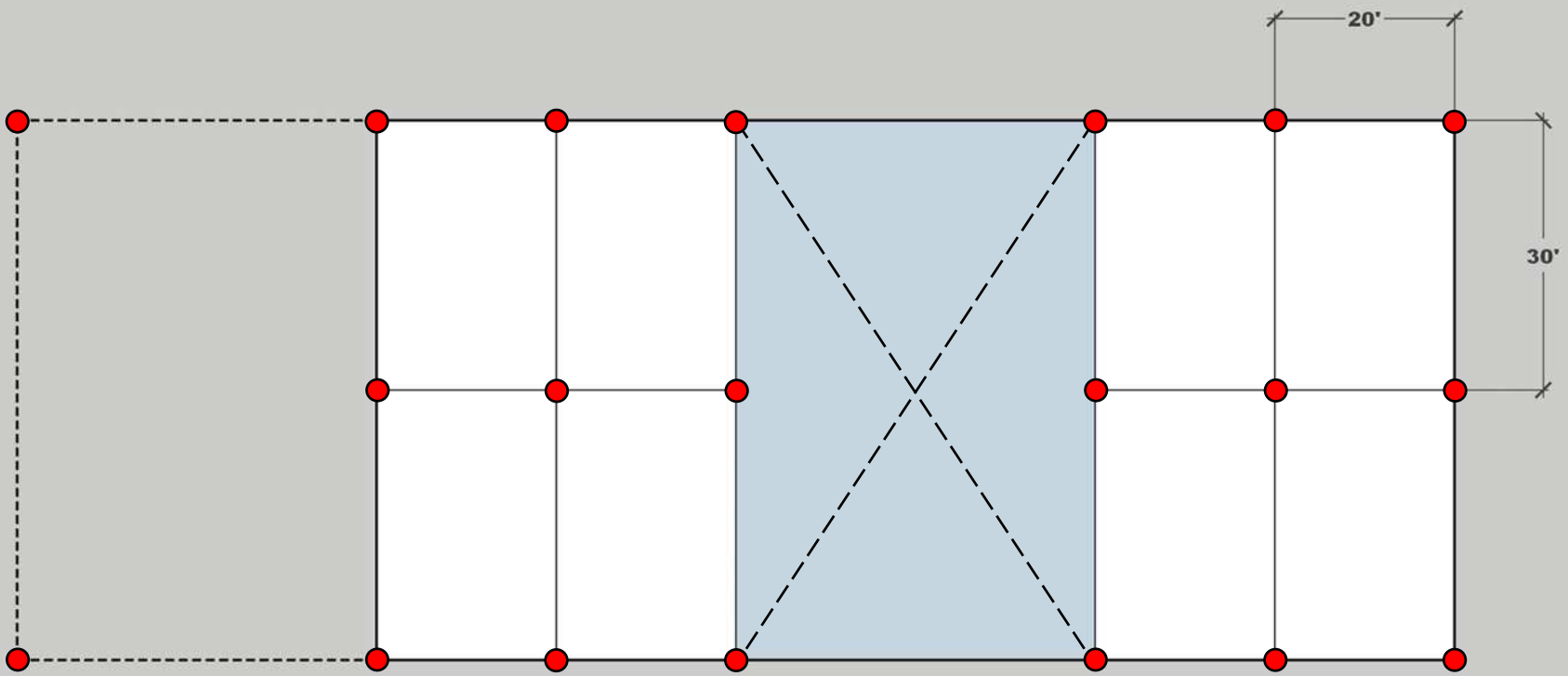
4060 R 2nd Floor Plan option selected*…What does this plan diagram tell us?*



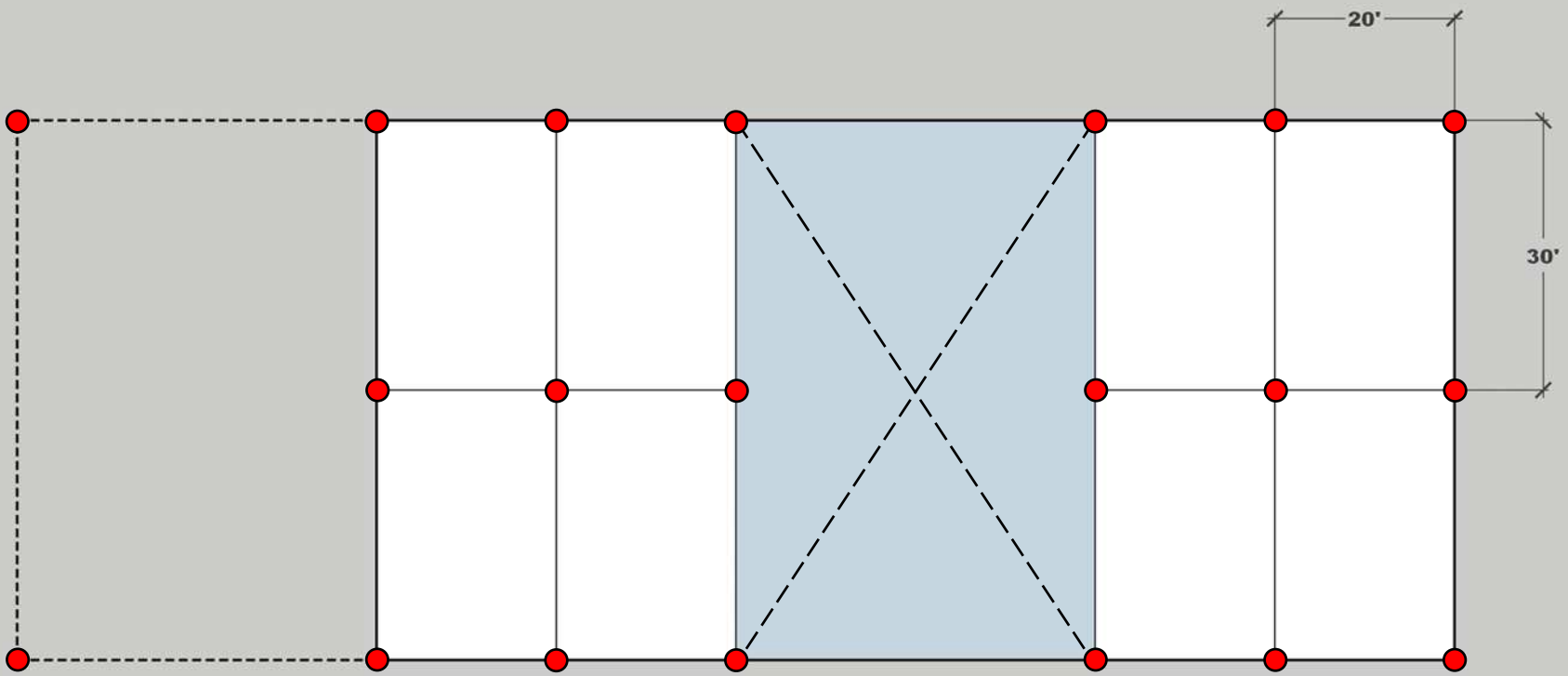
4060 R 2nd Floor Plan option selected*…Roof column locations for 4060 C*



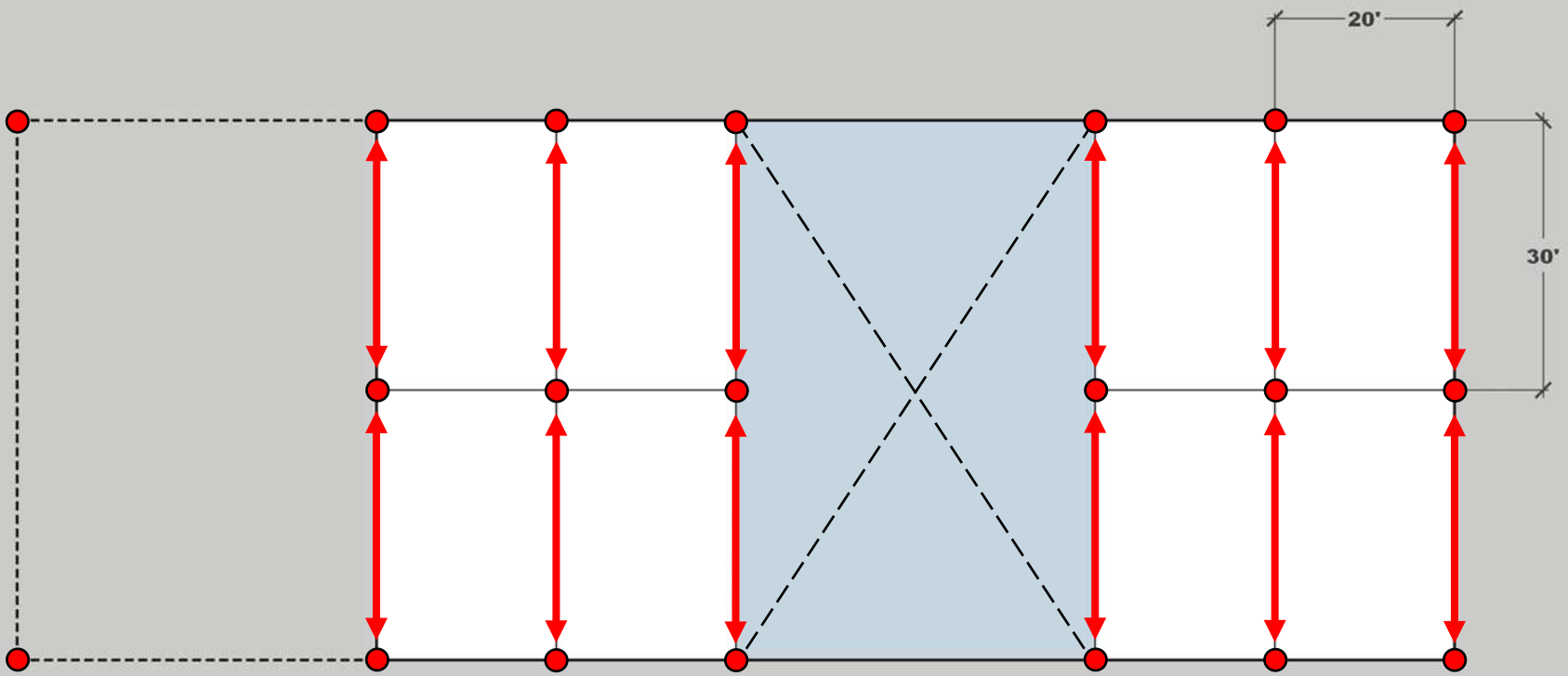
4060 R 2nd Floor Plan option selected…*2nd Floor area and open to below area*



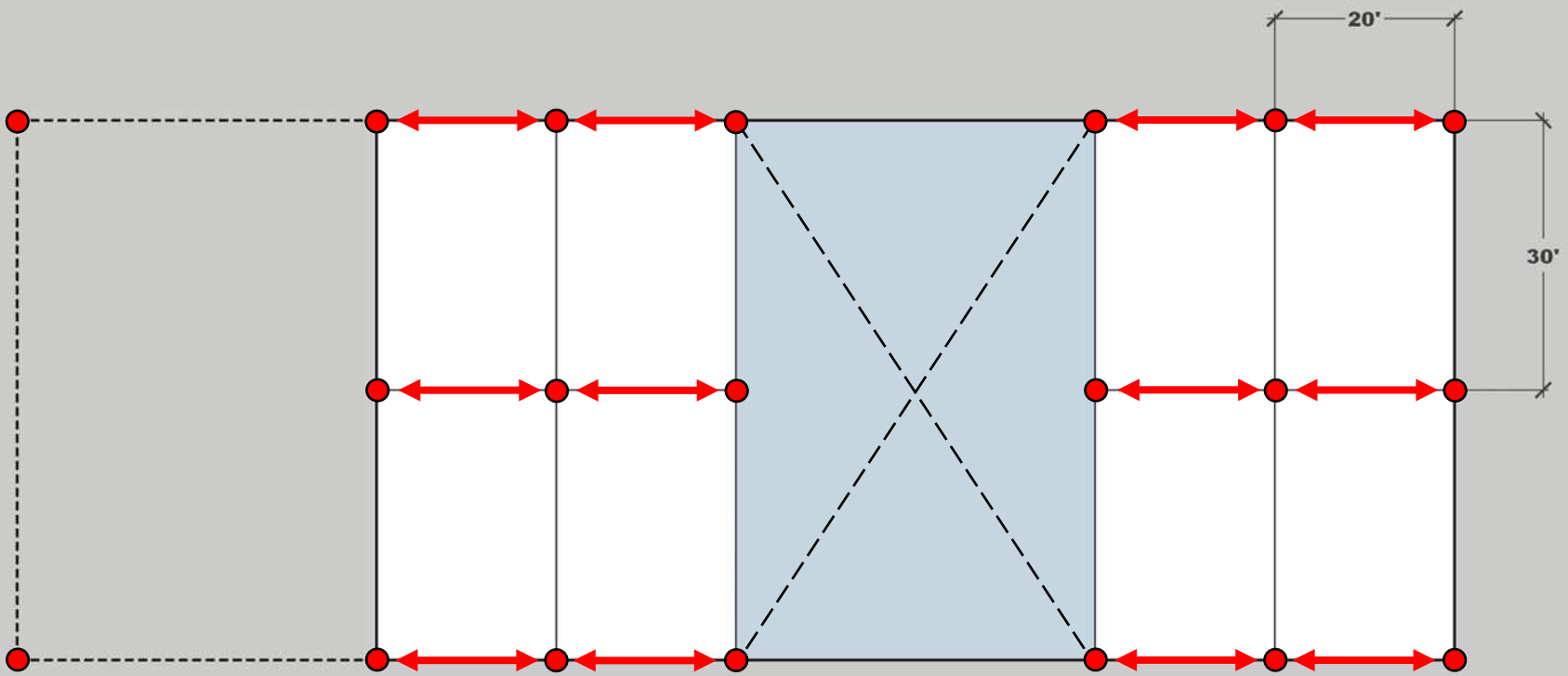
4060 R 2nd Floor Plan option selected*…Column locations for Floor Structure*



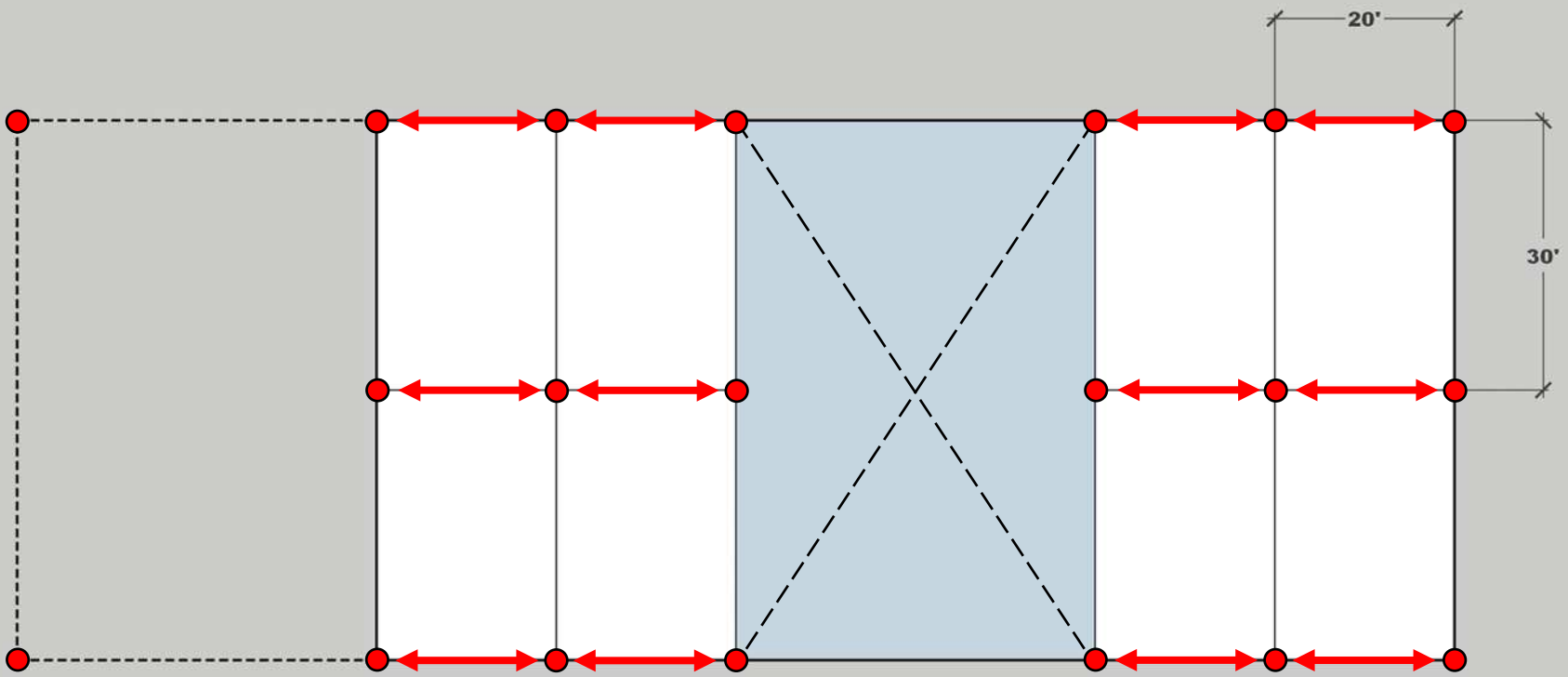
4060 R 2nd Floor Plan option selected*…Which way should the Floor Girders span?*



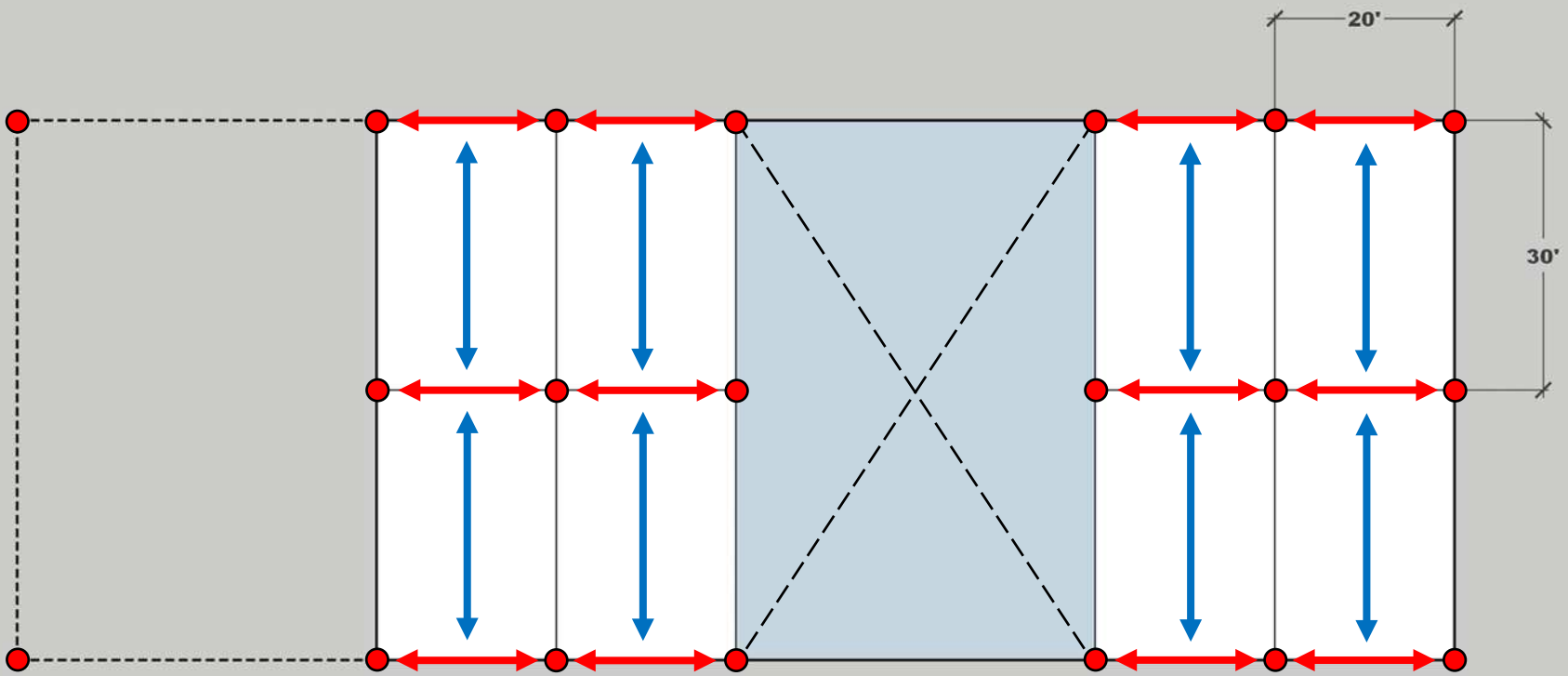
4060 R 2nd Floor Plan option selected*…Girders span the long direction?*



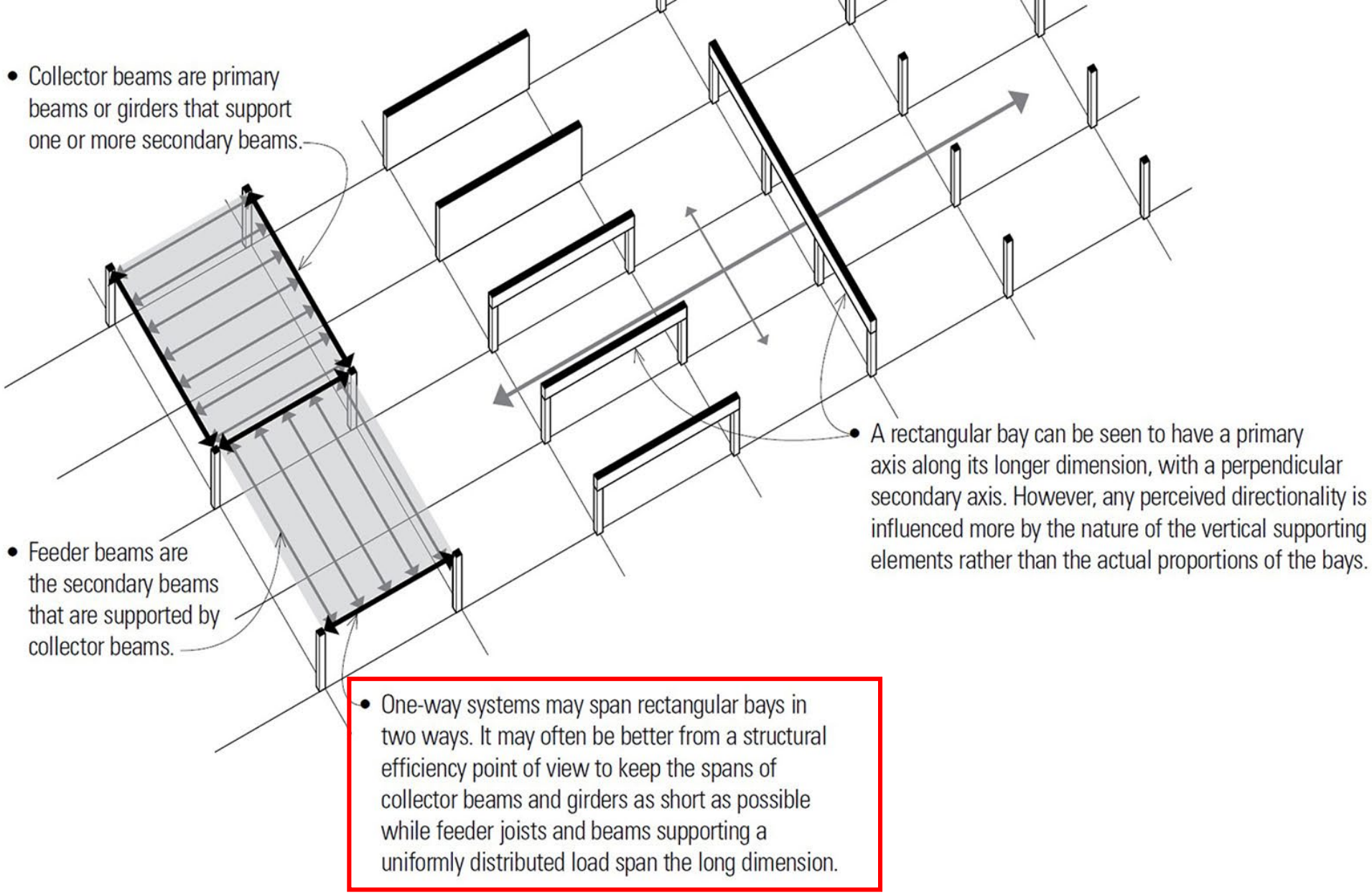
4060 R **2nd Floor Plan option selected**…*Girders typically span the short direction*



4060 R 2nd Floor Plan option selected*…Which way do the Beams span?*



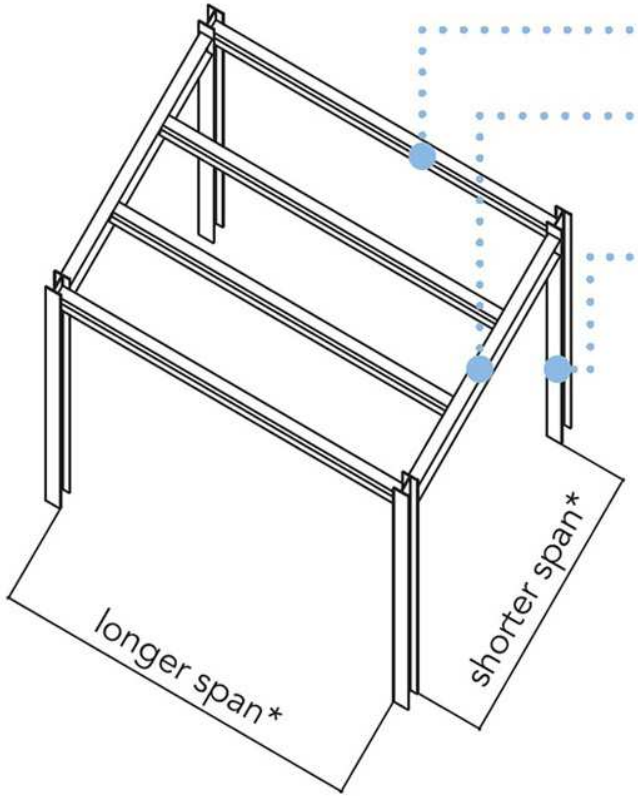
4060 R 2nd Floor Plan option selected*…Beams typically span the longer direction*



GIRDERS (Collector Beams) **& BEAM** (Feeder Beams) **in *Building Structures Illustrated***

STRUCTURAL STEEL DESIGN TIPS

Quick Estimates and Rules of Thumb for Architecture Students



Beam (framing to transfer floor load to girder)

Girder (a beam that transfers the load of several beams to columns)

Column (vertical support that transfers girder loads to the foundation)

*The most efficient and economical bays have a 1:1.25–1.5 width proportion and an area of 750–1,250 ft^2 with infill beams spanning the long direction to have closer girder and beam depths. Spacing should consider the type of deck and spans.

AISC (American Institute of Steel Construction) resources for Architecture students

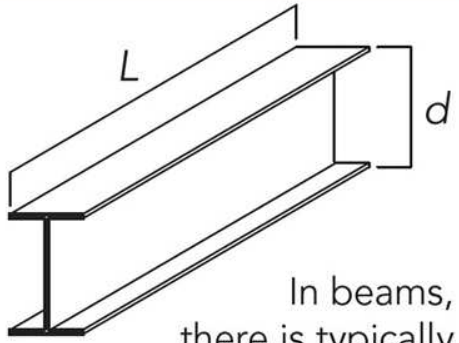
STRUCTURAL STEEL DESIGN TIPS

Quick Estimates and Rules of Thumb for Architecture Students

BEAM DEPTHS

W-shape beam depth estimate

Beam	½ in. depth (*d*) per foot of span (*L*), so 30 ft span means 15 in. beam depth (W16)
Girder or Beam with heavy loads	Use above estimate, but round up one size (W18)
Roof Purlin	Use above estimate for beams, but round down one size (W14)



In beams, there is typically more vertical depth than width

System	Typ. Span Range	Spacing	Typ. Shapes
Steel Girder	20–40 ft	—	W12–W30
Steel Beam	25–45 ft	10–15 ft	W12–W24

AISC (American Institute of Steel Construction) resources for Architecture students

FLOOR DECK

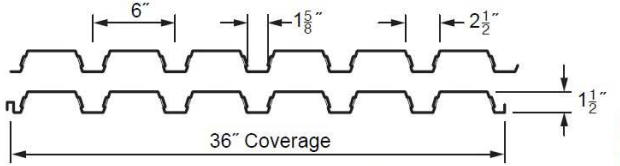
3 ½” Slab Depth (total depth)
20-gauge steel Composite Deck
with normal-weight concrete
(Composite steel deck self-weight: 2 psf)
(Concrete weight: 31 psf)

1.5CD, 1.5CDI Composite Deck

NORMAL-WEIGHT CONCRETE

PROPERTIES

Gage	Thickness (in)	Cover (in)	Weight (psf)
22	0.0295	3.6	1.63
20	0.0358		1.98
18	0.0474		2.62
16	0.0598		3.30



ICC-ES APPROVED
REFER TO ESR-3818

ASD SECTION PROPERTIES

DESIGN STRENGTHS (No Concrete Fill)

Gage	F_y (ksi)	A_s (in²/ft)	I_D 1 Span (in⁴/ft)	I_D 2+ Span (in⁴/ft)	I_p (in⁴/ft)	I_n (in⁴/ft)	S_p (in³/ft)	S_n (in³/ft)	Mn,p/Ω (in-lb/ft)	Mn,n/Ω (in-lb/ft)	Vn/Ω (lb/ft)	*Rbe/Ω (lb/ft)	*Rbi/Ω (lb/ft)
22	40	0.480	0.163	0.175	0.157	0.175	0.176	0.183	4215	4391	2009	654	1181
20	40	0.582	0.205	0.213	0.201	0.213	0.224	0.235	5375	5624	2428	932	1696
18	40	0.770	0.280	0.281	0.279	0.281	0.304	0.315	7291	7553	3192	1557	2858
16	40	0.971	0.355	0.355	0.355	0.355	0.390	0.395	9332	9454	3996	2381	4396

CONSTRUCTION CLEAR SPANS

Total Slab Depth (in)	Gage	Concrete Weight (psf)	Maximum Construction Clear Span (ft - in) Single	Double	Triple	Cantilever	Concrete Volume (ft³/ft²)
3 1/2	22	31	5 - 9	6 - 9	6 - 10	2 - 0	0.213
	20		6 - 9	8 - 0	8 - 1	2 - 5	
	18		8 - 4	9 - 8	9 - 11	3 - 1	
	16		9 - 8	10 - 9	11 - 1	3 - 8	

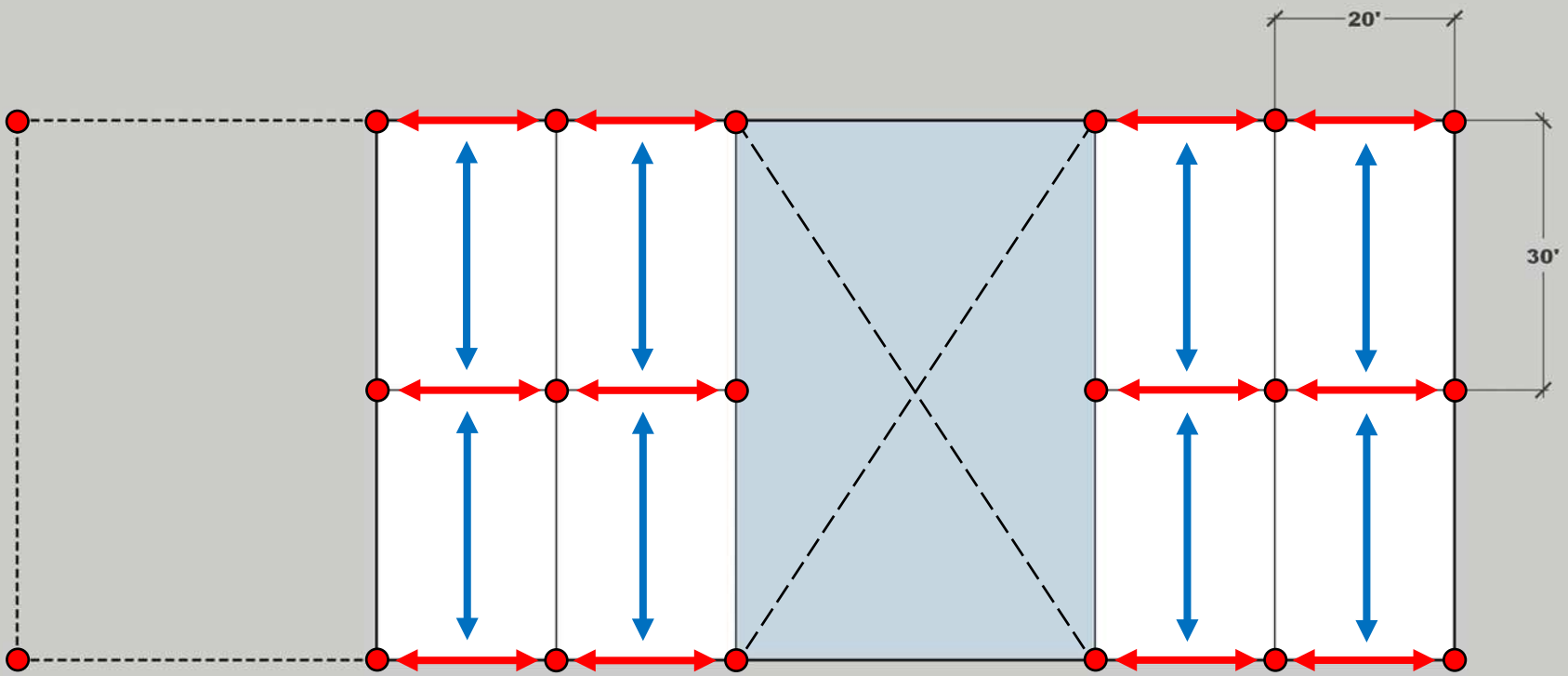
SLAB DESIGN - ALLOWABLE SUPERIMPOSED UNIFORM LOADS AND MINIMUM SDI SLAB REINFORCEMENT

Normal-Weight Concrete (γc = 145 pcf, f'c = 3000 psi, n = 9)

Total Slab Depth (in)	Gage	Allowable Superimposed Uniform Load (psf) Clear Span (ft - in) 6 - 0	6 - 6	7 - 0	7 - 6	8 - 0	8 - 6	9 - 0	10 - 0	11 - 0	Recommended Welded Wire Fabric	Wire Area (in²/ft)
3 1/2	22	294	246	207	176	151	130	113	85	65	6x6 - W1.4xW1.4	0.028
	20	354	297	251	215	185	160	139	106	82		
	18	400	386	328	281	243	212	185	139	105		
	16	400	400	400	349	301	251	212	154	116		
4	22	370	310	262	223	191	165	143	109	83	6x6 - W1.4xW1.4	0.028
	20	400	375	318	272	234	203	177	136	105		
	18	400	400	400	357	309	269	236	183	145		
	16	400	400	400	400	385	336	295	229	172		
4 1/2	22	400	377	319	272	234	202	175	133	103	6x6 - W1.4xW1.4	0.028
	20	400	400	387	331	286	248	216	167	130		
	18	400	400	400	400	378	330	289	225	178		
	16	400	400	400	400	400	400	363	285	228		
5	22	400	400	378	322	277	240	208	159	123	6x6 - W2.0xW2.0	0.040
	20	400	400	400	393	339	295	258	199	156		
	18	400	400	400	400	400	392	344	269	213		
	16	400	400	400	400	400	400	400	341	273		
5 1/2	22	400	400	400	374	322	279	242	186	144	6x6 - W2.0xW2.0	0.040
	20	400	400	400	400	394	343	300	232	182		
	18	400	400	400	400	400	400	400	314	249		
	16	400	400	400	400	400	400	400	398	319		
6	22	400	400	400	400	367	318	277	212	165	4x4 - W1.4xW1.4	0.042
	20	400	400	400	400	400	392	342	265	208		
	18	400	400	400	400	400	400	400	360	286		
	16	400	400	400	400	400	400	400	400	366		

Notes:
7. Welded wire fabric shown in the table above is the SDI minimum required for temperature and shrinkage (0.00075 x concrete area above top of deck).
8. Allowable Superimposed Uniform Loads shown in table above are based on the following criteria:
   - Unfactored service level loads, determined using SDI design method per ANSI/SDI C-2017 Standard for Composite Steel Floor Deck-Slabs.
   - Single span conditions without negative bending reinforcing over supports.
   - The presence of shear studs have not been considered, design strength may be increased if shear studs are used.
   - Slab deflection is limited to minimum of Clear Span/360 or 1" under service level superimposed loading.

Floor Deck to be used for STEEL BUILDING



4060 R 2nd Floor Plan option selected…*What is the o.c. spacing of the Beams?*

FLOOR DECK

3 ½” Slab Depth (total depth)
20-gauge steel Composite Deck
with normal-weight concrete
(Composite steel deck self-weight: 2 psf)
(Concrete weight: 31 psf)

LIBRARY FLOOR LOAD

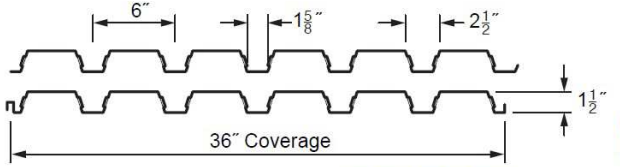
150 psf: Live Load
15 psf: Dead Load:
2 psf: Deck self-weight
31 psf: Concrete weight
198 psf: Floor Load on Floor Deck

1.5CD, 1.5CDI Composite Deck

NORMAL-WEIGHT CONCRETE

PROPERTIES

Gage	Thickness (in)	Cover (in)	Weight (psf)
22	0.0295	3.6	1.63
20	0.0358		1.98
18	0.0474		2.62
16	0.0598		3.30



ICC-ES APPROVED
REFER TO ESR-3818

ASD SECTION PROPERTIES

DESIGN STRENGTHS (No Concrete Fill)

Gage	F_y (ksi)	A_s (in²/ft)	I_D 1 Span (in⁴/ft)	I_D 2+ Span (in⁴/ft)	I_p (in⁴/ft)	I_n (in⁴/ft)	S_p (in³/ft)	S_n (in³/ft)	Mn,p/Ω (in-lb/ft)	Mn,n/Ω (in-lb/ft)	Vn/Ω (lb/ft)	*Rbe/Ω (lb/ft)	*Rbi/Ω (lb/ft)
22	40	0.480	0.163	0.175	0.157	0.175	0.176	0.183	4215	4391	2009	654	1181
20	40	0.582	0.205	0.213	0.201	0.213	0.224	0.235	5375	5624	2428	932	1696
18	40	0.770	0.280	0.281	0.279	0.281	0.304	0.315	7291	7553	3192	1557	2858
16	40	0.971	0.355	0.355	0.355	0.355	0.390	0.395	9332	9454	3996	2381	4396

CONSTRUCTION CLEAR SPANS

Total Slab Depth (in)	Gage	Concrete Weight (psf)	Maximum Construction Clear Span (ft - in)				Concrete Volume (ft³/ft²)
			Single	Double	Triple	Cantilever	
3 1/2	22	31	5 - 9	6 - 9	6 - 10	2 - 0	0.213
	20		6 - 9	8 - 0	8 - 1	2 - 5	
	18		8 - 4	9 - 8	9 - 11	3 - 1	
	16		9 - 8	10 - 9	11 - 1	3 - 8	

SLAB DESIGN - ALLOWABLE SUPERIMPOSED UNIFORM LOADS AND MINIMUM SDI SLAB REINFORCEMENT

Normal-Weight Concrete (γc = 145 pcf, f'c = 3000 psi, n = 9)

Total Slab Depth (in)	Gage	Allowable Superimposed Uniform Load (psf) Clear Span (ft - in)									Recommended Welded Wire Fabric	Wire Area (in²/ft)
		6 - 0	6 - 6	7 - 0	7 - 6	8 - 0	8 - 6	9 - 0	10 - 0	11 - 0		
3 1/2	22	294	246	207	176	151	130	113	85	65	6x6 - W1.4xW1.4	0.028
	20	354	297	251	215	185	160	139	106	82		
	18	400	386	328	281	243	212	185	139	105		
	16	400	400	400	349	301	251	212	154	116		
4	22	370	310	262	223	191	165	143	109	83	6x6 - W1.4xW1.4	0.028
	20	400	375	318	272	234	203	177	136	105		
	18	400	400	400	357	309	269	236	183	145		
	16	400	400	400	400	385	336	295	229	172		
4 1/2	22	400	377	319	272	234	202	175	133	103	6x6 - W1.4xW1.4	0.028
	20	400	400	387	331	286	248	216	167	130		
	18	400	400	400	400	378	330	289	225	178		
	16	400	400	400	400	400	400	363	285	228		
5	22	400	400	378	322	277	240	208	159	123	6x6 - W2.0xW2.0	0.040
	20	400	400	400	393	339	295	258	199	156		
	18	400	400	400	400	400	392	344	269	213		
	16	400	400	400	400	400	400	400	341	273		
5 1/2	22	400	400	400	374	322	279	242	186	144	6x6 - W2.0xW2.0	0.040
	20	400	400	400	400	394	343	300	232	182		
	18	400	400	400	400	400	400	400	314	249		
	16	400	400	400	400	400	400	400	398	319		
6	22	400	400	400	400	367	318	277	212	165	4x4 - W1.4xW1.4	0.042
	20	400	400	400	400	400	392	342	265	208		
	18	400	400	400	400	400	400	400	360	286		
	16	400	400	400	400	400	400	400	400	366		

Notes:
7. Welded wire fabric shown in the table above is the SDI minimum required for temperature and shrinkage (0.00075 x concrete area above top of deck).
8. Allowable Superimposed Uniform Loads shown in table above are based on the following criteria:
   - Unfactored service level loads, determined using SDI design method per ANSI/SDI C-2017 Standard for Composite Steel Floor Deck-Slabs.
   - Single span conditions without negative bending reinforcing over supports.
   - The presence of shear studs have not been considered, design strength may be increased if shear studs are used.
   - Slab deflection is limited to minimum of Clear Span/360 or 1" under service level superimposed loading.

FLOOR DECK

3 ½” Slab Depth (total depth)
20-gauge steel Composite Deck
with normal-weight concrete
(Composite steel deck self-weight: 2 psf)
(Concrete weight: 31 psf)

LIBRARY FLOOR LOAD

150 psf: Live Load
15 psf: Dead Load:
2 psf: Deck self-weight
31 psf: Concrete weight
198 psf: Floor Load on Floor Deck

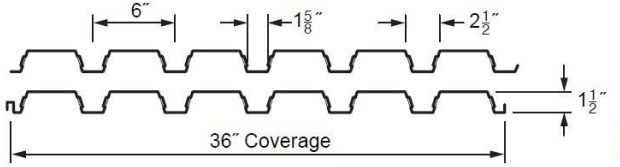
Composite Deck spans 7.5’ maximum (198 psf < 215 psf)

1.5CD, 1.5CDI Composite Deck

NORMAL-WEIGHT CONCRETE

PROPERTIES

Gage	Thickness (in)	Cover (in)	Weight (psf)
22	0.0295	3	1.63
20	0.0358		1.98
18	0.0474		2.62
16	0.0598		3.30



ICC-ES APPROVED
REFER TO ESR-3818

ASD SECTION PROPERTIES

DESIGN STRENGTHS (No Concrete Fill)

Gage	F_y (ksi)	A_s (in²/ft)	I_D 1 Span (in⁴/ft)	I_D 2+ Span (in⁴/ft)	I_p (in⁴/ft)		S_p (in³/ft)	S_n (in³/ft)	Mn,p/Ω (in-lb/ft)	Mn,n/Ω (in-lb/ft)	Vn/Ω (lb/ft)	*Rbe/Ω (lb/ft)	*Rbi/Ω (lb/ft)
22	40	0.480	0.163	0.175	0.157		0.176	0.183	4215	4391	2009	654	1181
20	40	0.582	0.205	0.213	0.201		0.224	0.235	5375	5624	2428	932	1696
18	40	0.770	0.280	0.281	0.279		0.304	0.315	7291	7553	3192	1557	2858
16	40	0.971	0.355	0.355	0.355		0.390	0.395	9332	9454	3996	2381	4396

CONSTRUCTION CLEAR SPANS

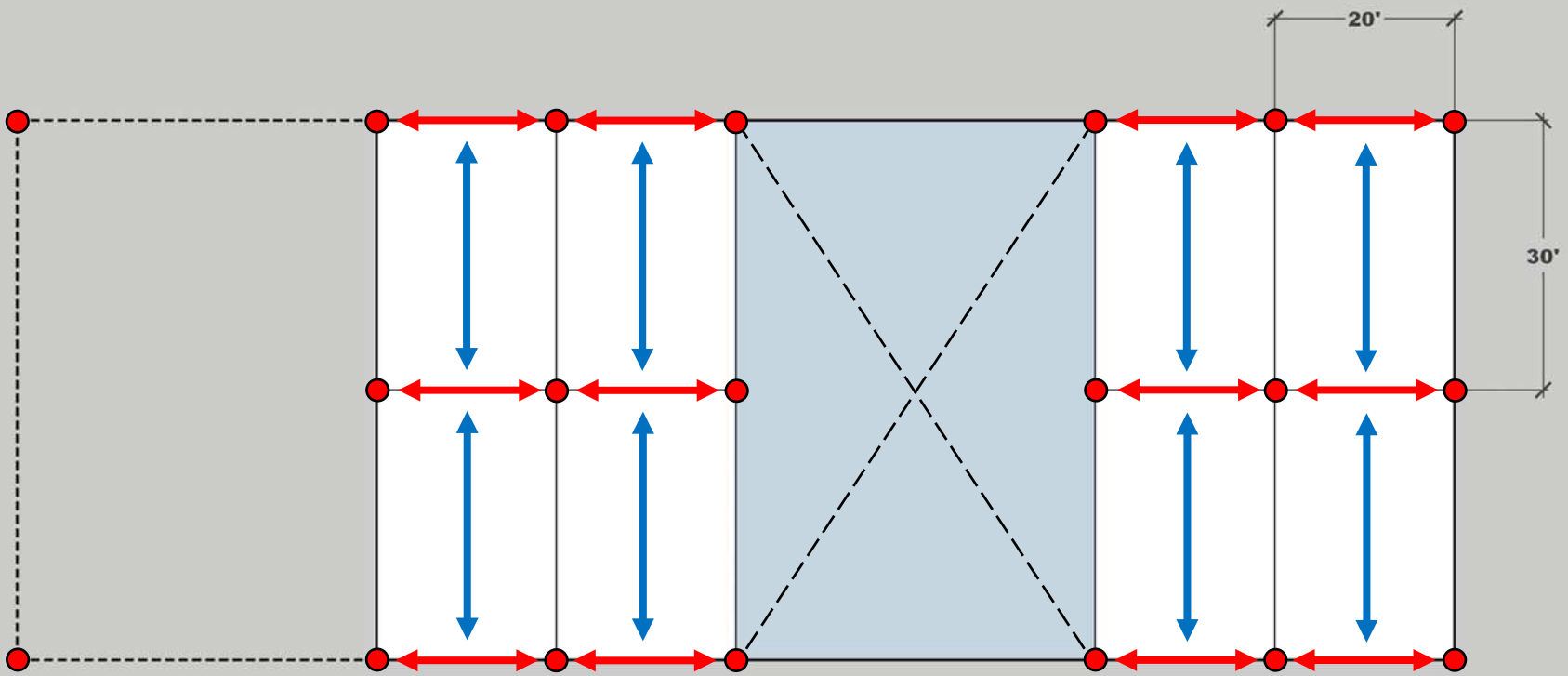
Total Slab Depth (in)	Gage	Concrete Weight (psf)	Maximum Construction Clear Span (ft - in)				Concrete Volume (ft³/ft²)
			Single	Double	Triple	Cantilever	
3 1/2	22	31	5 - 9	6 - 9	6 - 10	2 - 0	0.213
	20		6 - 9	8 - 0	8 - 1	2 - 5	
	18		8 - 4	9 - 8	9 - 11	3 - 1	
	16		9 - 8	10 - 9	11 - 1	3 - 8	

SLAB DESIGN - ALLOWABLE SUPERIMPOSED UNIFORM LOADS AND MINIMUM SDI SLAB REINFORCEMENT

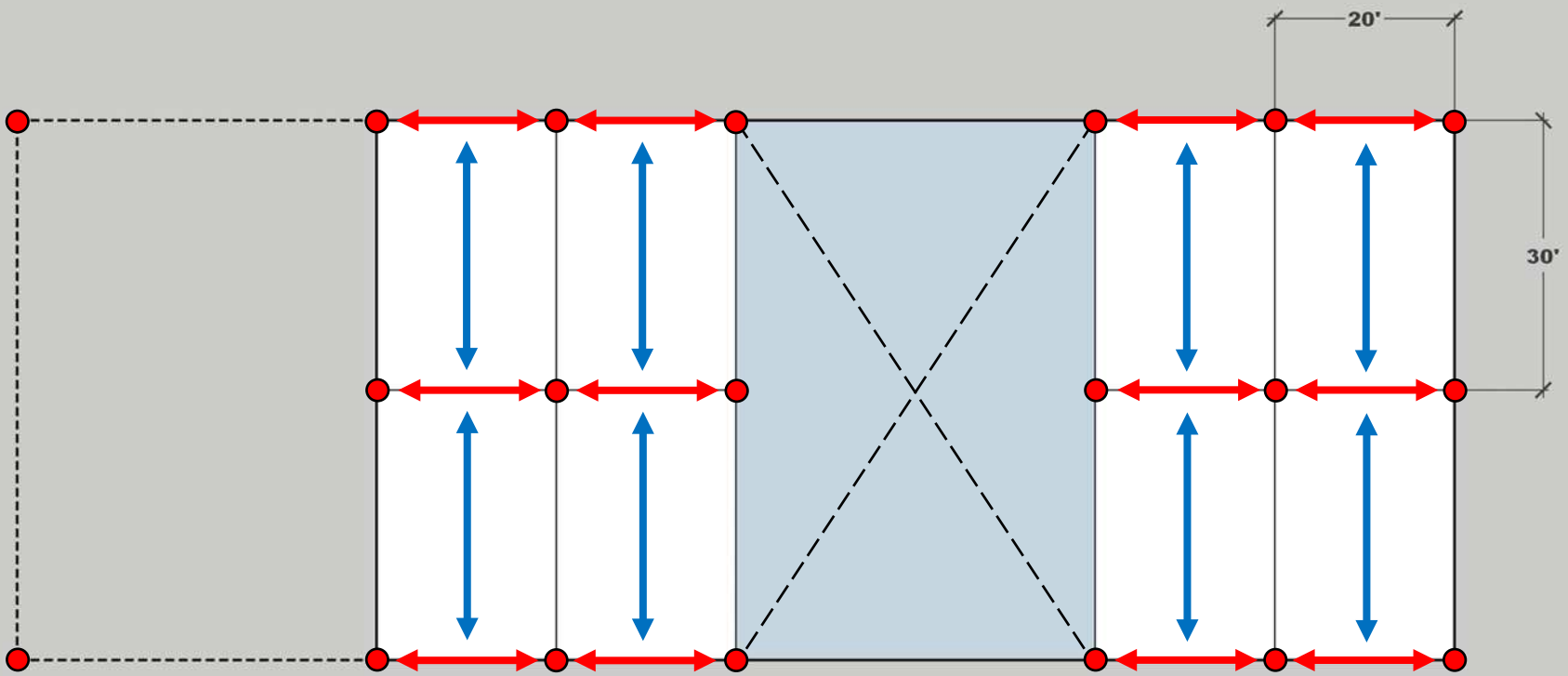
Normal-Weight Concrete (γc = 145 pcf, f'c = 3000 psi, n = 9)

Total Slab Depth (in)	Gage	Allowable Superimposed Uniform Load (psf) Clear Span (ft - in)									Recommended Welded Wire Fabric	Wire Area (in²/ft)
		6 - 0	6 - 6	7 - 0	7 - 6	8 - 0	8 - 6	9 - 0	10 - 0	11 - 0		
3 1/2	22	294	246	207	176	151	130	113	85	65	6x6 - W1.4xW1.4	0.028
	20	354	297	251	215	185	160	139	106	82		
	18	400	386	328	281	243	212	185	139	105		
	16	400	400	400	349	301	251	212	154	116		
4	22	370	310	262	223	191	165	143	109	83	6x6 - W1.4xW1.4	0.028
	20	400	375	318	272	234	203	177	136	105		
	18	400	400	400	357	309	269	236	183	145		
	16	400	400	400	400	385	336	295	229	172		
4 1/2	22	400	377	319	272	234	202	175	133	103	6x6 - W1.4xW1.4	0.028
	20	400	400	387	331	286	248	216	167	130		
	18	400	400	400	400	378	330	289	225	178		
	16	400	400	400	400	400	400	363	285	228		
5	22	400	400	378	322	277	240	208	159	123	6x6 - W2.0xW2.0	0.040
	20	400	400	400	393	339	295	258	199	156		
	18	400	400	400	400	400	392	344	269	213		
	16	400	400	400	400	400	400	400	341	273		
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	18	400	400	400	400	400	400	400	314	249		
	16	400	400	400	400	400	400	400	398	319		
6	22	400	400	400	400	367	318	277	212	165	4x4 - W1.4xW1.4	0.042
	20	400	400	400	400	400	392	342	265	208		
	18	400	400	400	400	400	400	400	360	286		
	16	400	400	400	400	400	400	400	400	366		

Notes:
7. Welded wire fabric shown in the table above is the SDI minimum required for temperature and shrinkage (0.00075 x concrete area above top of deck).
8. Allowable Superimposed Uniform Loads shown in table above are based on the following criteria:
   - Unfactored service level loads, determined using SDI design method per ANSI/SDI C-2017 Standard for Composite Steel Floor Deck-Slabs.
   - Single span conditions without negative bending reinforcing over supports.
   - The presence of shear studs have not been considered, design strength may be increased if shear studs are used.
   - Slab deflection is limited to minimum of Clear Span/360 or 1" under service level superimposed loading.



4060 R 2nd Floor Plan option selected…*What is the o.c. spacing of the Beams?*



Options

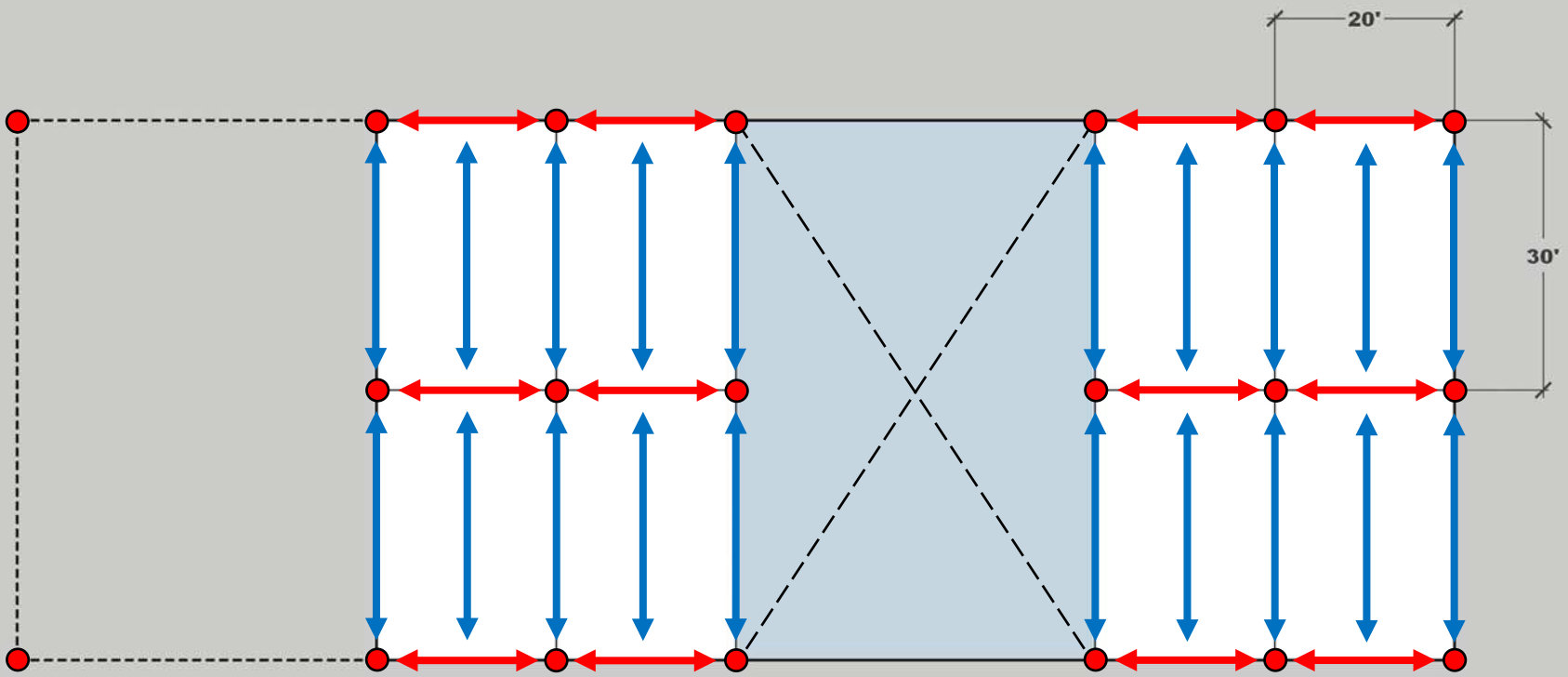
20’ / 7.5’ = 2.67 spaces

40’ / 7.5’ = 5.33 spaces

20’ / 3 spaces = 6.67’

20’ / 2 spaces = 10’

4060 R **2nd Floor Plan option selected***…What is the o.c. spacing of the Beams?*



Options

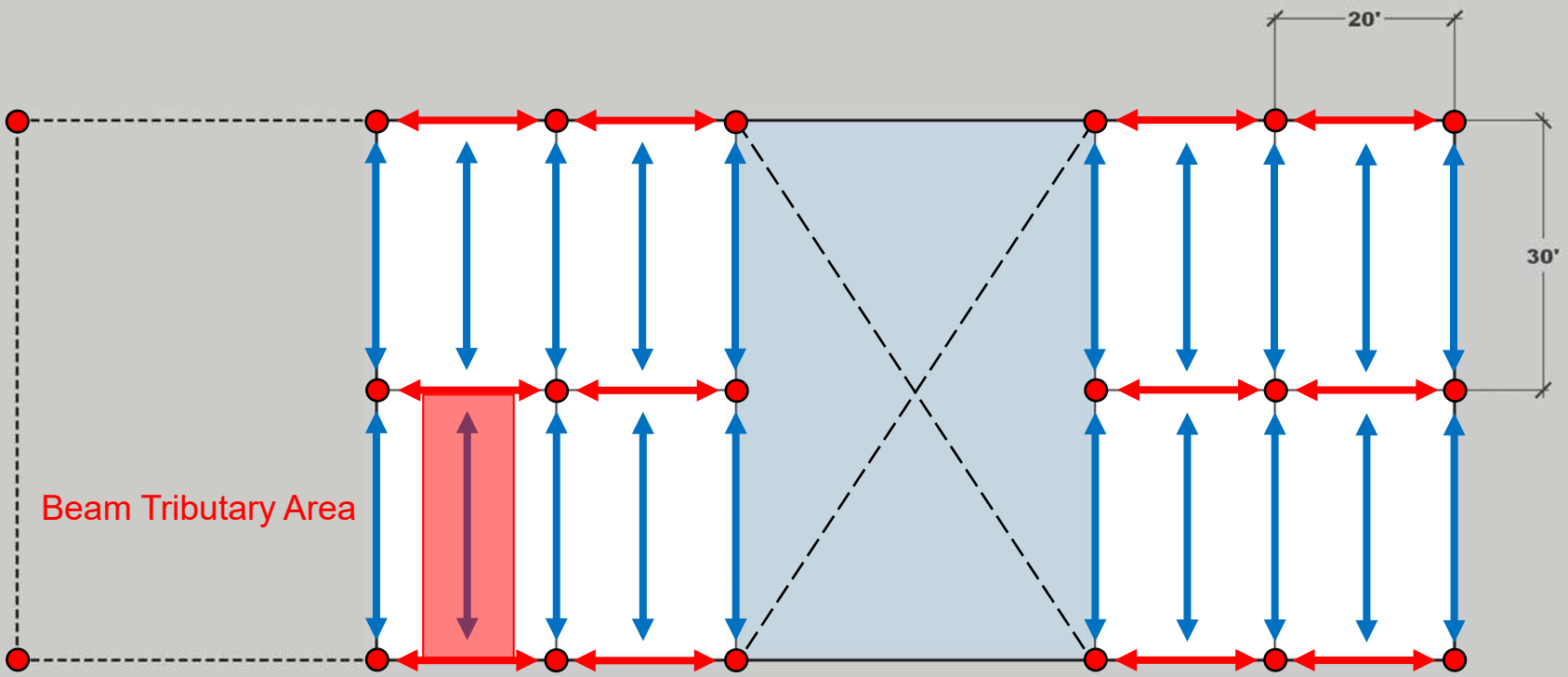
20’ / 7.5’ = 2.67 spaces

40’ / 7.5’ = 5.33 spaces

20’ / 3 spaces = 6.67’

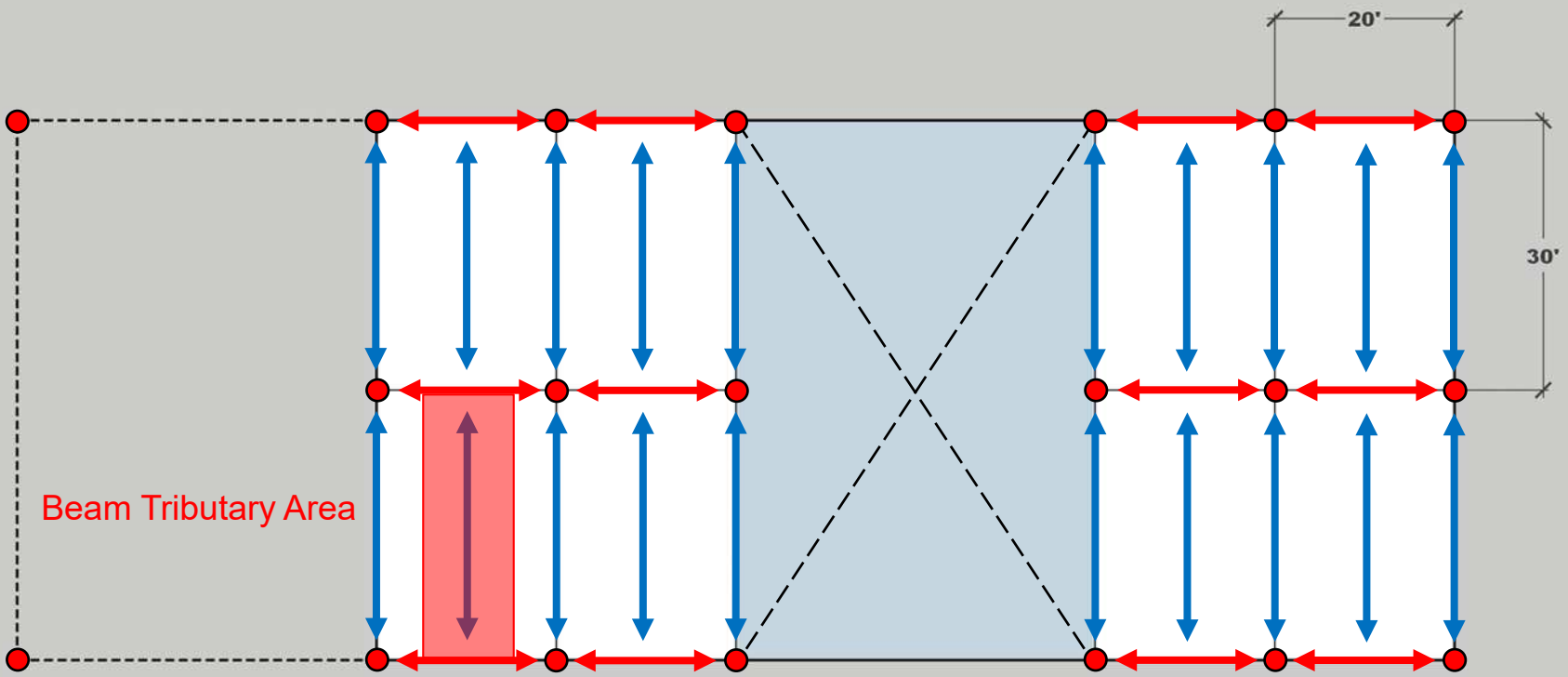
20’ / 2 spaces = 10’

4060 R 2nd Floor Plan option selected…*Beams spaced at 6.67’ o.c.*



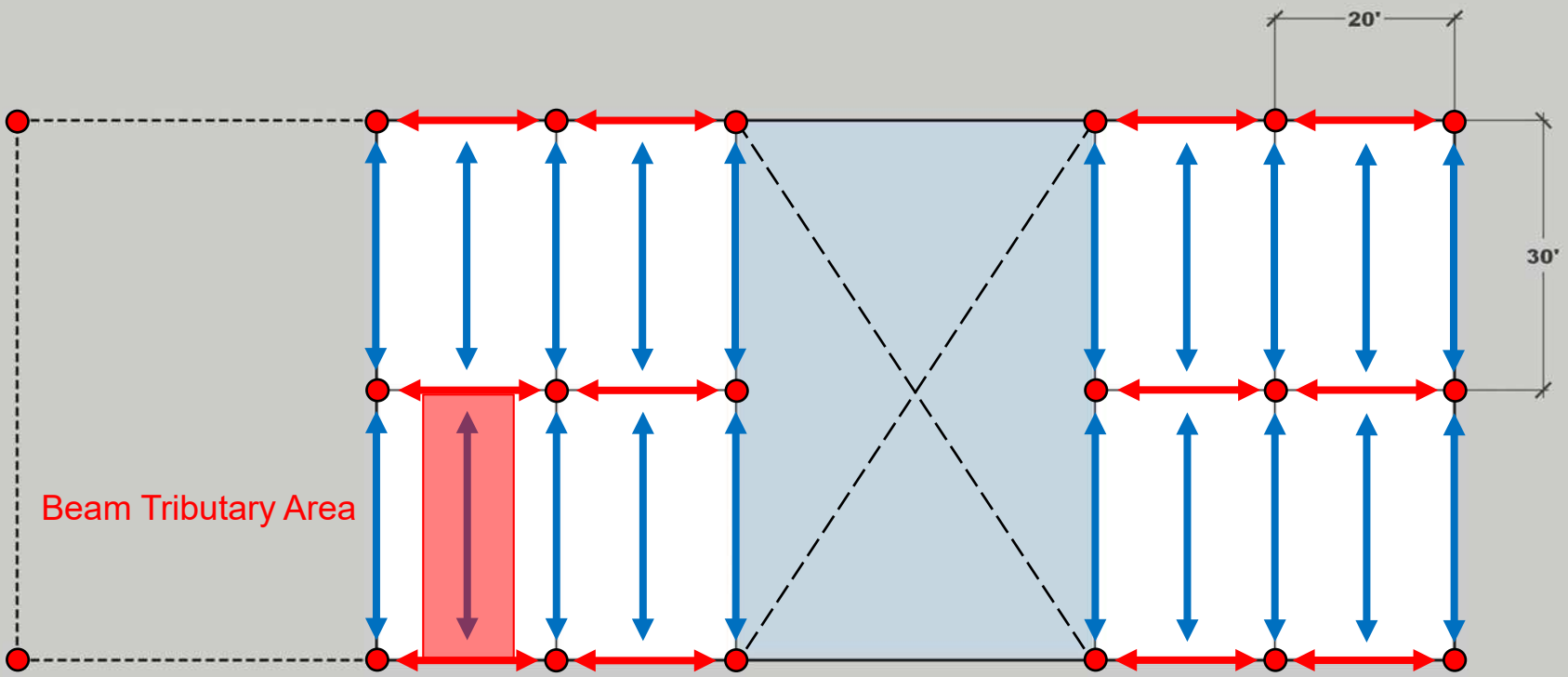
Beams spaced at 6.67’ o.c.
Beams span 30’ – 0”

4060 R 2nd Floor Plan option selected…*What is the load on the Beam in plf?*



Beams spaced at 6.67’ o.c.
Beams span 30’ – 0”
Beam Load Calculation: (198 psf) (6.67’) = 1,320.66 plf
Beam Load = 1,321 plf

4060 R **2nd Floor Plan option selected*…Beam Load is 1,321 plf***



Beams spaced at 6.67’ o.c.
Beams span 30’ – 0”
Beam Load Calculation: (198 psf) (6.67’) = 1,320.66 plf
Beam Load = 1,321 plf

4060 R 2nd Floor Plan option selected…*What is the size of the Beam?*

PROMPT

Calculate the size of the lightest weight steel wide flange beam that spans 30 feet with a uniformly distributed load of 1,321 plf. The live load portion of the load is 1,000 plf. Limit Live load deflection to L/360.

Find Beam size using Google Gemini

To determine the lightest standard ASTM A992 ($F_y = 50$ ksi) steel wide flange beam (W-shape), we evaluate both the **live load deflection limit** and the **flexural strength requirements** (assuming continuous lateral support to the compression flange).

Step 1: Deflection Limit Criterion

Span (L): $30 \text{ ft} = 360 \text{ in}$

Modulus of Elasticity (E): $29,000 \text{ ksi}$

Live Load (w_L): $1,000 \text{ plf} = 1.000 \text{ k/ft} = \frac{1.0}{12} \text{ k/in}$

Allowable Live Load Deflection ($\Delta_{L,\text{allow}}$):

$$\Delta_{L,\text{allow}} = \frac{L}{360} = \frac{360 \text{ in}}{360} = 1.00 \text{ in}$$

Using the maximum deflection formula for a simple beam under uniform load:

$$\Delta_L = \frac{5 w_L L^4}{384 E I_x} \leq 1.00 \text{ in}$$

Solving for the required moment of inertia ($I_{x,\text{req}}$):

$$I_{x,\text{req}} = \frac{5 w_L L^4}{384 E \Delta_{L,\text{allow}}} = \frac{5 \left(\frac{1.0}{12} \text{ k/in}\right) (360 \text{ in})^4}{384 (29,000 \text{ ksi})(1.00 \text{ in})} = 628.45 \text{ in}^4$$

Step 2: Bending Strength Requirement

Dead Load (w_D): $1,321 \text{ plf} - 1,000 \text{ plf} = 321 \text{ plf} = 0.321 \text{ k/ft}$

Total Service Load (w): 1.321 k/ft

ASD Method:

$$M_a = \frac{wL^2}{8} = \frac{1.321 \text{ k/ft} \times (30 \text{ ft})^2}{8} = 148.61 \text{ kip-ft} = 1,783.35 \text{ kip-in}$$

$$Z_{x,\text{req}} = \frac{M_a \cdot \Omega}{F_y} = \frac{1,783.35 \times 1.67}{50 \text{ ksi}} = 59.56 \text{ in}^3$$

LRFD Method:

$$w_u = 1.2w_D + 1.6w_L = 1.2(0.321) + 1.6(1.0) = 1.9852 \text{ k/ft}$$

$$M_u = \frac{w_u L^2}{8} = \frac{1.9852 \text{ k/ft} \times (30 \text{ ft})^2}{8} = 223.34 \text{ kip-ft} = 2,680.02 \text{ kip-in}$$

$$Z_{x,\text{req}} = \frac{M_u}{\phi F_y} = \frac{2,680.02}{0.9 \times 50 \text{ ksi}} = 59.56 \text{ in}^3$$

Follow the process, to learn

The lightest Wide-Flange Section:
W18 x 44 *(or)* **W21 x 44**

NOTE:
AISC does not have W18 x 44

Find Result

Step 3: Selection of Lightest W-Shape

We need a section meeting:

1. $I_x \geq 628.45 \text{ in}^4$
2. $Z_x \geq 59.56 \text{ in}^3$

Beam Size	Weight (lb/ft)	I_x (in^4)	Z_x (in^3)	Deflection (Δ_L)
W18x44	**44**	**896**	**84.8**	$0.70 \text{ in} \leq 1.00 \text{ in}$
W21x44	**44**	**843**	**95.4**	$0.75 \text{ in} \leq 1.00 \text{ in}$
W16x50	50	659	92.0	$0.95 \text{ in} \leq 1.00 \text{ in}$
W18x40	40	612	78.4	$1.03 \text{ in} > 1.00 \text{ in}$

Final Recommendation

Both **W18x44** and **W21x44** tie for the lightest section at **44 lb/ft.**

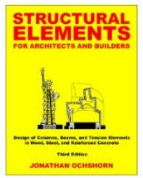
W18x44 is the preferred choice if depth is restricted (depth = 17.8 in) and provides lower live load deflection ($\Delta_L = 0.70 \text{ in}$).

W21x44 provides a higher plastic moment capacity ($Z_x = 95.4 \text{ in}^3$).

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Structural elements calculators

Follow on Bluesky contact



Laterally-Supported Steel Floor Beam Calculator

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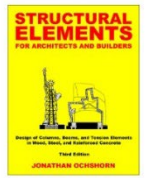
https://jonochshorn.com/scholarship/calculators-st/example8.6/index.html

Find Beam size using OCHSHORN

jonochshorn.com

Structural elements calculators

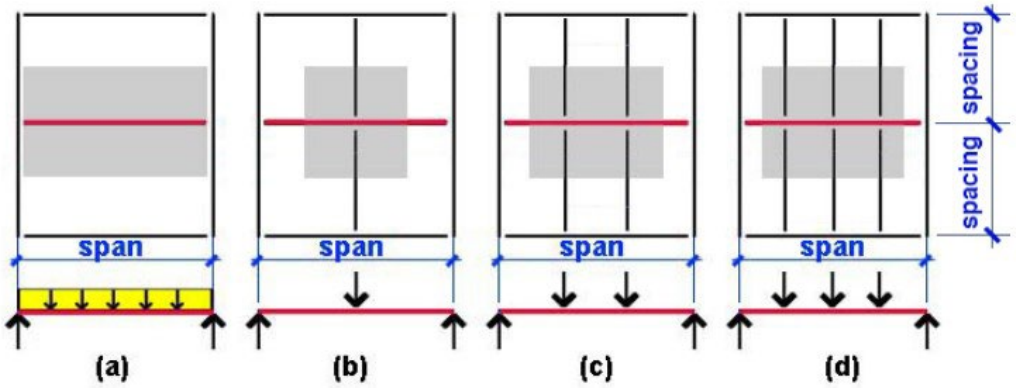
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Laterally-Supported Steel Floor Beam Calculator

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Review span types & review table

span (ft)	spacing	
30	30	ft

ASTM designation and yield stress	live load (psf)	dead load (psf)
A992: Fy=50 ksi	50	55

(not used)

60

Yield stress of 50 ksi has been selected.

allow. live load deflection = span/

360

allow. total load deflection = span/

240

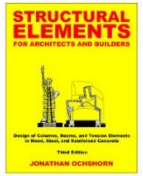
Load type:	Mmax (ft-kips)	Vmax (kips)	Lightest choice
a) uniformly-distributed load	**354**	**47.3**	***W24×62***
b) load at mid-span	**354**	**23.6**	***W24×62***
c) loads at third points	**315**	**31.5**	***W24×55***
d) loads at quarter points	**354**	**35.4**	***W24×62***

Design criteria:		reduction coeff:	live load used:
Use live load reduction?	no	**not used**	**50 psf**
Consider live load deflection?	no		
Consider total load deflection?	no		
Consider shear stress?	yes		
Consider bending stress?	yes		
Lateral support:	continuous		

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Structural elements calculators

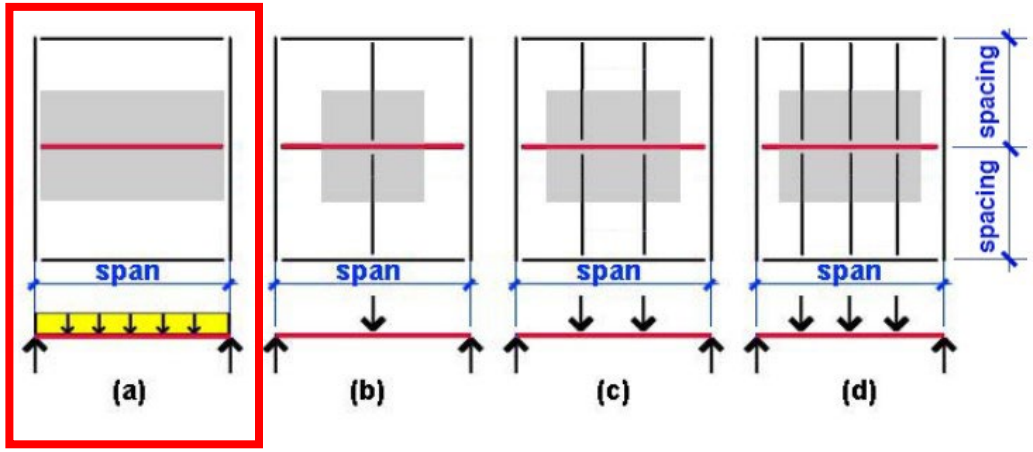
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Laterally-Supported Steel Floor Beam Calculator

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Uniformly-distributed Load

Enter span & spacing

span (ft)	spacing	
30	6.67	ft

ASTM designation and yield stress	live load (psf)	dead load (psf)
A992: Fy=50 ksi	150	48

(not used)

60

Yield stress of 50 ksi has been selected.

allow. live load deflection = span/

360

allow. total load deflection = span/

240

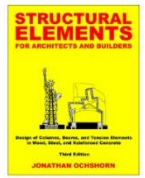
Load type:	Mmax (ft-kips)	Vmax (kips)	Lightest choice
a) uniformly-distributed load	**149**	**19.8**	***W21×44***
b) load at mid-span	**149**	**9.9**	***W18×35***
c) loads at third points	**132**	**13.2**	***W18×40***
d) loads at quarter points	**149**	**14.9**	***W18×40***

Design criteria:		reduction coeff:	live load used:
Use live load reduction?	no	**not used**	**150 psf**
Consider live load deflection?	yes		
Consider total load deflection?	no		
Consider shear stress?	yes		
Consider bending stress?	yes		
Lateral support:	continuous		

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Structural elements calculators

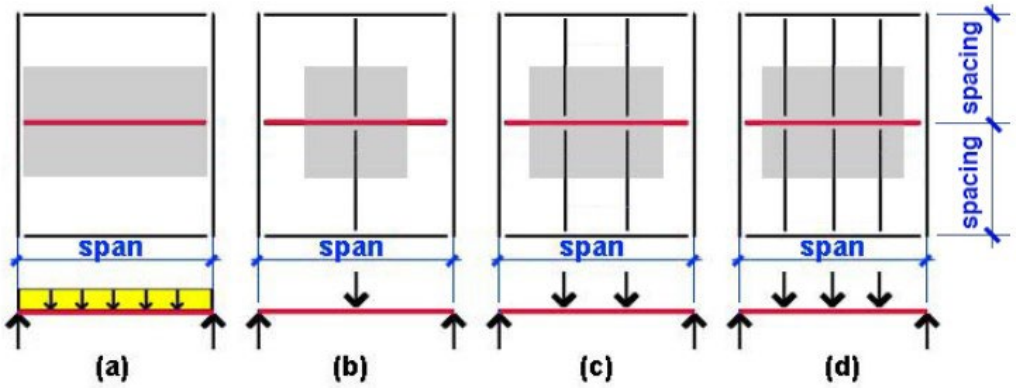
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Laterally-Supported Steel Floor Beam Calculator

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Enter live load & dead load

span (ft)	spacing	
30	6.67	ft

ASTM designation and yield stress	live load (psf)	dead load (psf)
A992: Fy=50 ksi	150	48

(not used)

60

Yield stress of 50 ksi has been selected.

allow. live load deflection = span/

360

allow. total load deflection = span/

240

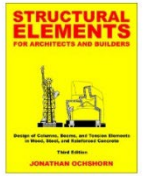
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c) loads at third points	**132**	**13.2**	***W18×40***
d) loads at quarter points	**149**	**14.9**	***W18×40***

Design criteria:		reduction coeff:	live load used:
Use live load reduction?	no	**not used**	**150 psf**
Consider live load deflection?	yes		
Consider total load deflection?	no		
Consider shear stress?	yes		
Consider bending stress?	yes		
Lateral support:	continuous		

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Structural elements calculators

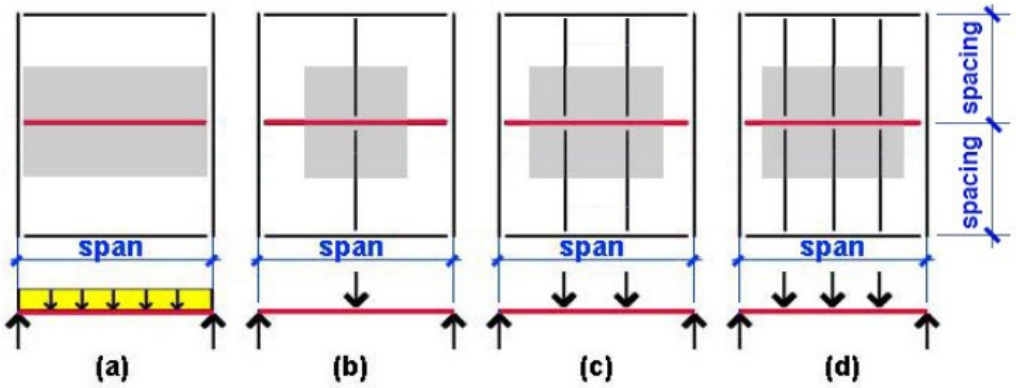
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Laterally-Supported Steel Floor Beam Calculator

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Enter deflection Live Load L/360

span (ft)	spacing	
30	6.67	ft

ASTM designation and yield stress	live load (psf)	dead load (psf)
A992: Fy=50 ksi	150	48

(not used)

60

Yield stress of 50 ksi has been selected.

allow. live load deflection = span/

360

allow. total load deflection = span/

240

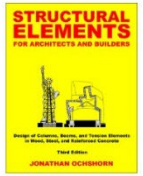
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c) loads at third points	**132**	**13.2**	***W18×40***
d) loads at quarter points	**149**	**14.9**	***W18×40***

Design criteria:		reduction coeff:	live load used:
Use live load reduction?	no	**not used**	**150 psf**
Consider live load deflection?	yes		
Consider total load deflection?	no		
Consider shear stress?	yes		
Consider bending stress?	yes		
Lateral support:	continuous		

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Structural elements calculators

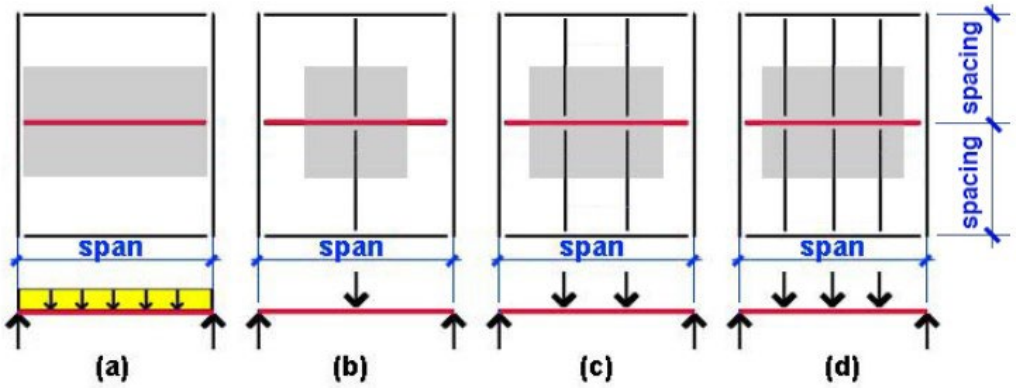
Follow on Bluesky contact



Laterally-Supported Steel Floor Beam Calculator

Jonathan Ochshorn

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span (ft)	spacing	
30	6.67	ft

ASTM designation and yield stress	live load (psf)	dead load (psf)
A992: Fy=50 ksi	150	48

(not used)

60

Yield stress of 50 ksi has been selected.

allow. live load deflection = span/

360

allow. total load deflection = span/

240

Load type:	Mmax (ft-kips)	Vmax (kips)	Lightest choice
a) uniformly-distributed load	**149**	**19.8**	***W21×44***
b) load at mid-span	**149**	**9.9**	***W18×35***
c) loads at third points	**132**	**13.2**	***W18×40***
d) loads at quarter points	**149**	**14.9**	***W18×40***

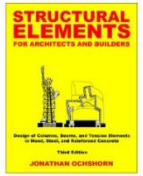
Design criteria:		reduction coeff:	live load used:
Use live load reduction?	no	**not used**	**150 psf**
Consider live load deflection?	yes		
Consider total load deflection?	no		
Consider shear stress?	yes		
Consider bending stress?	yes		
Lateral support:	continuous		

Check yes for Consider LL Deflection

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Structural elements calculators

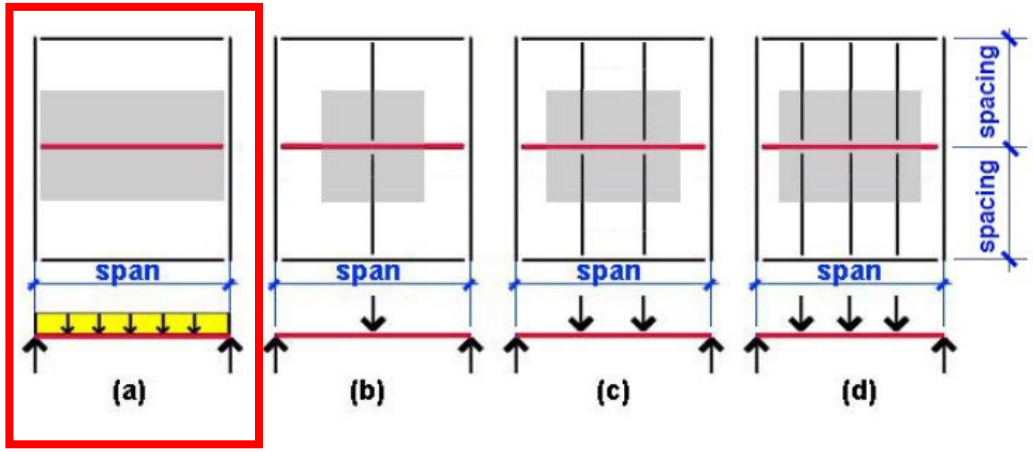
Follow on Bluesky contact



Laterally-Supported Steel Floor Beam Calculator

Jonathan Ochshorn

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Uniformly-distributed Load

Beam Results = W21 x 44

span (ft)	spacing	
30	6.67	ft

ASTM designation and yield stress	live load (psf)	dead load (psf)
A992: Fy=50 ksi	150	48

(not used)

60

Yield stress of 50 ksi has been selected.

allow. live load deflection = span/

360

allow. total load deflection = span/

240

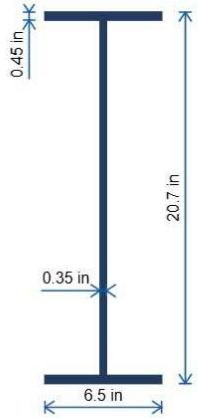
Load type:	Mmax (ft-kips)	Vmax (kips)	Lightest choice
a) uniformly-distributed load	**149**	**19.8**	***W21×44***
b) load at mid-span	**149**	**9.9**	***W18×35***
c) loads at third points	**132**	**13.2**	***W18×40***
d) loads at quarter points	**149**	**14.9**	***W18×40***

Design criteria:		reduction coeff:	live load used:
Use live load reduction?	no	**not used**	**150 psf**
Consider live load deflection?	yes		
Consider total load deflection?	no		
Consider shear stress?	yes		
Consider bending stress?	yes		
Lateral support:	continuous		

W21x44 Dimensions

I Beam Design Software

The steel beam dimensions for the W21x44 section can be seen in the table. The depth of the section is 20.7 in. The width of the section is 6.5 in.



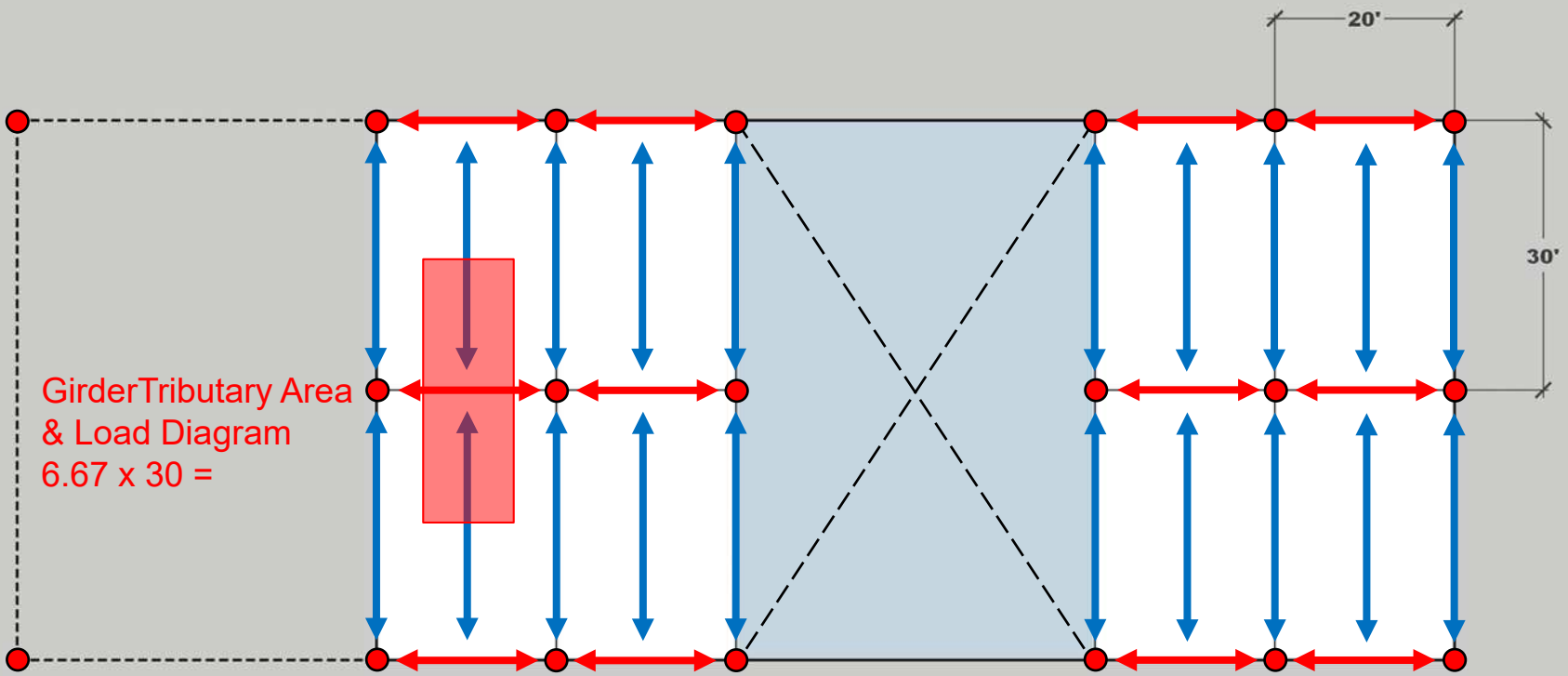
Dimension	Value
Depth	20.7 in
Top Width	6.5 in
Top Thickness	0.45 in
Bottom Width	6.5 in
Bottom Thickness	0.45 in
Web Thickness	0.35 in
Fillet Radius	0.5 in

W21x44 Section Properties

The W21x44 section has an area of 13in^2. The below table outlines the W21x44 section properties including the torsion constant as well as moment of inertia and plastic modulus properties.

A	J	Iyp	Izp	Alpha	Cy	Cz	Iw	Plastic Mod Yp	Plastic Mod Xp
13 in^2	0.77 in^4	20.7 in^4	843 in^4	0 deg	0 in	0 in	2110 in^6	10.2 in^3	95.4 in^3

Find dimensions, size in Revit, add to model and note in framing plan



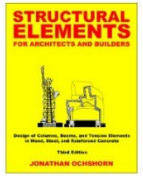
Girders spaced at 30’ o.c.
Girders span 20’ – 0”
Girder Load Calculation: Live Load 150 psf + Dead Load 48 psf + DL beam 7 psf
Girder Load = 205 psf = LL 150 psf + DL 55 psf

4060 R 2nd Floor Plan option selected…*What is the size of the Girder?*

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Structural elements calculators

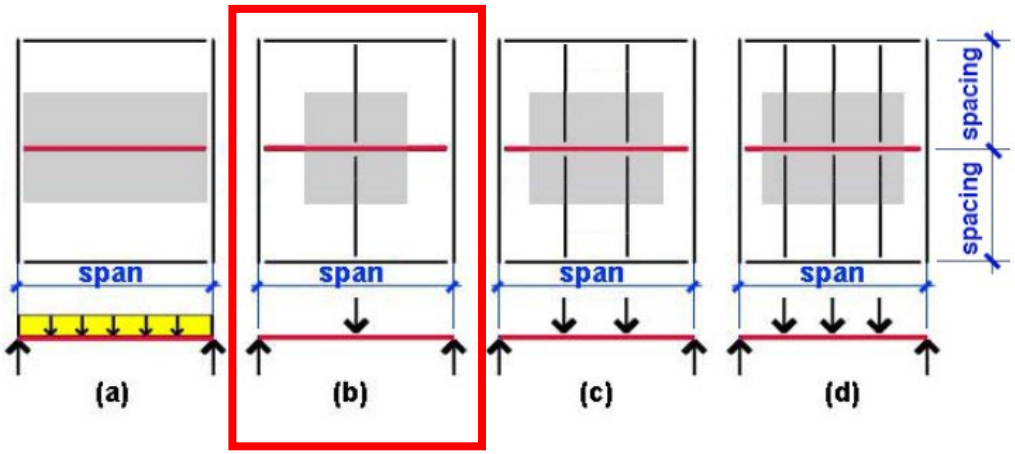
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Laterally-Supported Steel Floor Beam Calculator

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Girder Load at mid-span

span (ft)	spacing	
20	30	ft

ASTM designation and yield stress	live load (psf)	dead load (psf)
A992: Fy=50 ksi	150	55

(not used)

60

Yield stress of 50 ksi has been selected.

allow. live load deflection = span/

360

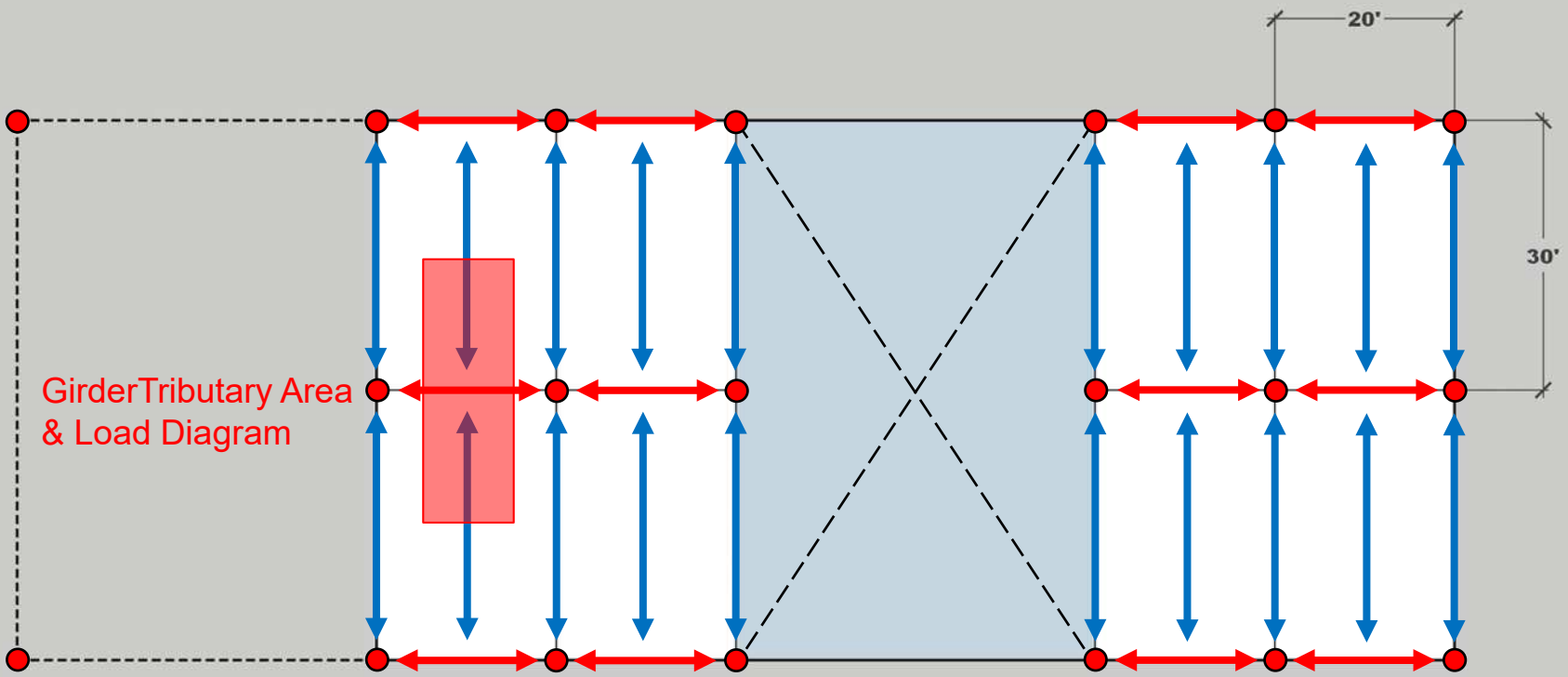
allow. total load deflection = span/

240

Load type:	Mmax (ft-kips)	Vmax (kips)	Lightest choice
a) uniformly-distributed load	**308**	**61.5**	***W21×55***
b) load at mid-span	**308**	**30.8**	***W21×55***
c) loads at third points	**273**	**41.0**	***W21×50***
d) loads at quarter points	**308**	**46.1**	***W21×55***

Design criteria:		reduction coeff:	live load used:
Use live load reduction?	no	**not used**	**150 psf**
Consider live load deflection?	yes		
Consider total load deflection?	no		
Consider shear stress?	yes		
Consider bending stress?	yes		
Lateral support:	continuous		

Girder Results = W21 x 55



Girders spaced at 30’ o.c.
Girders span 20’ – 0”
Girder Load = 205 psf = LL 150 psf + DL 55 psf
Girder Size: W21 x 55

4060 R **2nd Floor Plan option selected***…Girder is W21 x 55*

Search Sections...

Country Libraries

- America
- Australia
- Europe
- Canadian
- Britian/UK
- India
- Mexico
- South Africa
- New Zealand
- Singapore

Shape Libraries

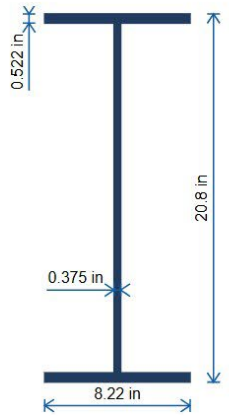
- AISC Steel Beams Tables
- American ADM Tables
- NDS® Timber Tables
- AISI Steel Tables
- CISC Steel Beam Tables
- CSSBI Steel Tables
- Australian 250 Grade Beam Tables
- Australian Wood Tables
- Australian Aluminum Tables
- British Steel Beam Tables
- British Wood Tables
- European Steel Beam Tables
- European Wood Tables

American › AISC › W Shapes › W21x55

W21x55 Dimensions

I Beam Design Software

The steel beam dimensions for the W21x55 section can be seen in the table. The depth of the section is 20.8 in. The width of the section is 8.22 in.



Dimension	Value
Depth	20.8 in
Top Width	8.22 in
Top Thickness	0.522 in
Bottom Width	8.22 in
Bottom Thickness	0.522 in
Web Thickness	0.375 in
Fillet Radius	0.498 in

W21x55 Section Properties

The W21x55 section has an area of 16.2in^2. The below table outlines the W21x55 section properties including the torsion constant as well as moment of inertia and plastic modulus properties.

A	J	Iyp	Izp	Alpha	Iw	Plastic Mod Yp	Plastic Mod Xp
16.2 in^2	1.24 in^4	48.4 in^4	1140 in^4	0 deg	4980 in^6	18.4 in^3	126 in^3

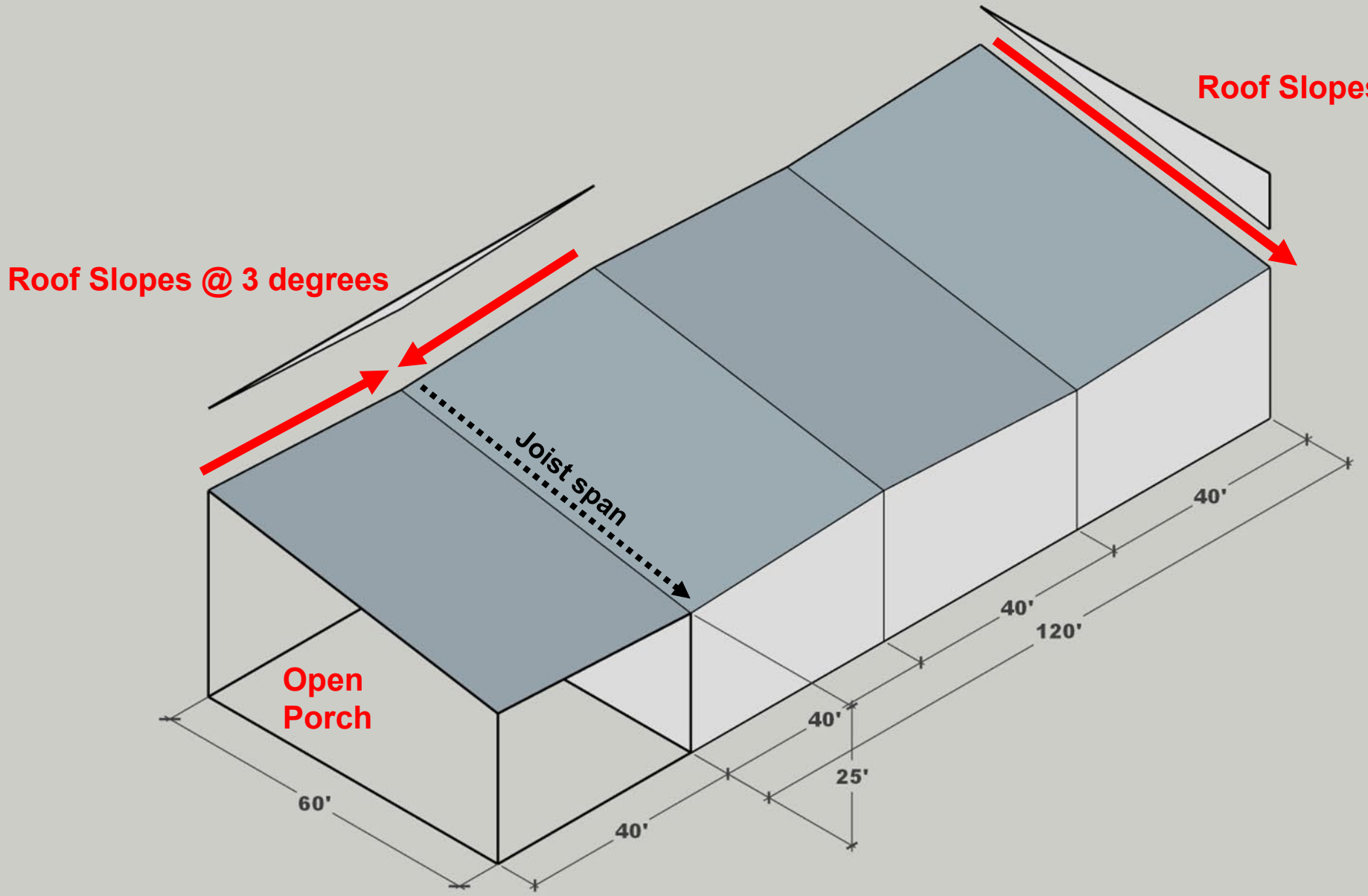
Find dimensions, size in Revit, add to model and note in framing plan

STEEL BUILDING as LIBRARY

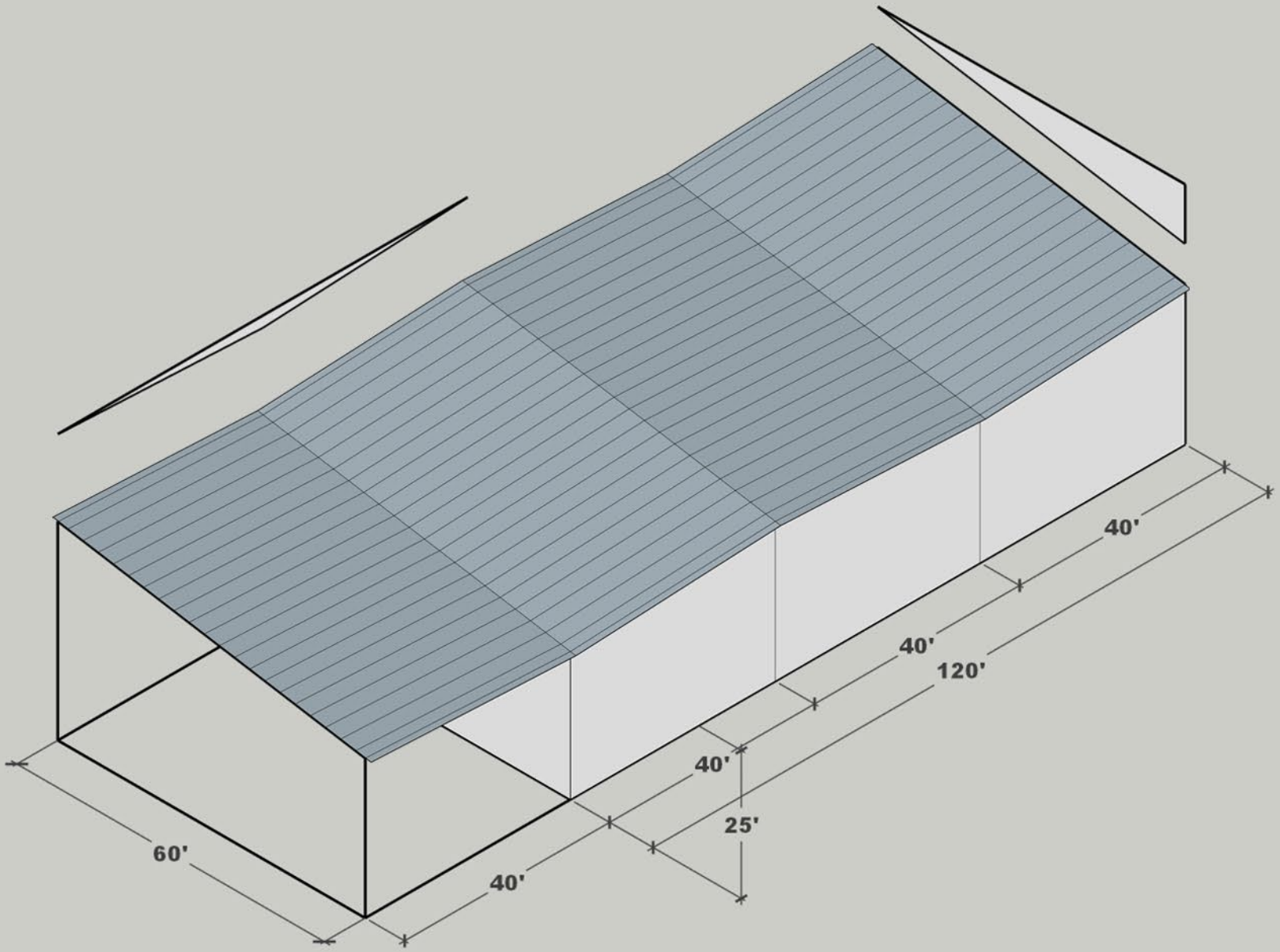
PROJECT PARAMETER SELECTIONS:

ROOF: MODEL 4060 C
FLOOR: MODEL 4060 R

…Update 3D model with sizes of Beam & Girder & Columns

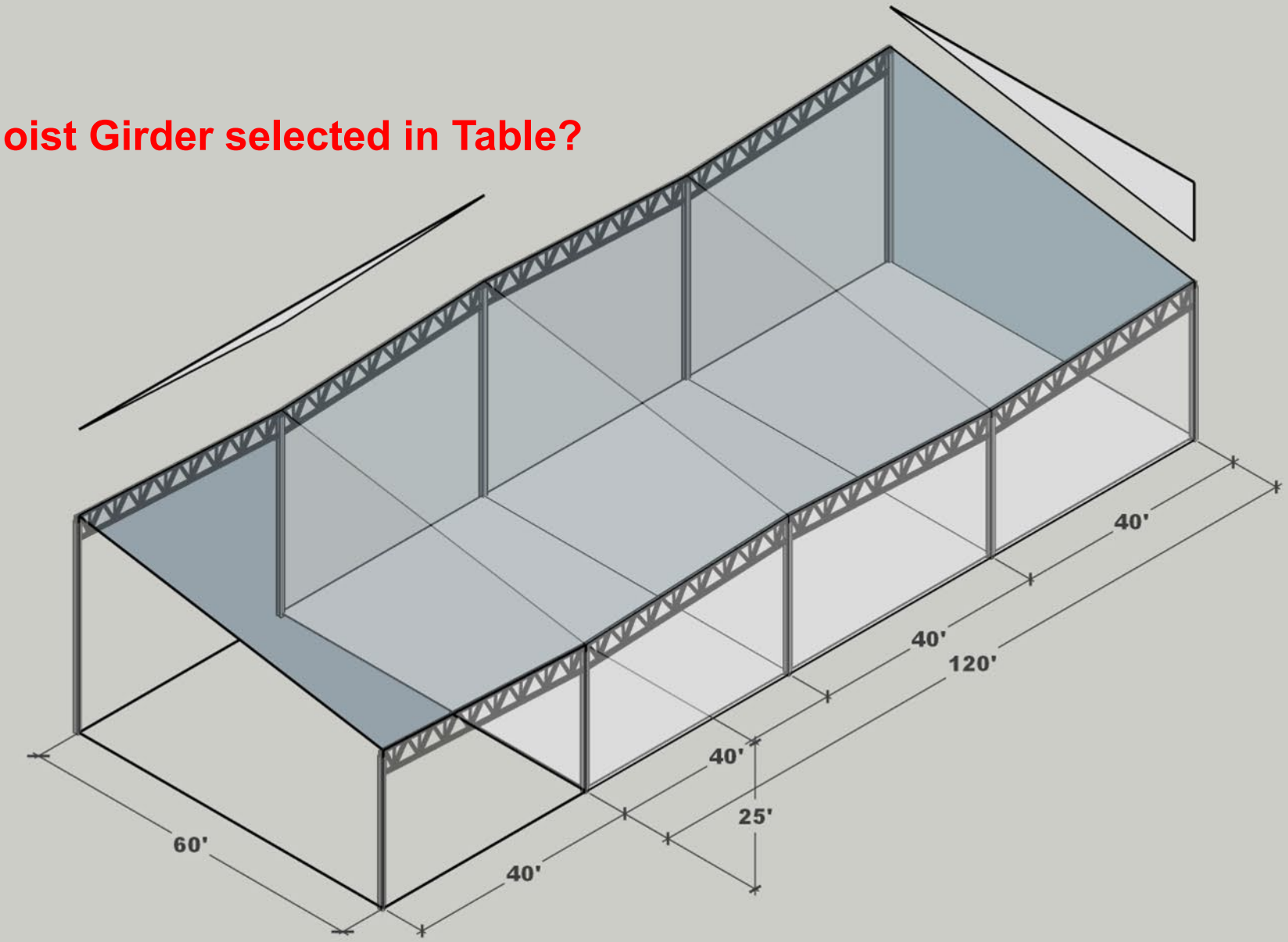


4060 C (40’ x 60’ Bays) **(Joists <u>must</u> span 60’)** LIBRARY 100 psf Snow Load



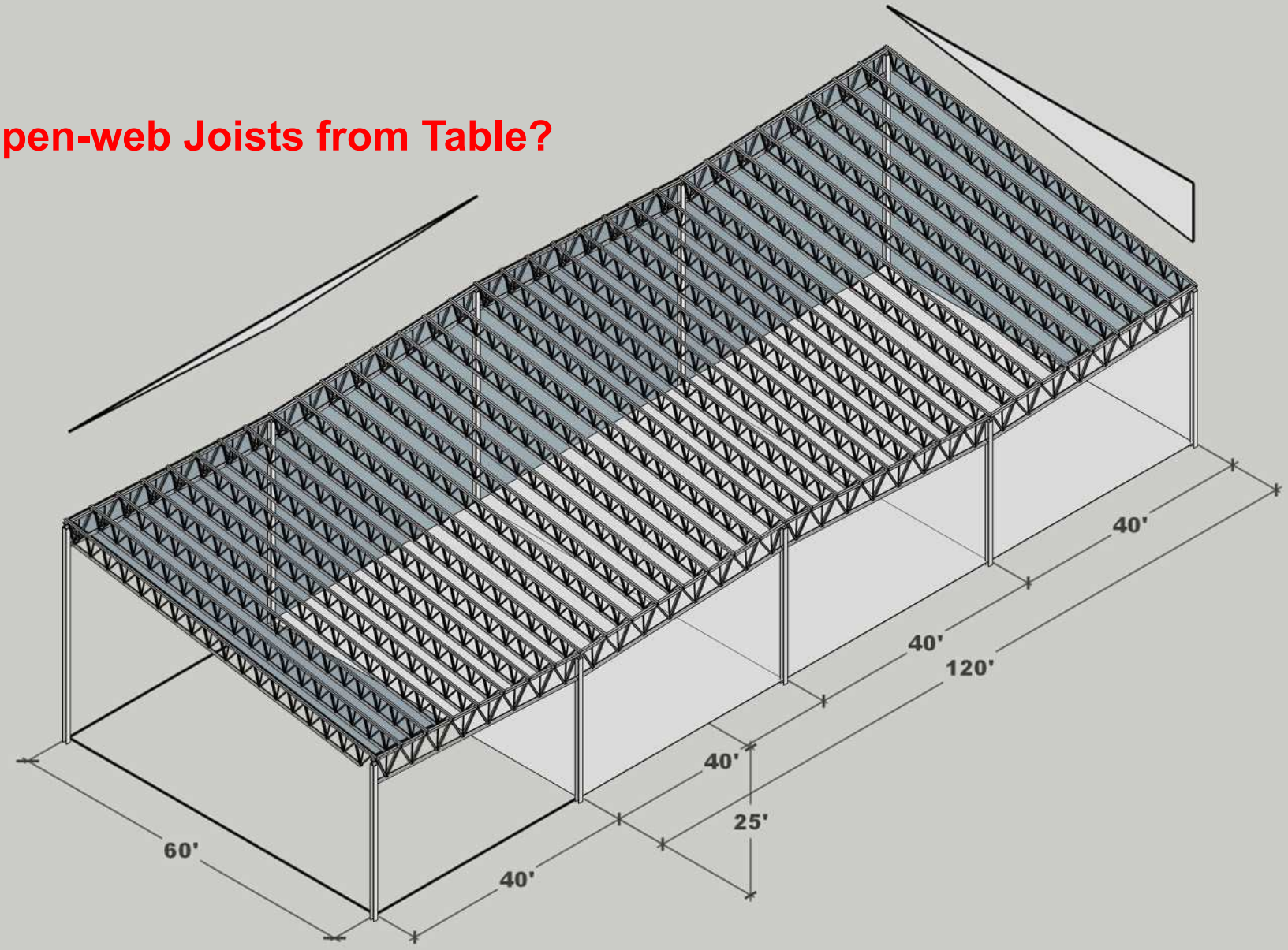
4060 C Type-B Roof Deck (1.5” x 36” wide panels)

Label specific Joist Girder selected in Table?

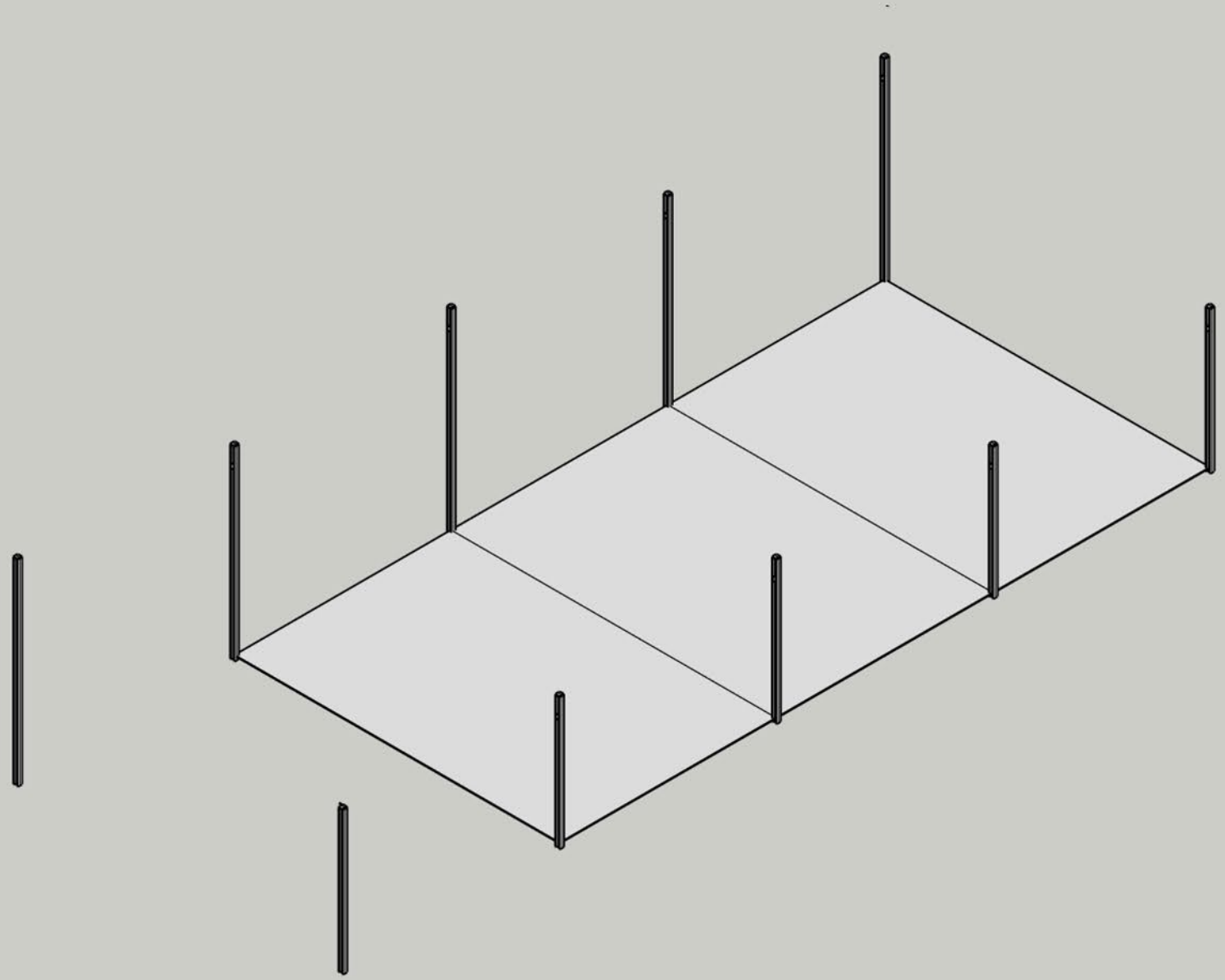


4060 C **Roof Columns and Joist Girders** (8 Panels @ 5’ wide = 40’ span)

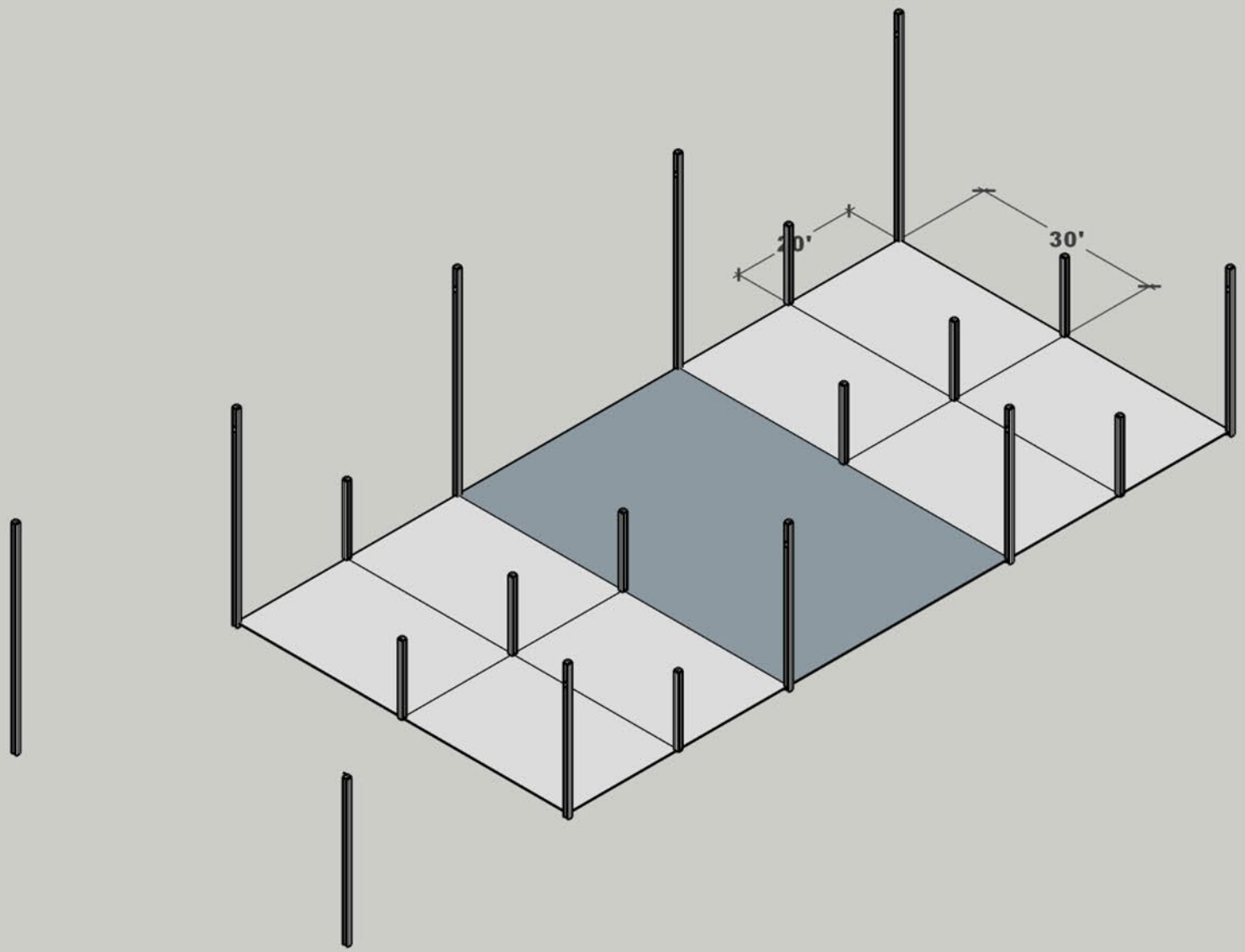
Label specific Open-web Joists from Table?



4060 C Roof Columns and Joists (60’ span, Joists spaced at 5’ o.c. to support Deck)

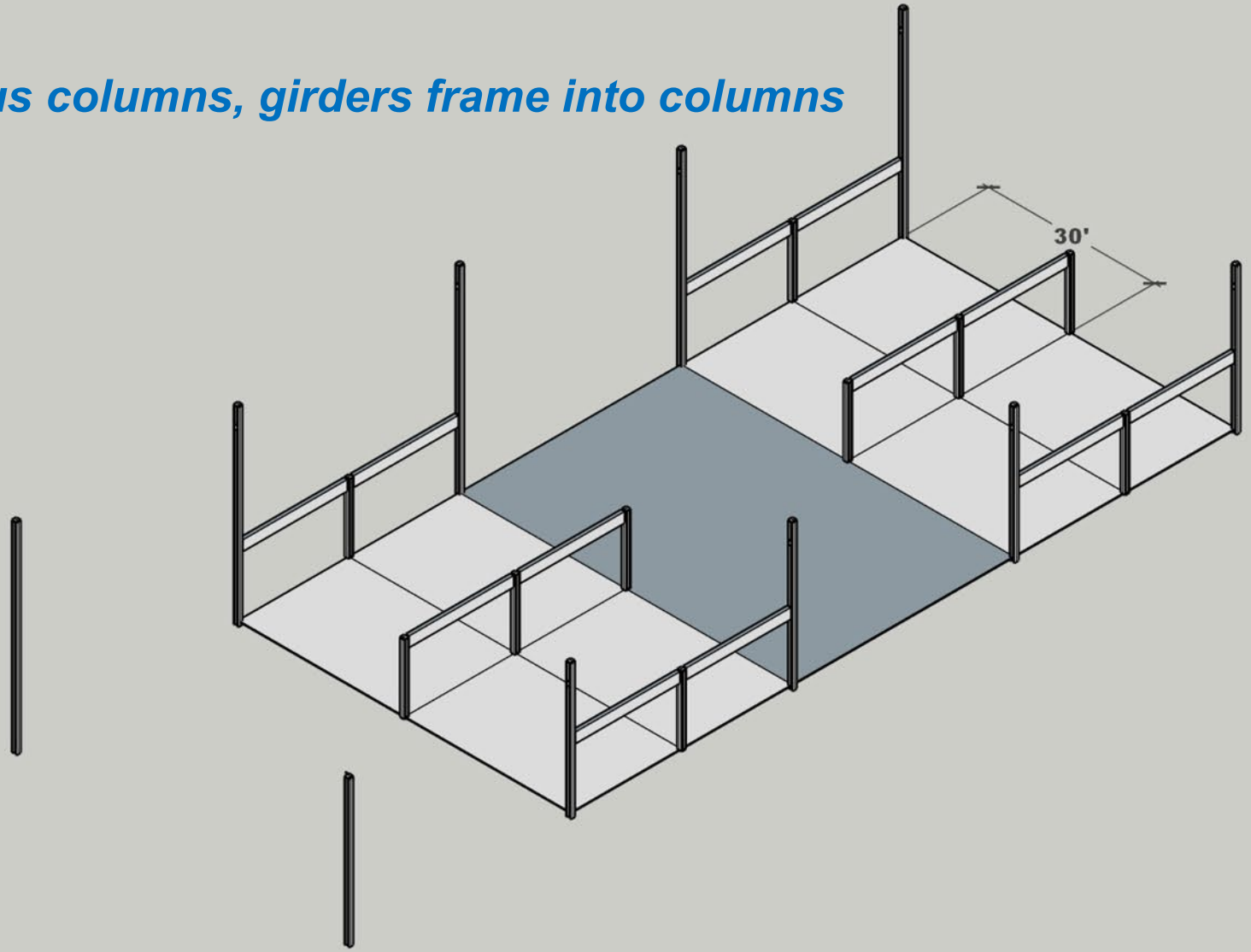


4060 C Roof Columns

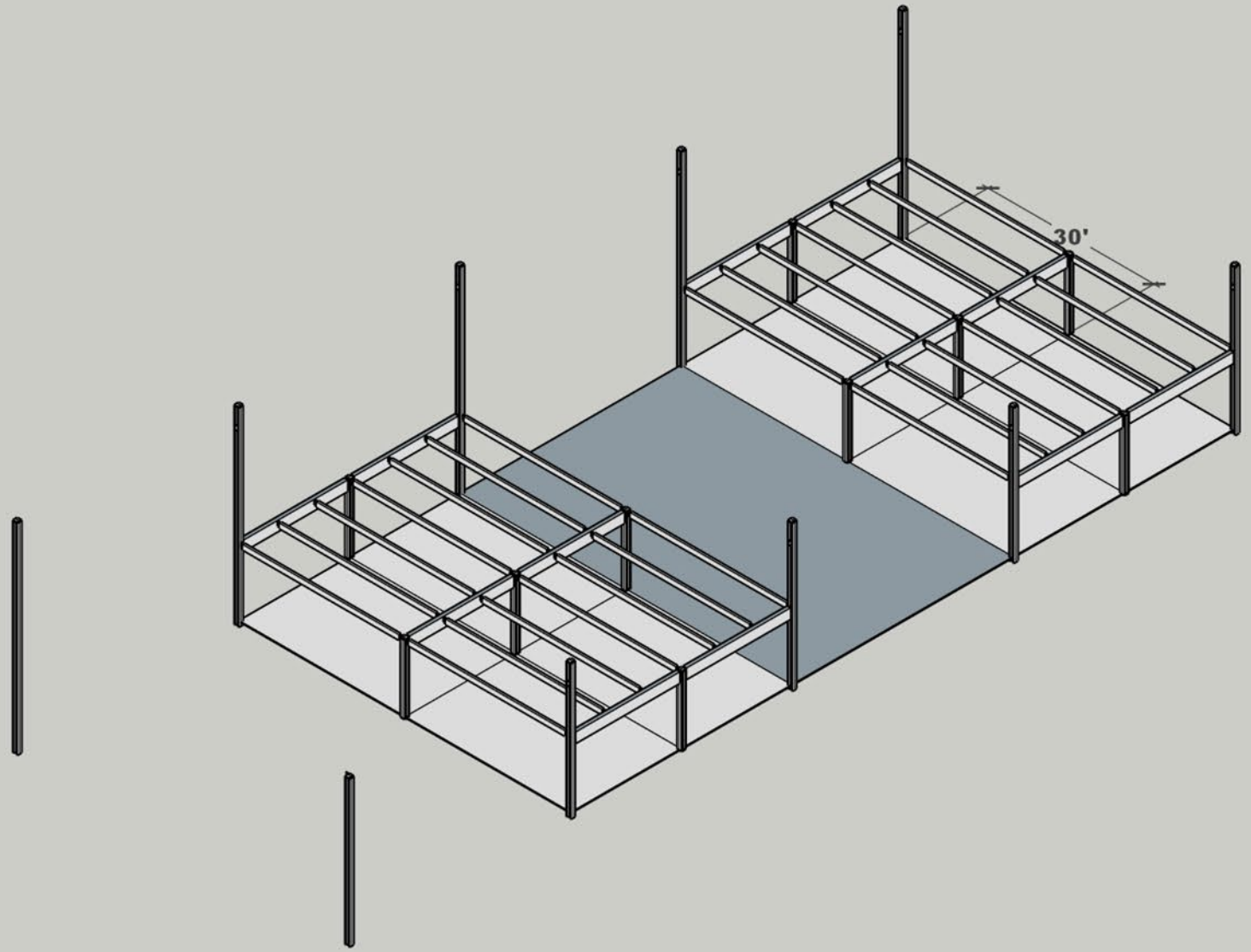


4060 C **Roof Columns and** **R** **Floor Columns** (20’ x 30’ Structural Floor Bays) 1:1.5 ratio

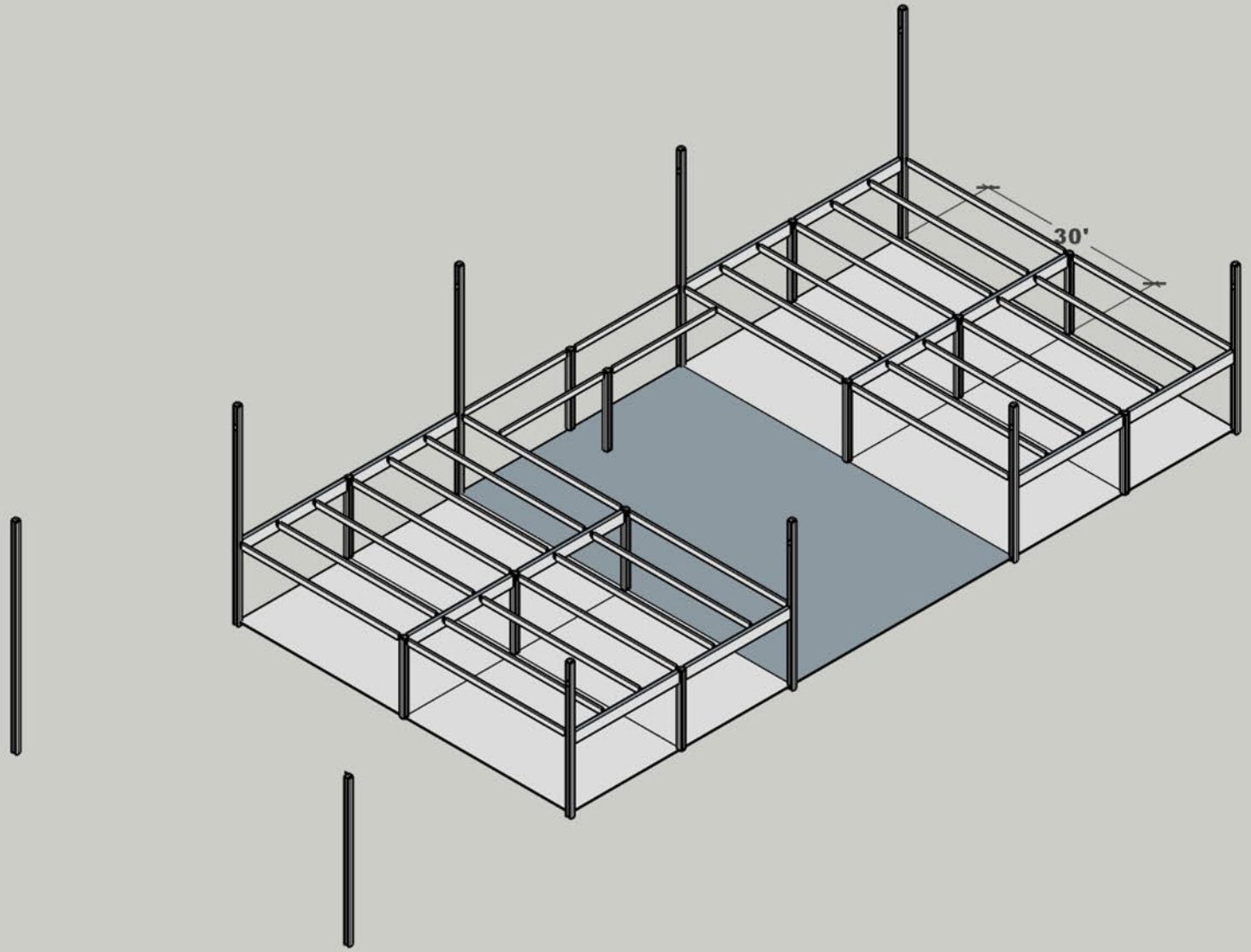
…model continuous columns, girders frame into columns



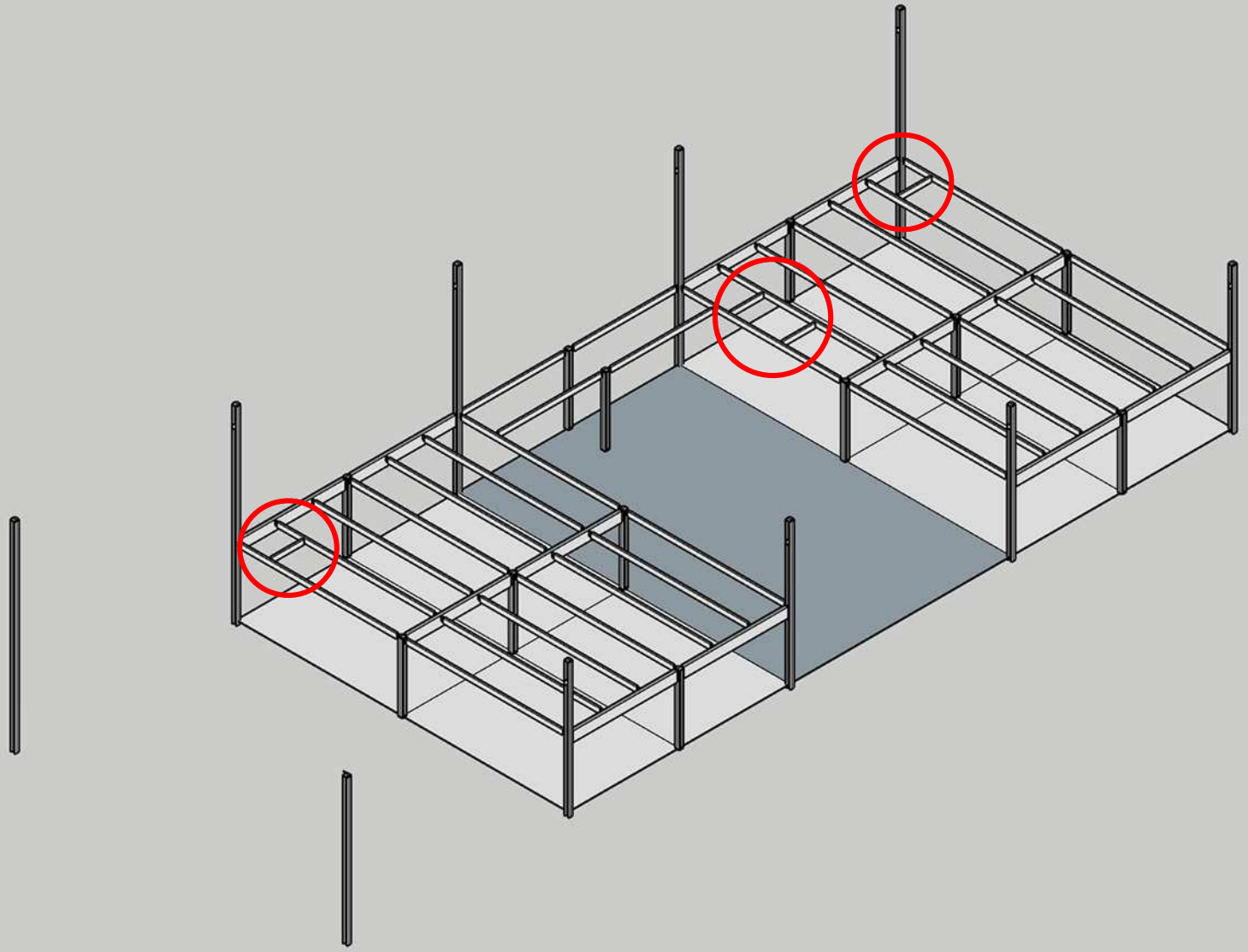
4060 R Columns and Wide-Flange Girders W21 x 55 (Girders span 20’ to columns)



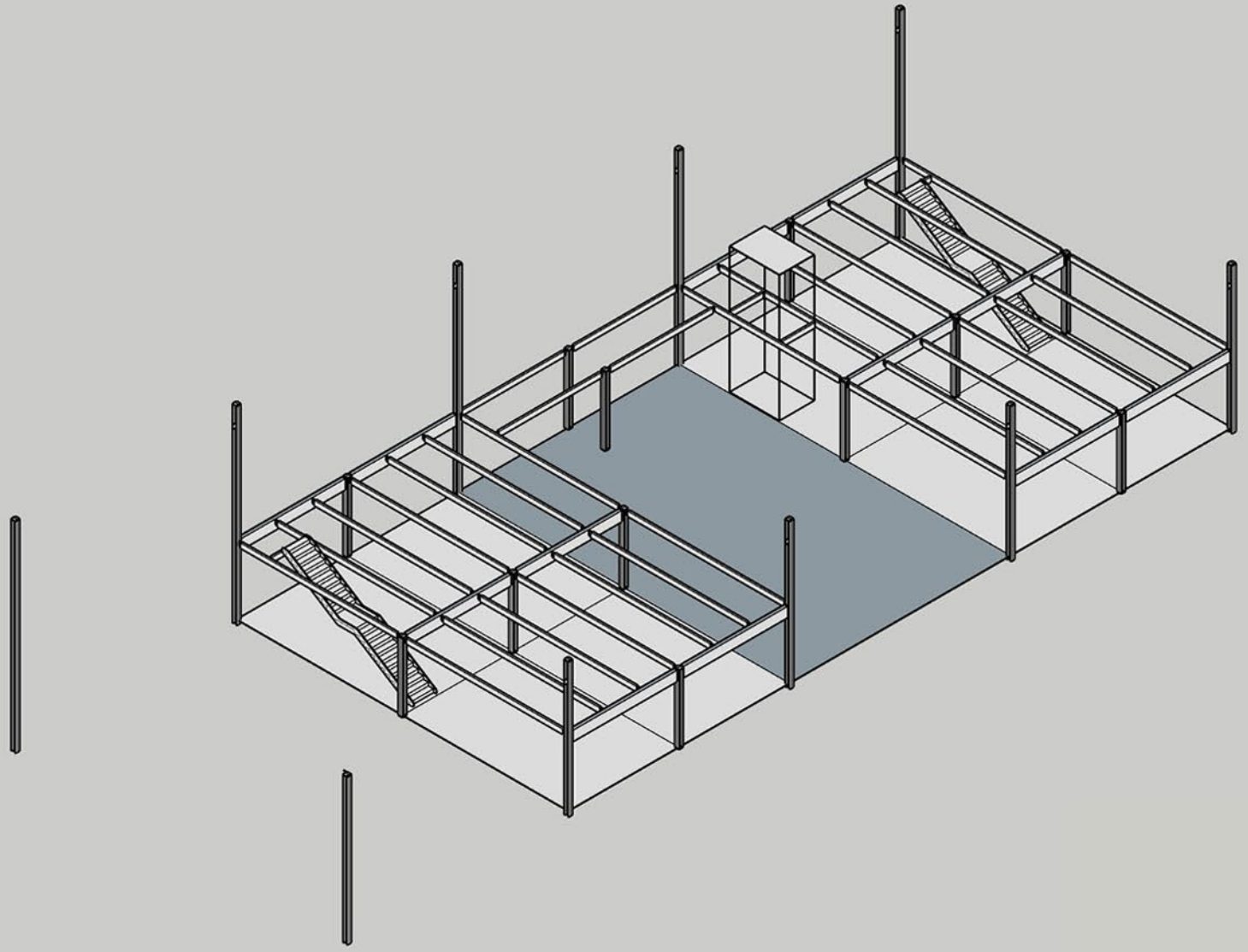
4060 R Columns, Girders and Beams W21 x 44 (Beams span 30’ to Girders (and Columns))



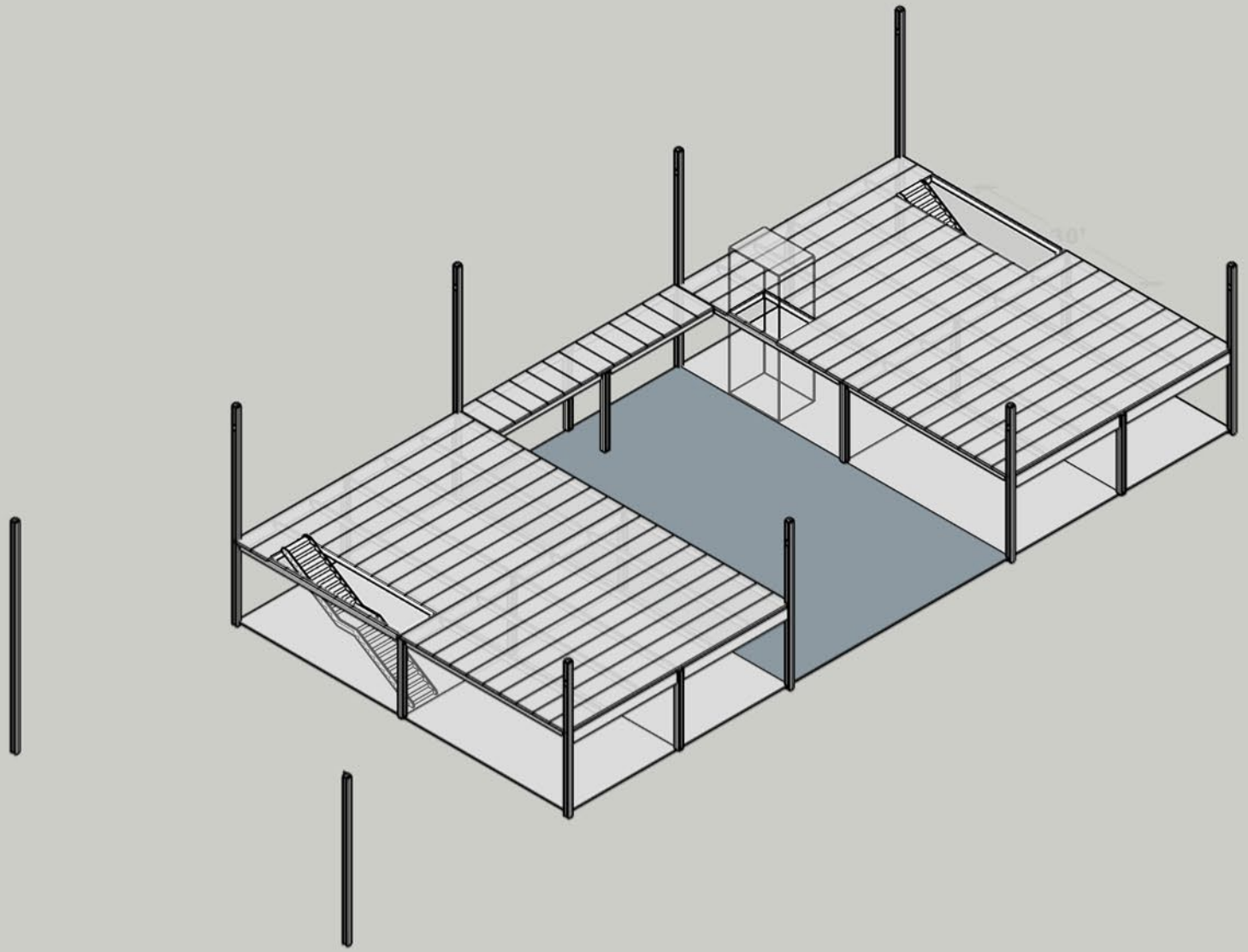
4060 R Add Columns and Beams for bridge? Make up to 7.5’ wide?



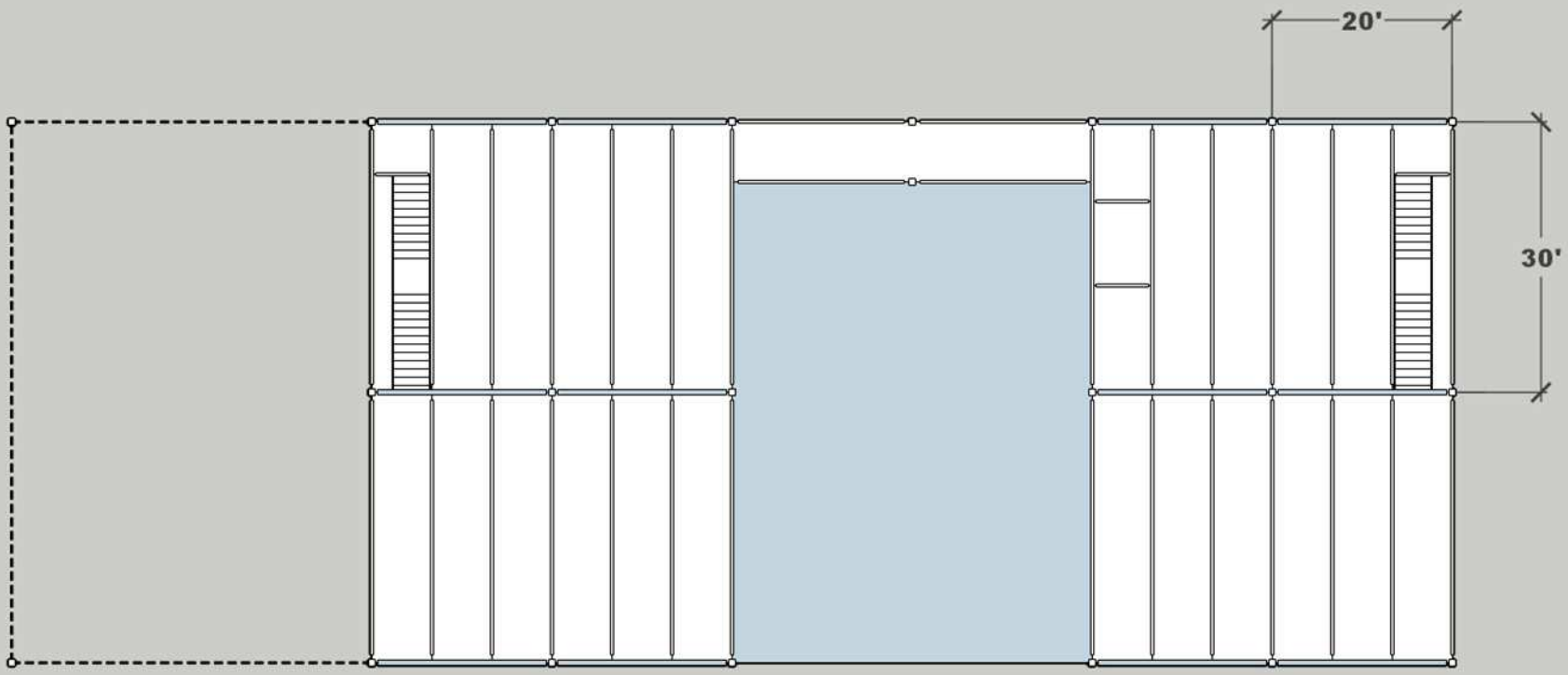
4060 R Add structural opening for stairs and opening for elevator (header beams)



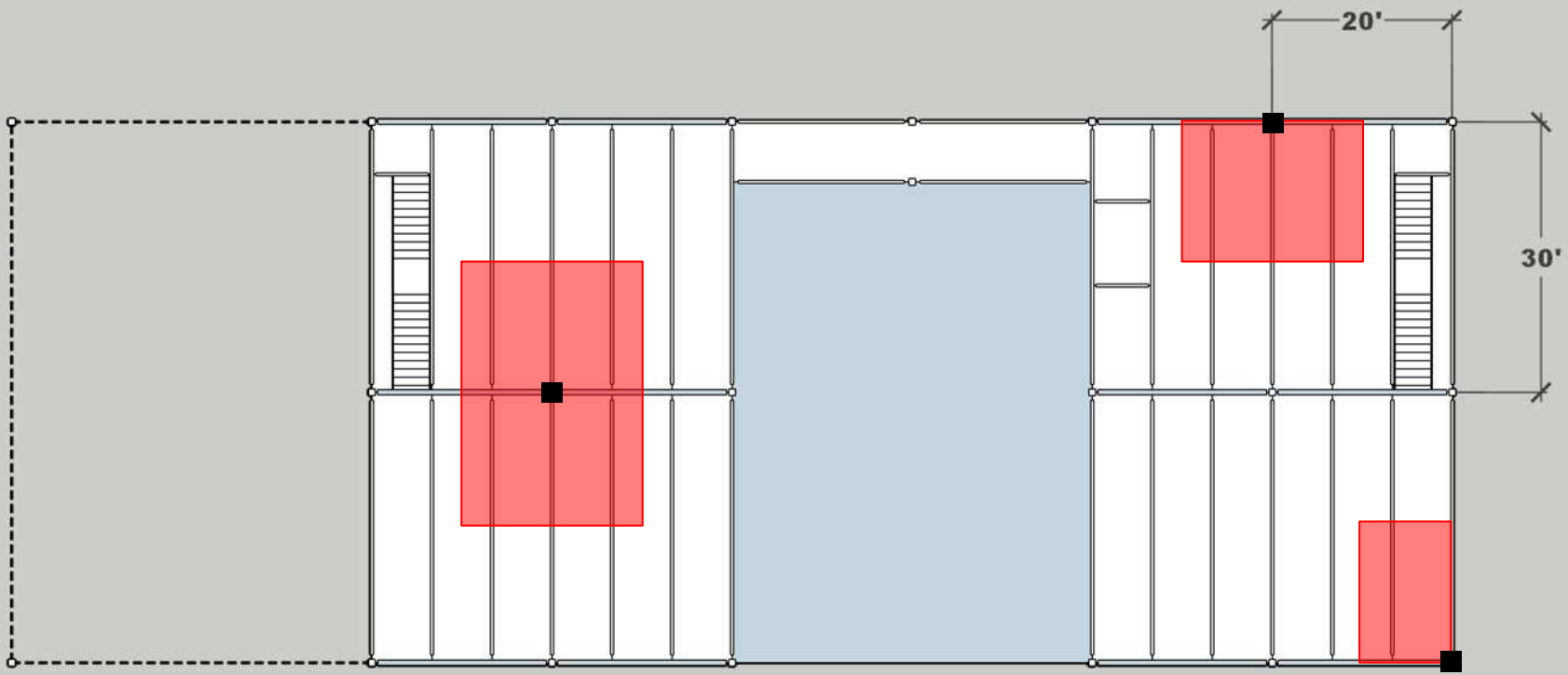
4060 R Add structural Stairs and elevator



4060 R Composite Deck spans 6.67’ over Beams (Deck corrugation in 3’ width)



4060 R Framing Plan @ Second Floor*…Label and Dimension?*



4060 C **Framing Plan @ Second Floor** ***…Find Load and size of three Columns?***

STEEL FLOOR

TERTIARY STRUCTURE

FLOOR LOADS

Floor Live Load: 150 psf
Floor Dead Load: 20 psf (MEP & flooring)
Composite Deck: 33 psf (31 psf + 2 psf)
Total Load: 203 psf

TERTIARY STRUCTURAL FLOOR DECK

3.5” Composite Deck comprised of
1.5” Steel Deck with integral Concrete slab.

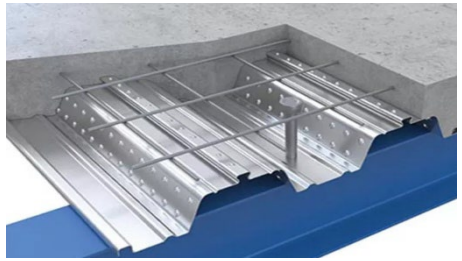
PRINCIPLE OF ECONOMY

Minimum of material to achieve a maximum of work

The 1.5” steel deck minimizes the use and weight of steel while accommodating acceptable construction spans and loads.
The 3.5” total depth of the Composite Deck minimizes the use of concrete, thereby minimizing floor deck weight and maximizing the use of material to achieve acceptable floor deck span and load capacity.



Composite Steel Decking sheet



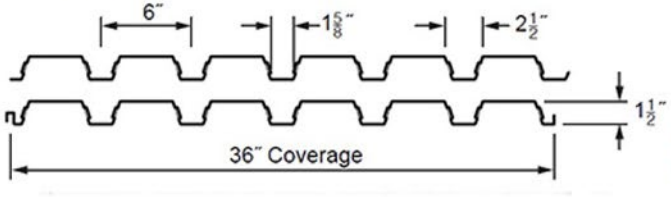
Composite Steel Deck Detail
Deck ribs and dimples to bond with concrete
Steel Studs welded to deck and beam
Welded Wire Fabric to reinforce concrete

1.5CD, 1.5CDI Composite Deck

NORMAL-WEIGHT CONCRETE

PROPERTIES

Gage	Thickness (in)	Cover (in)	Weight (psf)
22	0.0295	36	1.63
20	0.0358		1.98
18	0.0474		2.62
16	0.0598		3.30



ICC-ES APPROVED
REFER TO ESR-3818

CONSTRUCTION CLEAR SPANS

Total Slab Depth (in)	Gage	Concrete Weight (psf)	Maximum Construction Clear Span (ft - in) Single	Double	Triple	Cantilever	Concrete Volume (ft^3/ft^2)
3 1/2	22	31	5 - 9	6 - 9	6 - 10	2 - 0	0.213
	20		6 - 9	8 - 0	8 - 1	2 - 5	
	18		8 - 4	9 - 8	9 - 11	3 - 1	
	16		9 - 8	10 - 9	11 - 1	3 - 8	

SLAB DESIGN - ALLOWABLE SUPERIMPOSED UNIFORM LOADS AND MINIMUM SDI SLAB REINFORCEMENT

(f'c = 3000 psi, n = 9) Total Slab Depth (in)	Gage	Allowable Superimposed Uniform Load (psf) Clear Span (ft - in) 6 - 0	6 - 6	7 - 0	7 - 6	8 - 0	8 - 6	9 - 0	10 - 0	11 - 0	Recommended Welded Wire Fabric	Wire Area (in^2/ft)
3 1/2	22	294	246	207	176	151	130	113	85	65	6x6 - W1.4xW1.4	0.028
	20	354	297		215	185	160	139	106	82		
	18	400	386	328	281	243	212	185	139	105		
	16	400	400	400	349	301	251	212	154	116		

Composite Floor Deck (corrugated steel deck + concrete)
Total Floor Deck depth: 3.5”
Total Floor Load: 203 plf

Maximum allowable span for Composite Floor Deck
From the Composite Deck Table: 207 plf Allowable Uniform Load
7’-0” allowable Clear Span

Total Load of 203 plf < 207 plf allowable
7’-0” span is maximum allowable for the Floor Deck

STEEL FLOOR

SECONDARY HORIZONTAL STRUCTURE

BEAM SPACING

Beams are spaced at 6.67’ o.c. to optimize the floor deck to span as close to 7’ as possible, while conforming to the 20’ x 30’ structural bay for the floor system. The 20 x 30 bay is the preferred bay ratio of 1.5 to 1, with Beams spanning in the 30’ direction and Girders spanning 20’.

TRIBUTARY LOAD ON BEAM A

Composite Deck Floor Load: 203 psf
Deck span and Beam spacing: 6.67’
Uniformly Distributed Tributary Load on Beam calculation: 203 psf (6.67’) = 1,354 plf
LOAD ON BEAM A: 1,354 plf

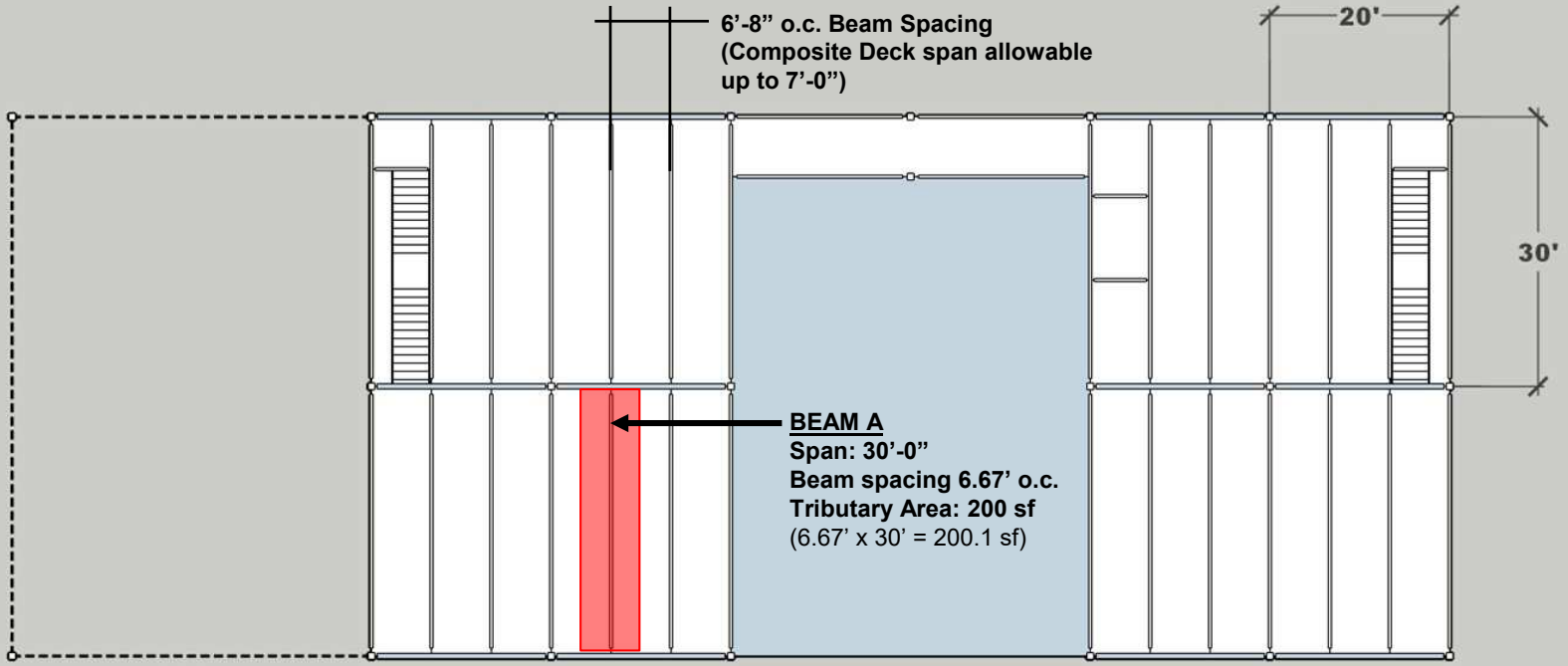
SIZING OF BEAM A

Sizing based on values entered and results indicated using the following online structural Steel Beam Calculator:
https://jonochshorn.com/scholarship/calculators-st/example8.6/index.html

Results indicated a wide-flange beam W18 x 35 is the lightest section to meet the load and beam span criteria. This beam is 17.7 inches deep and weighs 35 plf.
SELECTION FOR BEAM A: W18 X 35

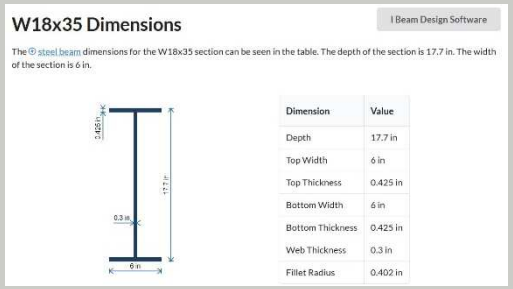
PRINCIPLE OF ECONOMY

Minimum of material to achieve a maximum of work
Typical floor beam sized for the lightest weight in pounds per lineal foot to carry to floor load. The lightest weight section uses the least amount of steel to perform the work of supporting the floor loads and structural floor deck.

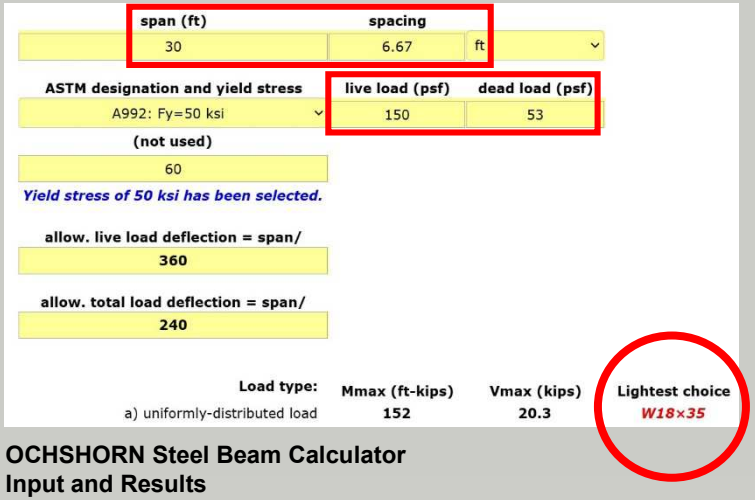


BEAM A

Total Tributary Load: 40.62 kips
(203 psf x 200 sf)
Uniform Distributed Load: 1,354 plf
(203 psf x 6.67’ = 1,354 plf)



W18X35 size



OCHSHORN Steel Beam Calculator Input and Results

STEEL ROOF

COMPOSITE DECK

Beams are spaced at 6.67’ o.c. to optimize the floor deck to span as close to 7’ as possible, while conforming to the 20’ x 30’ structural bay for the floor system. The 20 x 30 bay is the preferred bay ratio of 1.5 to 1, with Beams spanning in the 30’ direction and Girders spanning 20’.

OPEN-WEB JOISTS

Composite Deck Floor Load: 203 psf
Deck span and Beam spacing: 6.67’
Uniformly Distributed Tributary Load on Beam calculation: 203 psf (6.67’) = 1,354 plf
LOAD ON BEAM A: 1,354 plf

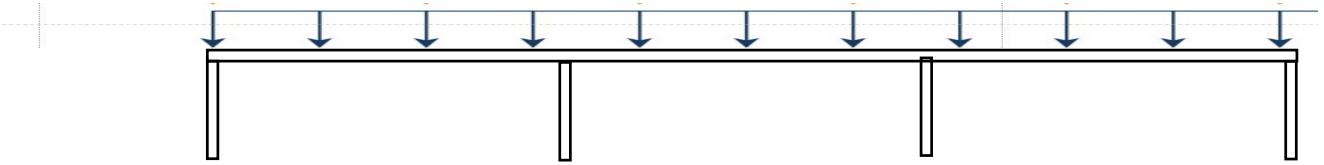
JOIST GIRDERS

Beams are spaced at 6.67’ o.c. to optimize the floor deck to span as close to 7’ as possible, while conforming to the 20’ x 30’ structural bay for the floor system. The 20 x 30 bay is the preferred bay ratio of 1.5 to 1, with Beams spanning in the 30’ direction and Girders spanning 20’.

Results indicated a wide-flange beam W18 x 35 is the lightest section to meet the load and beam span criteria. This beam is 17.7 inches deep and weighs 35 plf.
SELECTION FOR BEAM A: W18 X 35

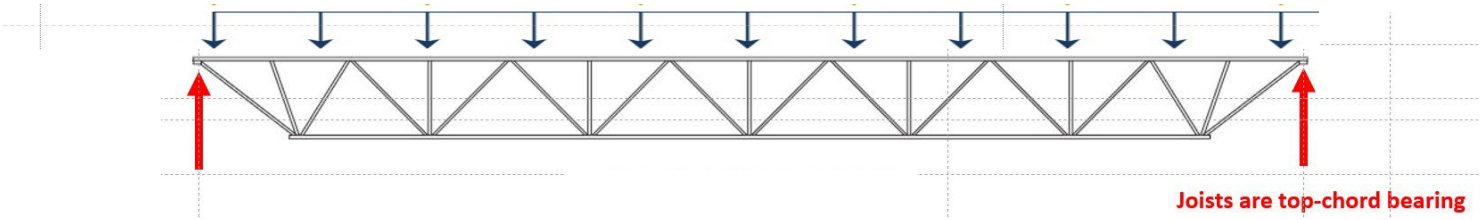
PRINCIPLE OF ECONOMY

Minimum of material to achieve a maximum of work
Typical floor beam sized for the lightest weight in pounds per lineal foot to carry to floor load. The lightest weight section uses the least amount of steel to perform the work of supporting the floor loads and structural floor deck.



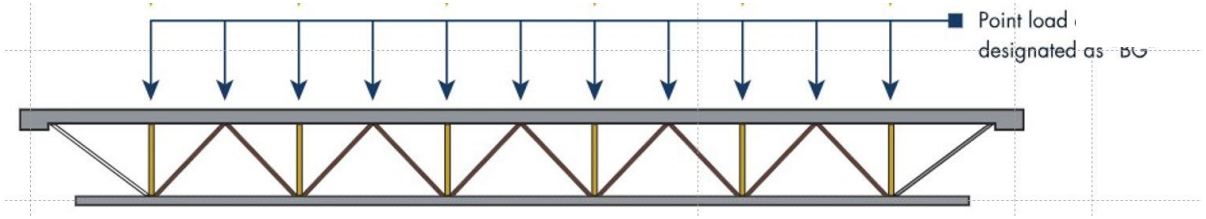
COMPOSITE DECK

Total Tributary Load: 40.62 kips
(203 psf x 200 sf)
Uniform Distributed Load: 1,354 plf
(203 psf x 6.67’ = 1,354 plf)



OPEN-WEB JOIST

Total Tributary Load: 40.62 kips
(203 psf x 200 sf)
Uniform Distributed Load: 1,354 plf
(203 psf x 6.67’ = 1,354 plf)



JOIST GIRDERS

Total Tributary Load: 40.62 kips
(203 psf x 200 sf)
Uniform Distributed Load: 1,354 plf
(203 psf x 6.67’ = 1,354 plf)


…Lateral Load-resistance

LATERAL LOAD resistance

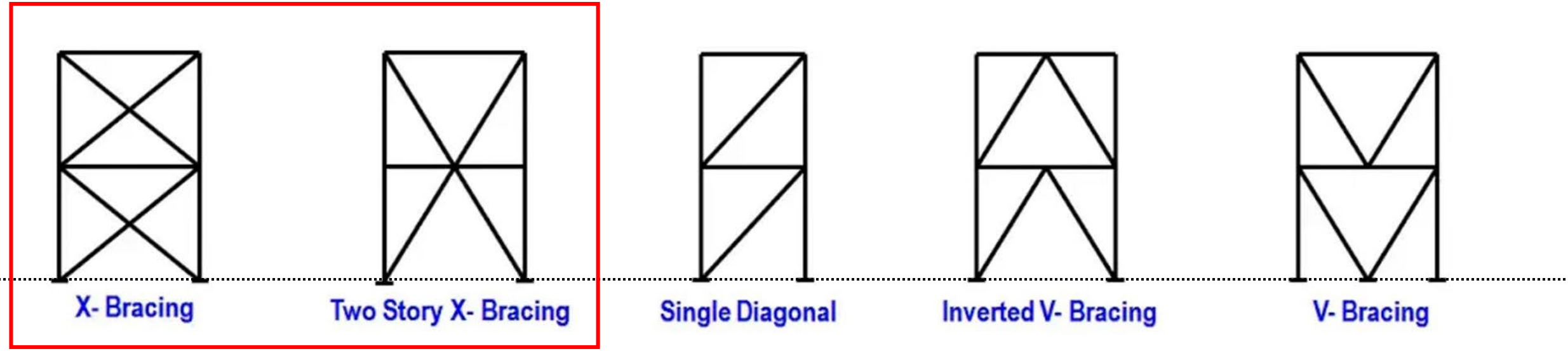
LATERAL LOAD RESISTANT TYPES

Diaphragms: Horizontal structural elements (typically floor structure and roof structure) that resist lateral loads by transferring the lateral load from the exterior walls or roof to vertical lateral-resisting elements. Floors and roofs act like large “beams” to transfer the lateral loads.

Braced Frames: Diagonal linear elements acting in tension or compression or both, that transfer lateral loads to vertical elements and down to the foundation and earth.

Moment Frames: Vertical and horizontal elements with rigid connections between them that resist rotation and allowing the transfer of lateral loads down to the foundation and earth.

Shear Walls: Vertical planar elements that resist lateral force through shear and bending, and transfer horizontal forces vertically to transfer down to the foundation and earth.



Concentric Braced Frames transfer lateral forces by connecting to columns and beams at their joints, forming a triangulated "truss." They are called concentric because the axes of the braces, columns, and beams align at the connections. They provide high stiffness and strength to resist lateral loads.

…any one of these geometries can be applied to our Steel Building.
…it must have triangulation.

CONCENTRIC BRACED FRAMES

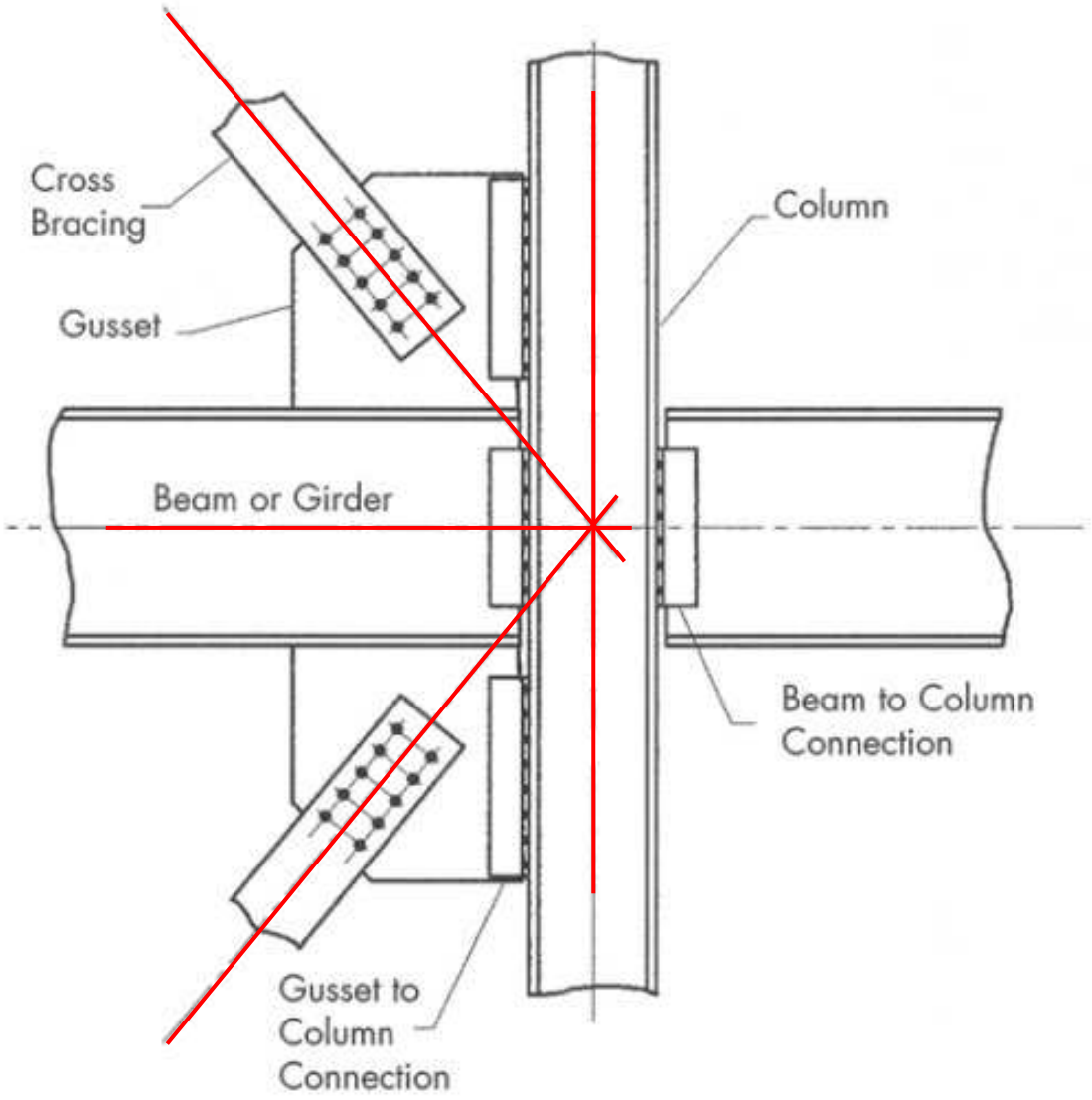
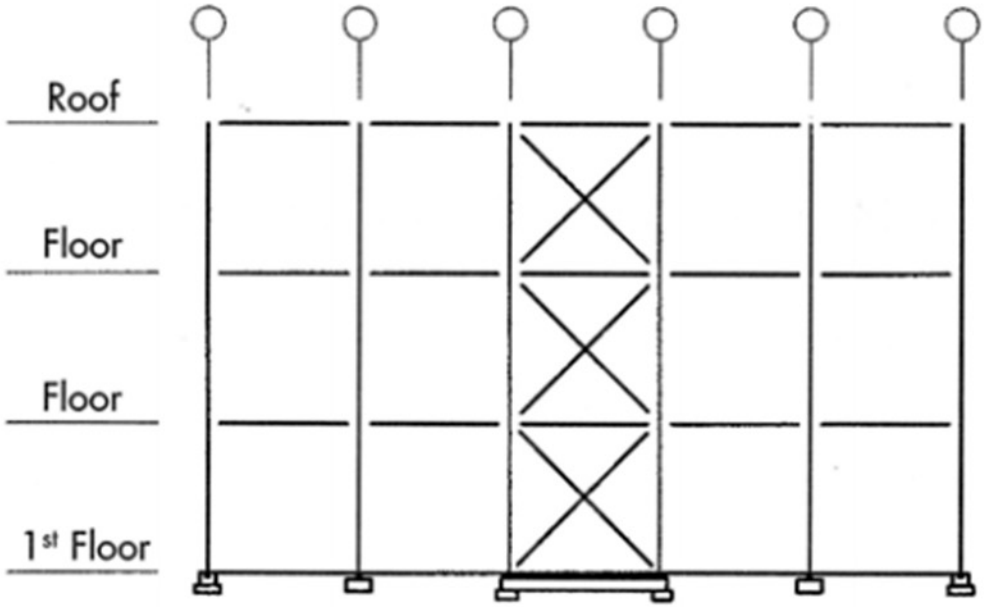
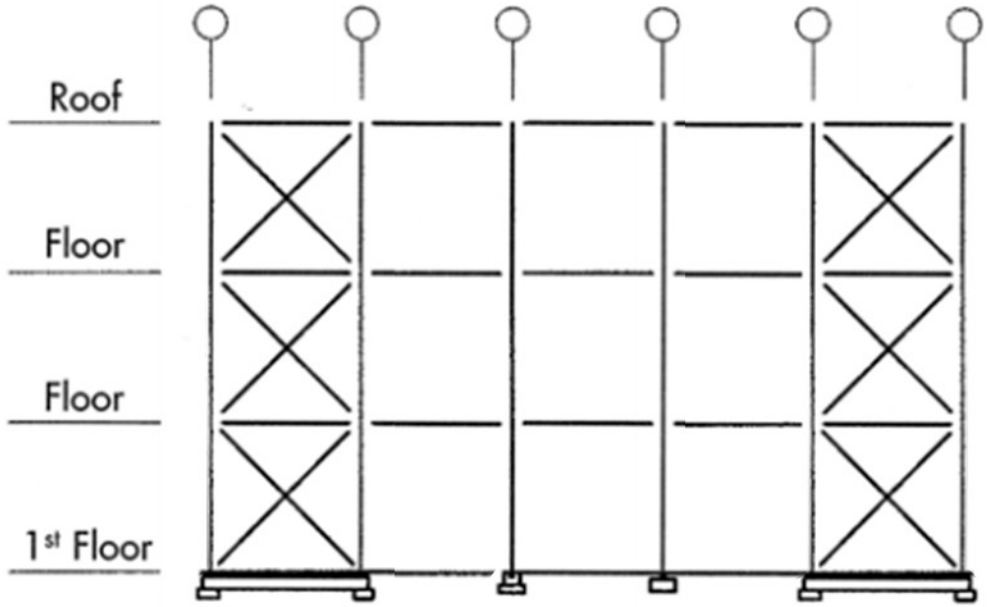


Fig. 6. Typical beam to column brace connections.

CONCENTRIC BRACED FRAMES axis always align at a concentric point (like truss nodes)

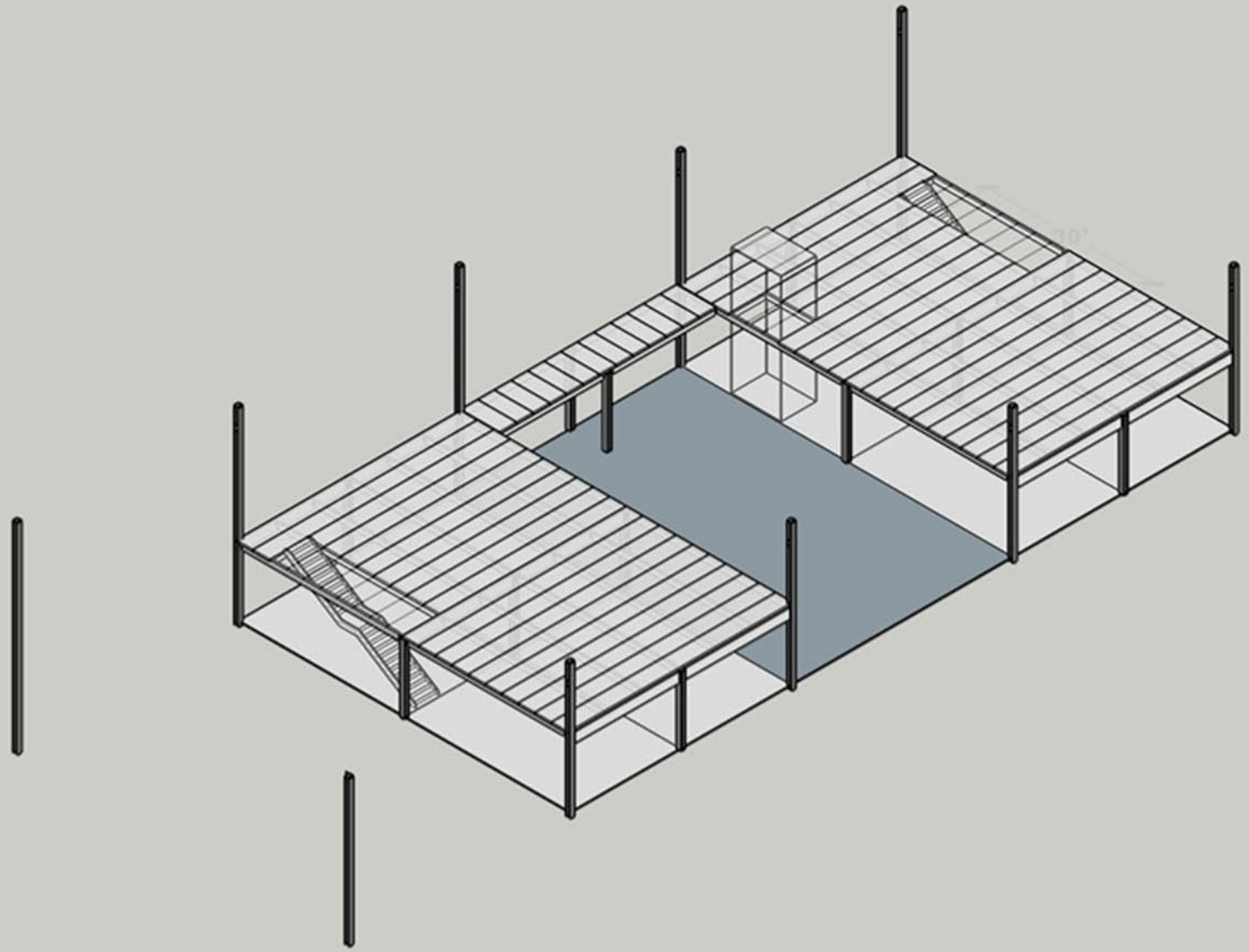


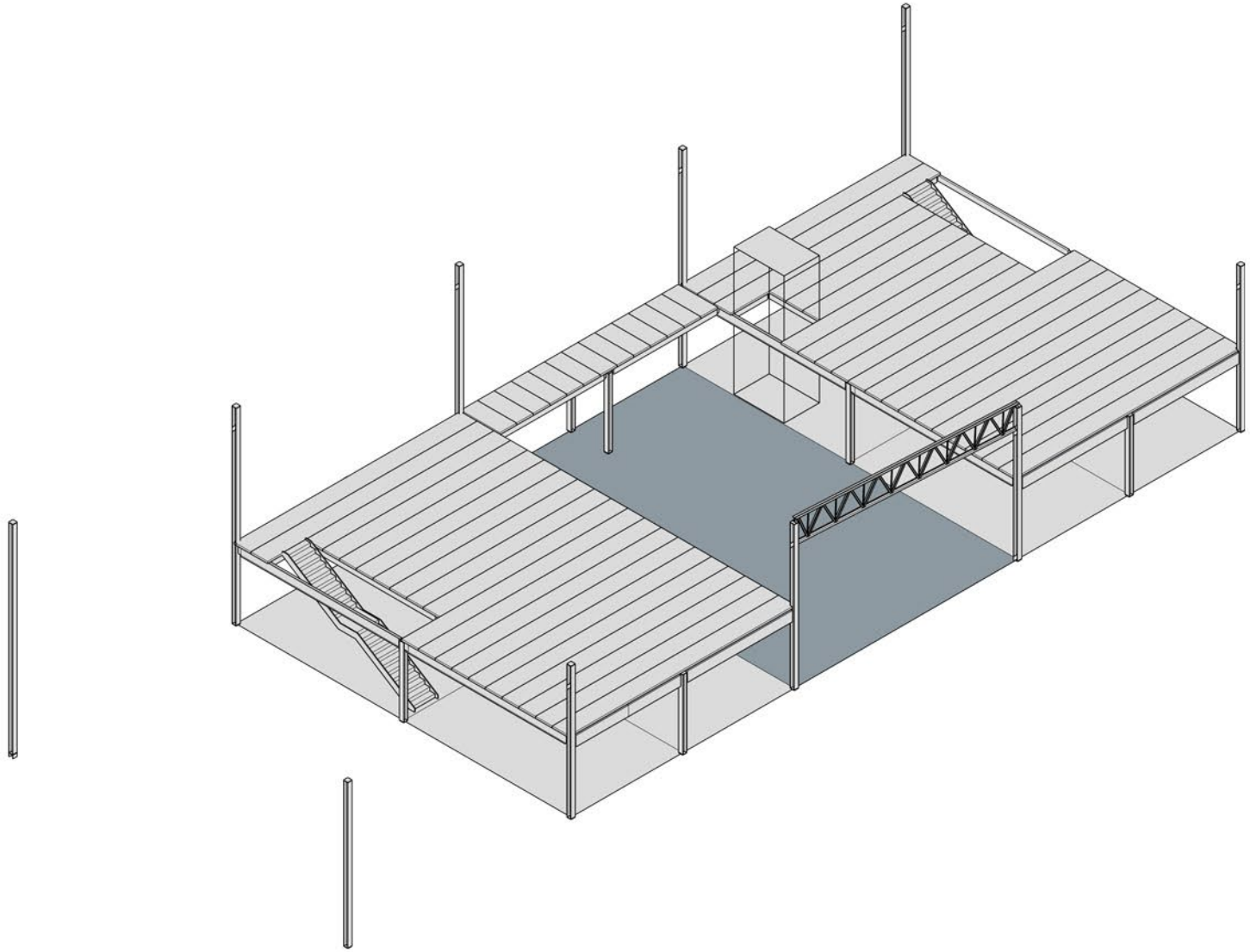
Across the central bay, both sides

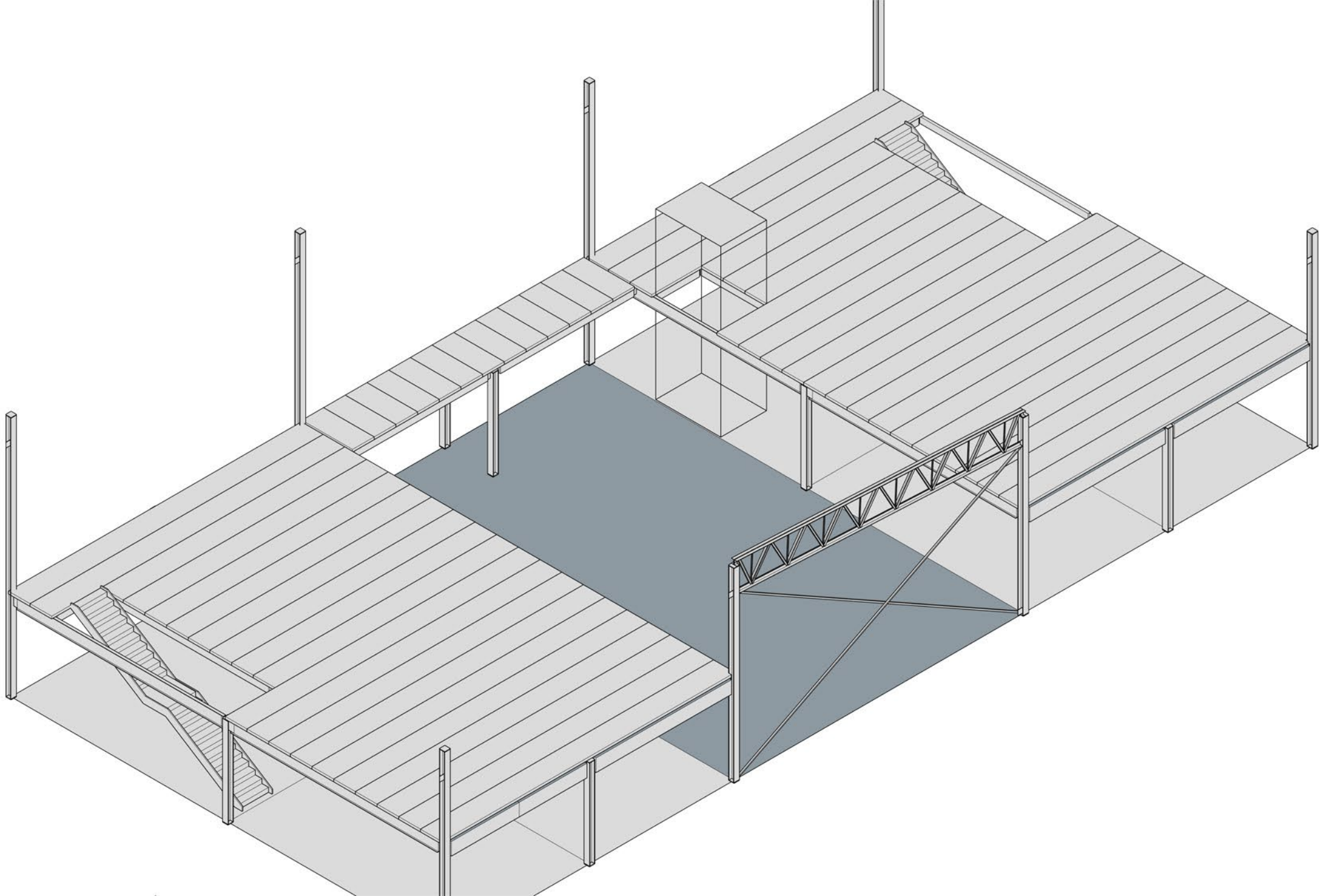


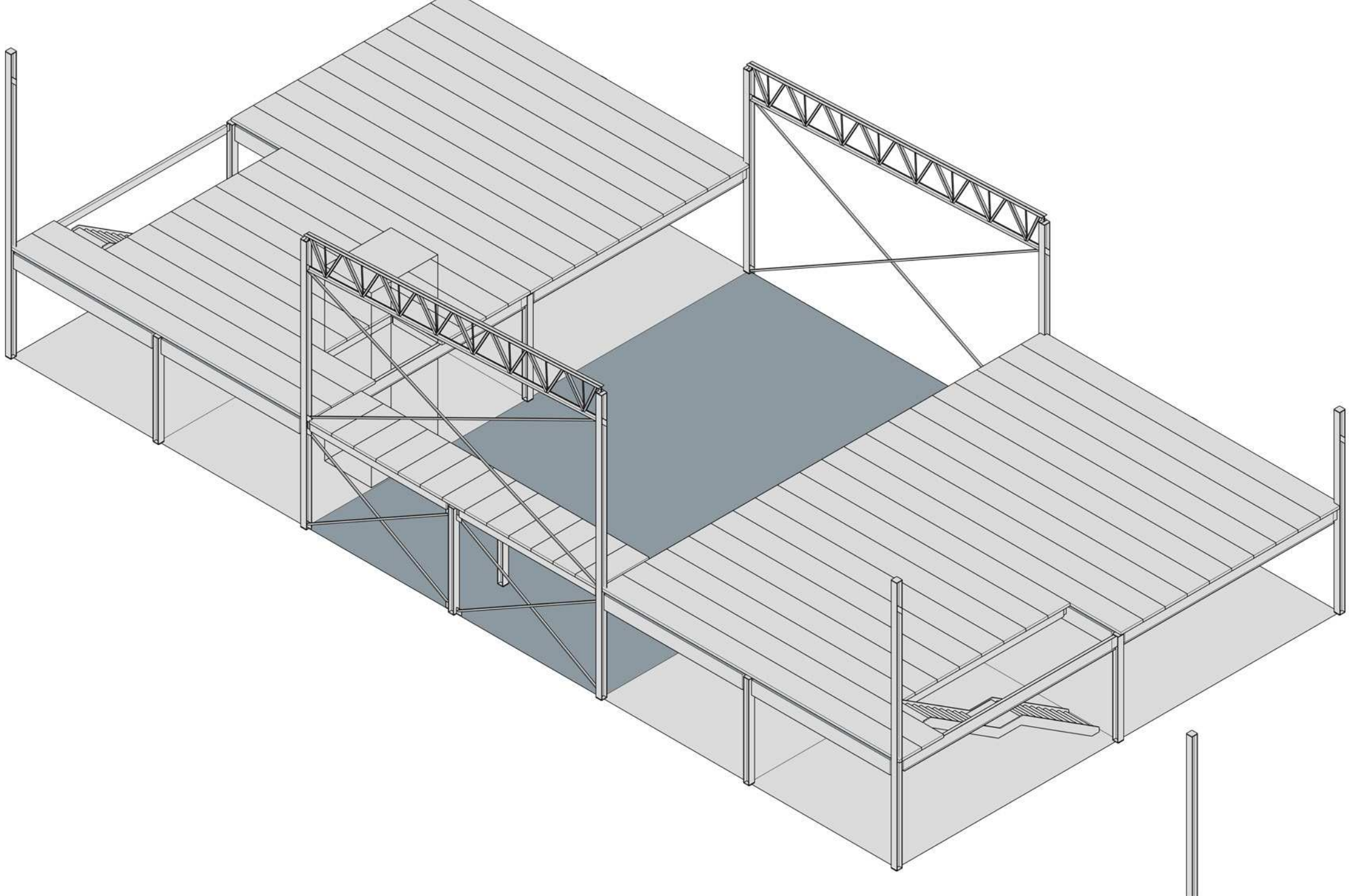
At the four corners

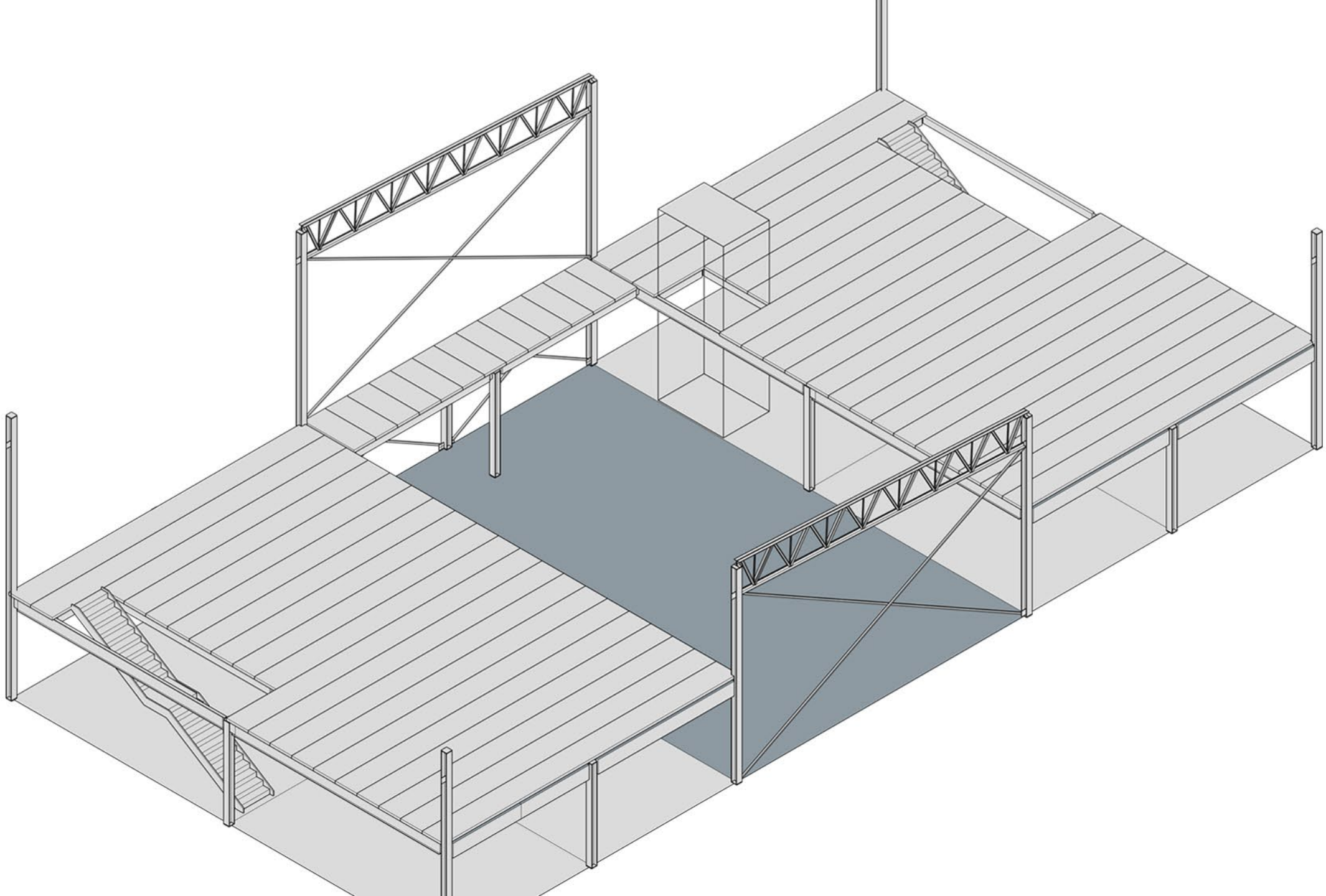
CONCENTRIC BRACED FRAMES common application options

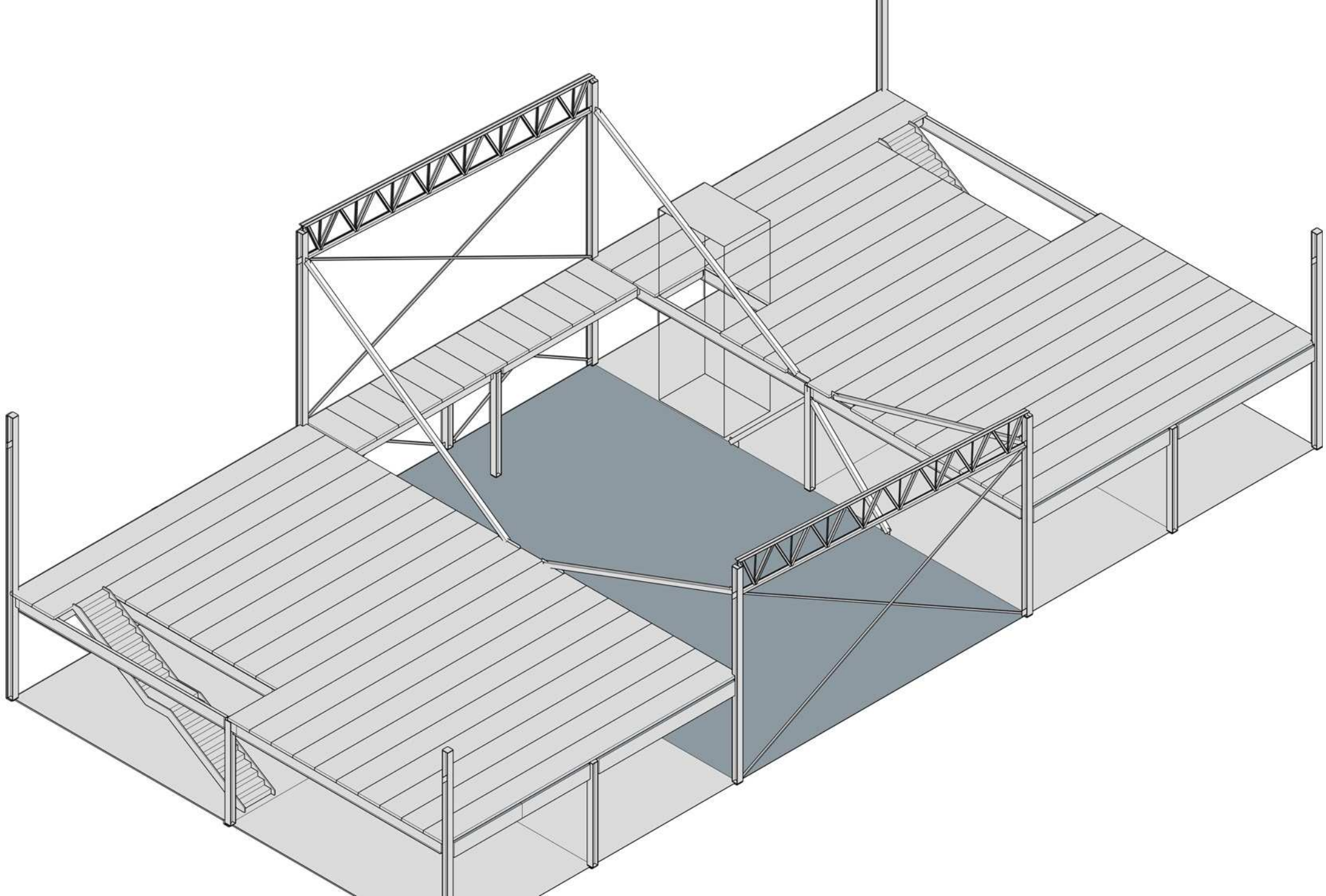


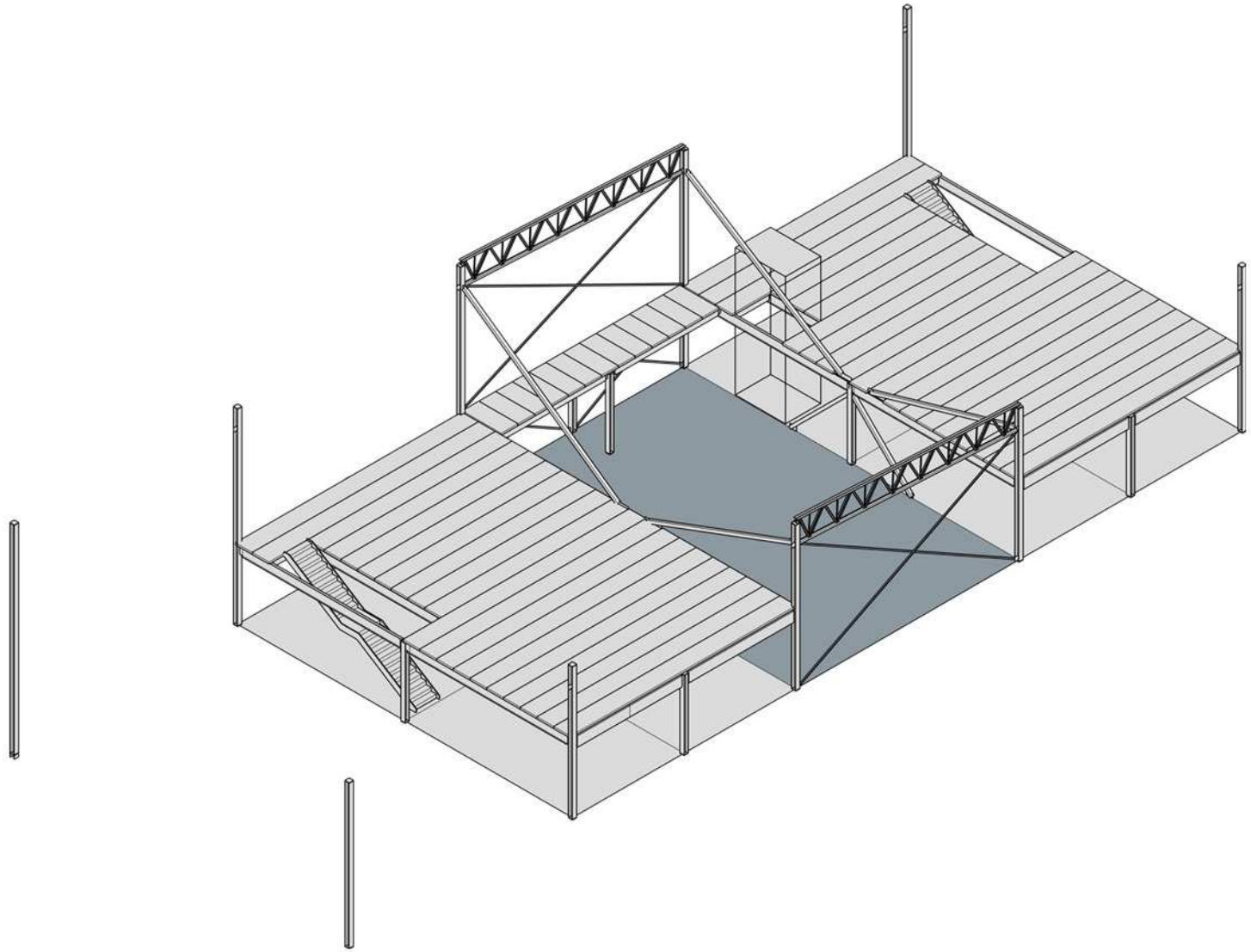


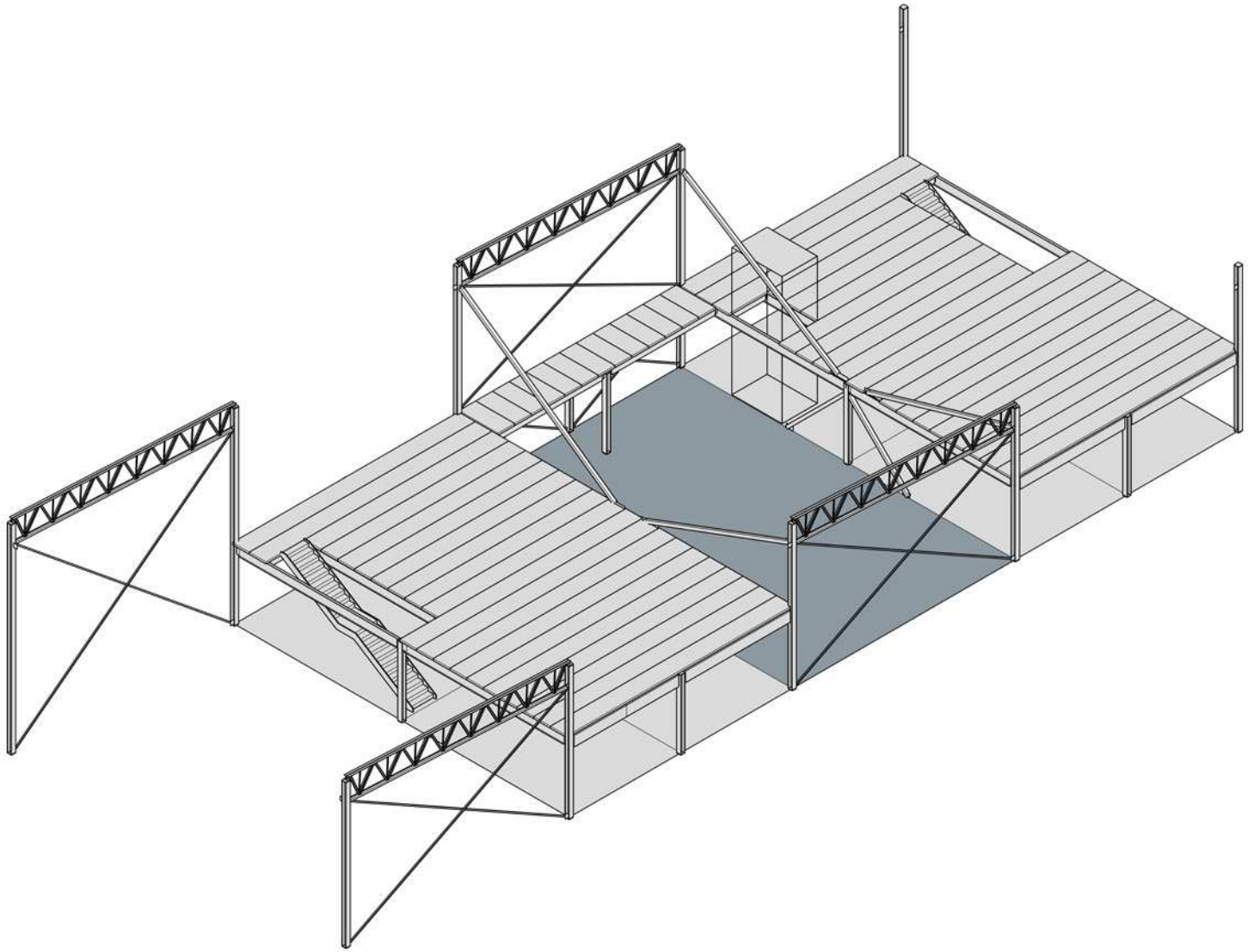


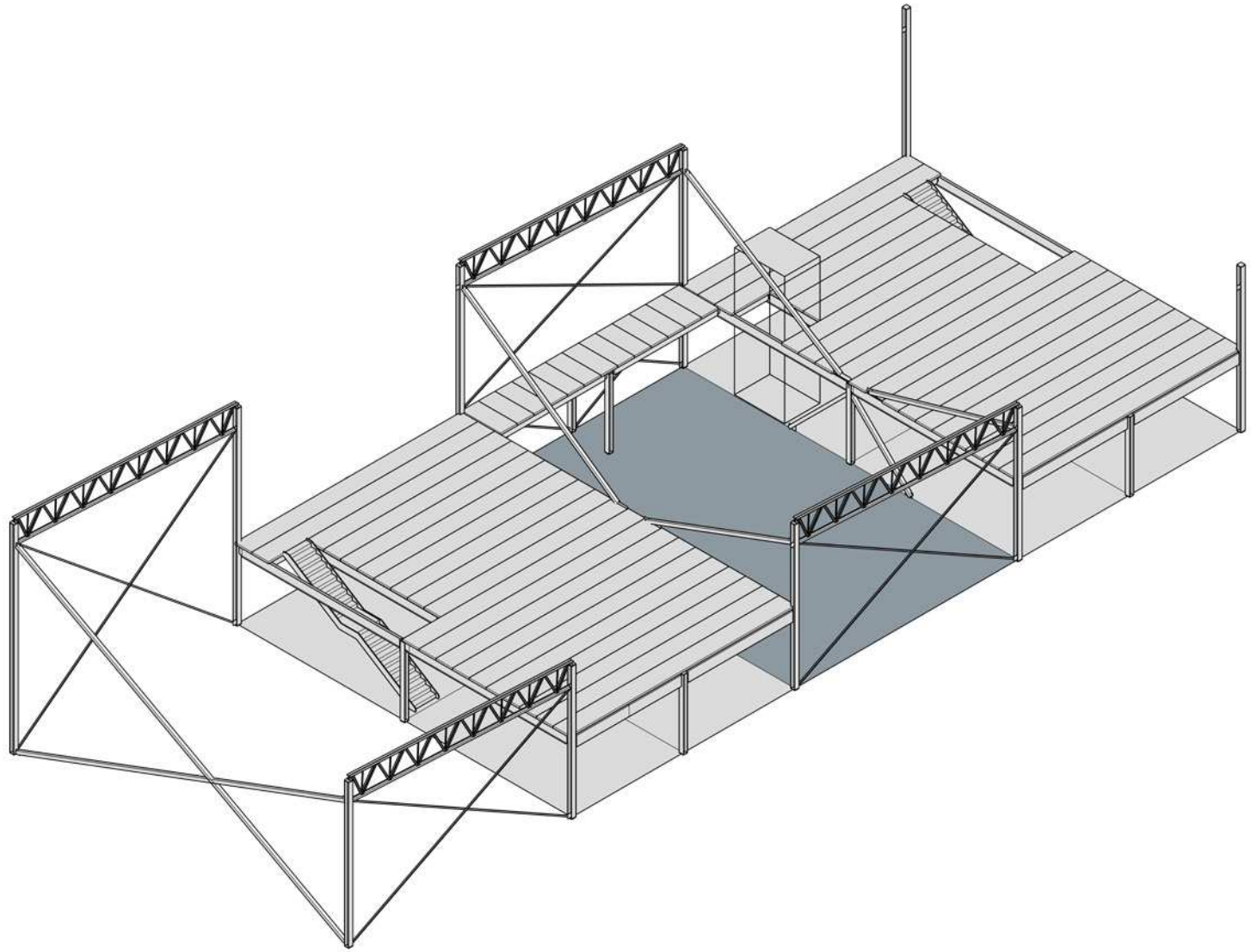


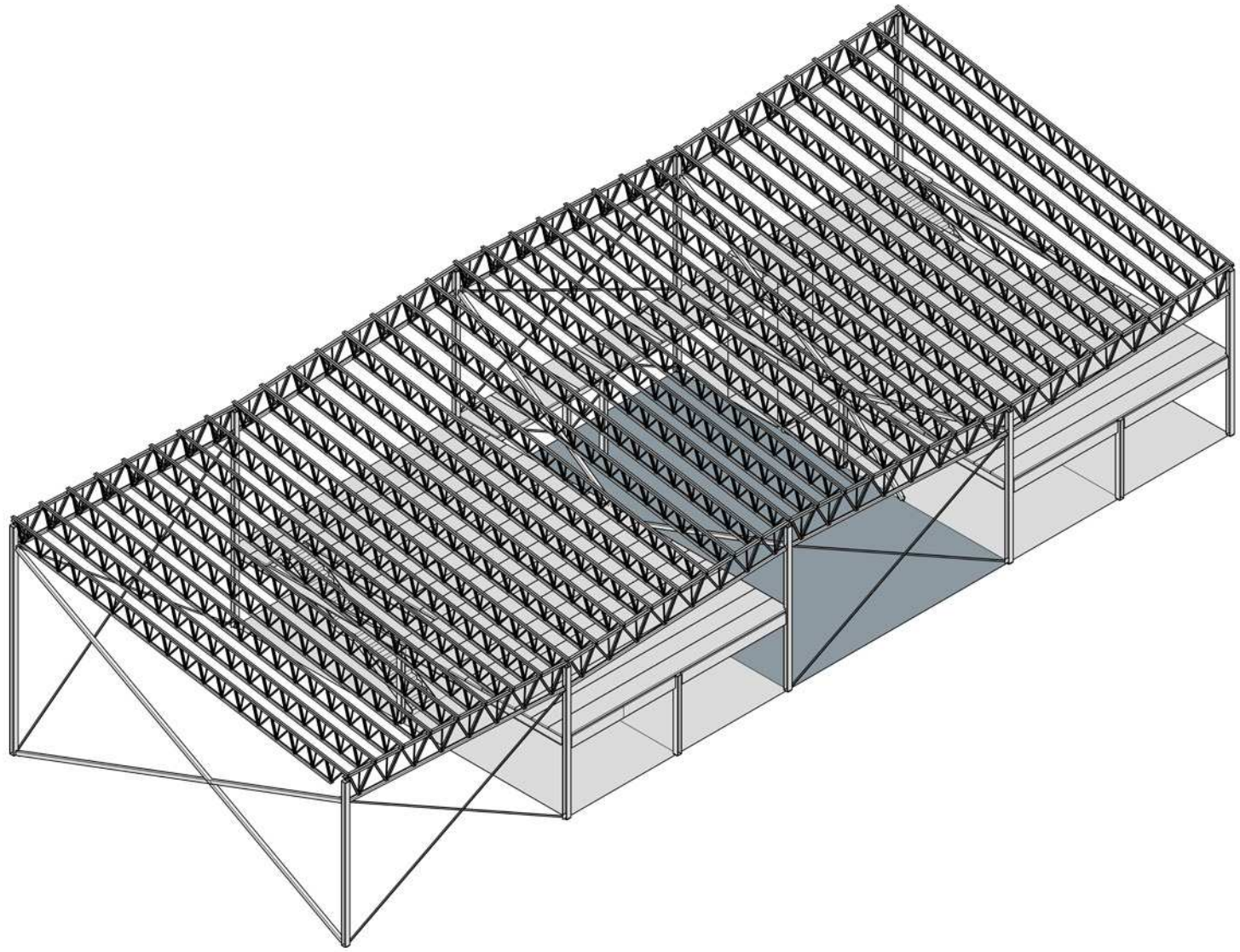


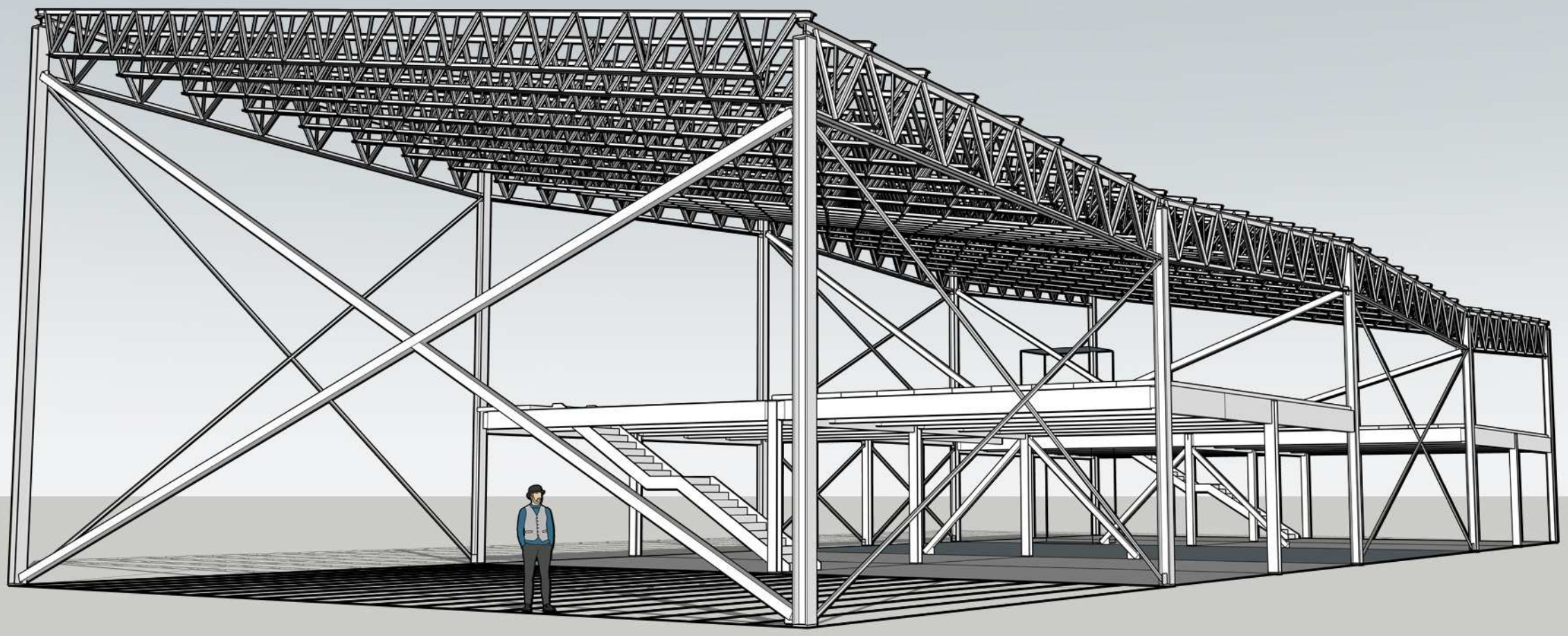


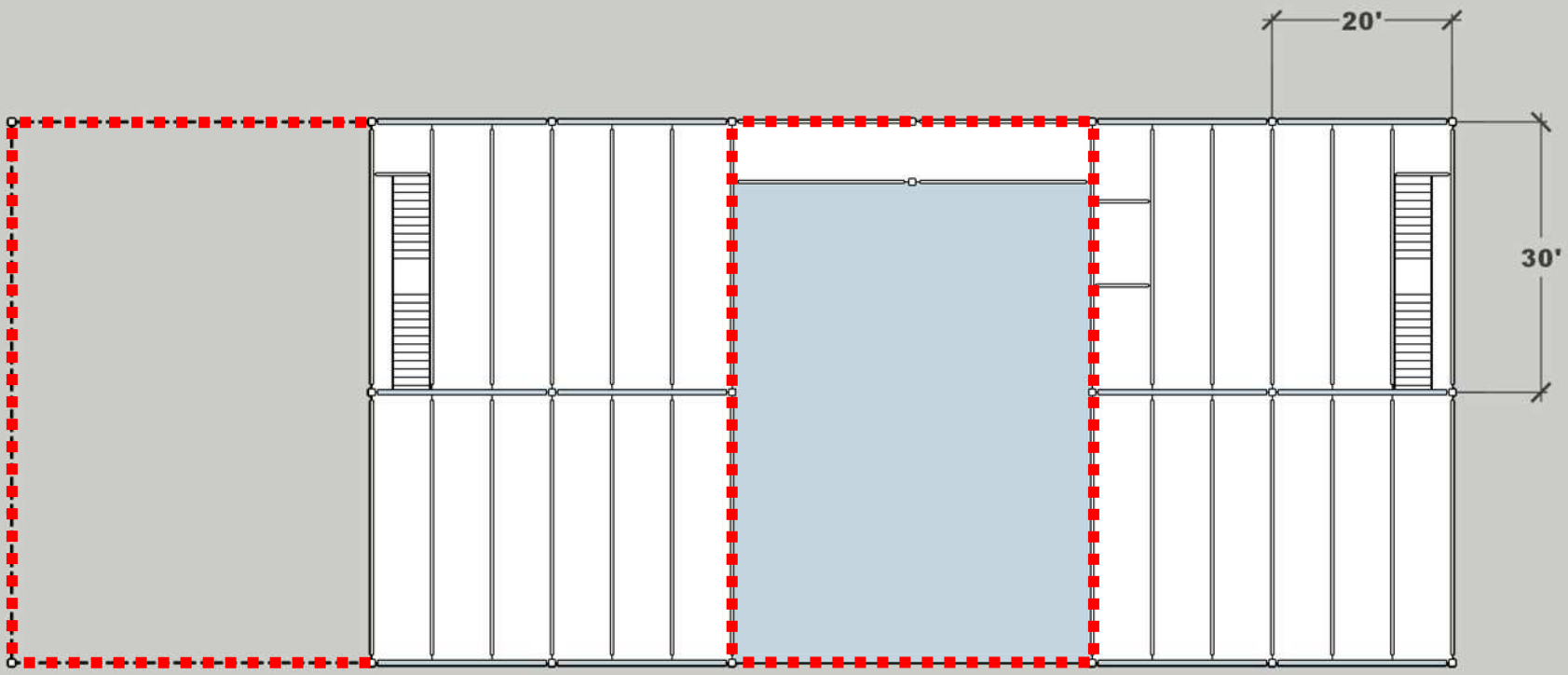




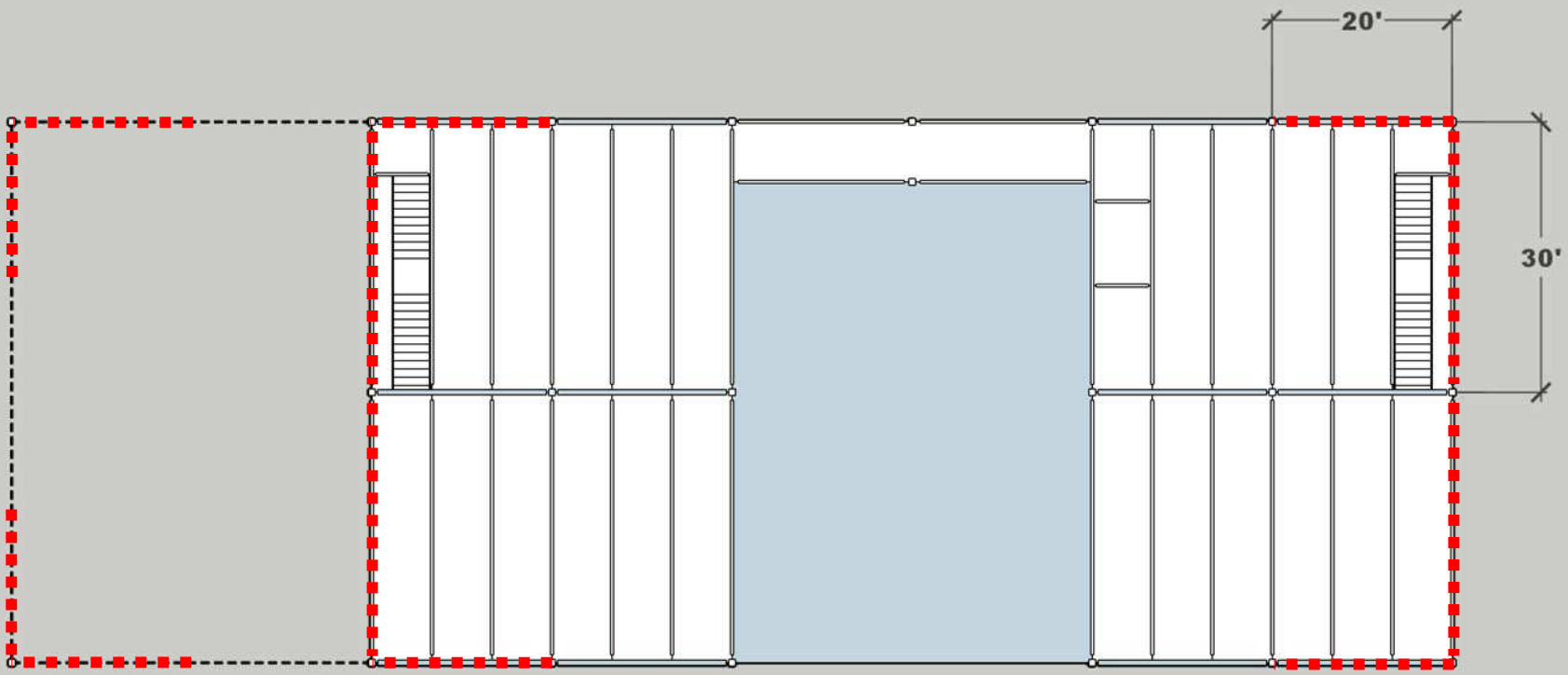




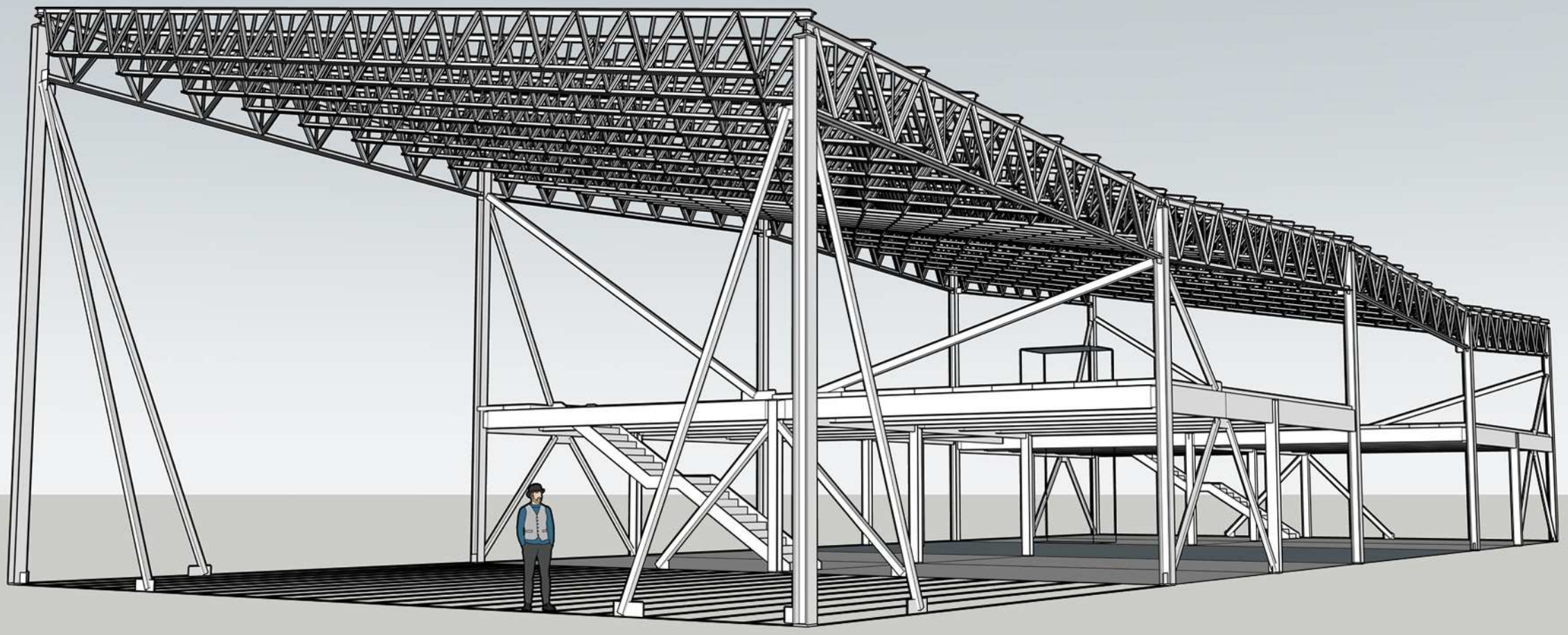




4060 C **Framing Plan with Lateral Bracing locations at central bay, and porch**



4060 C Framing Plan with Lateral Bracing locations at corners, and porch



STEEL BUILDING PROJECT

…questions on Floor or Lateral Structure?